



Location Privacy

Cyrus Shahabi, Ph.D.

*Professor of Computer Science, Electrical Engineering
& Spatial Sciences*

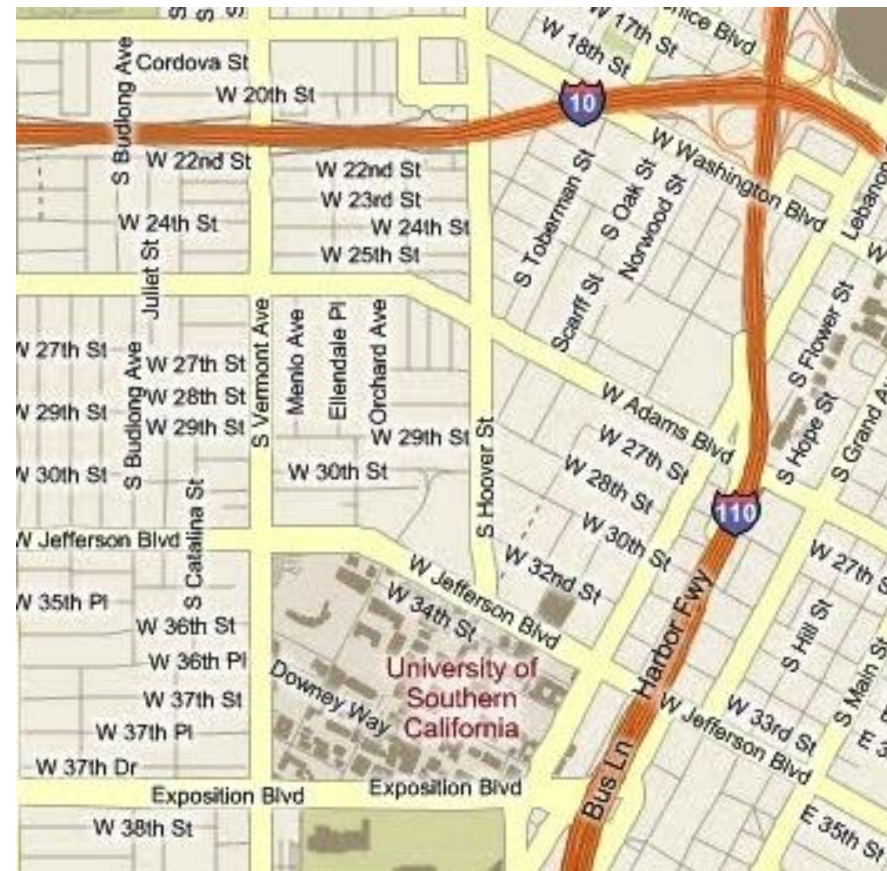
*Viterbi School of Engineering
University of Southern California*

Los Angeles, CA 900890781

shahabi@usc.edu



Privacy in Location Based Services





Privacy in Location Based Services



POI

Location
Server (LS)





Privacy in Location Based Services

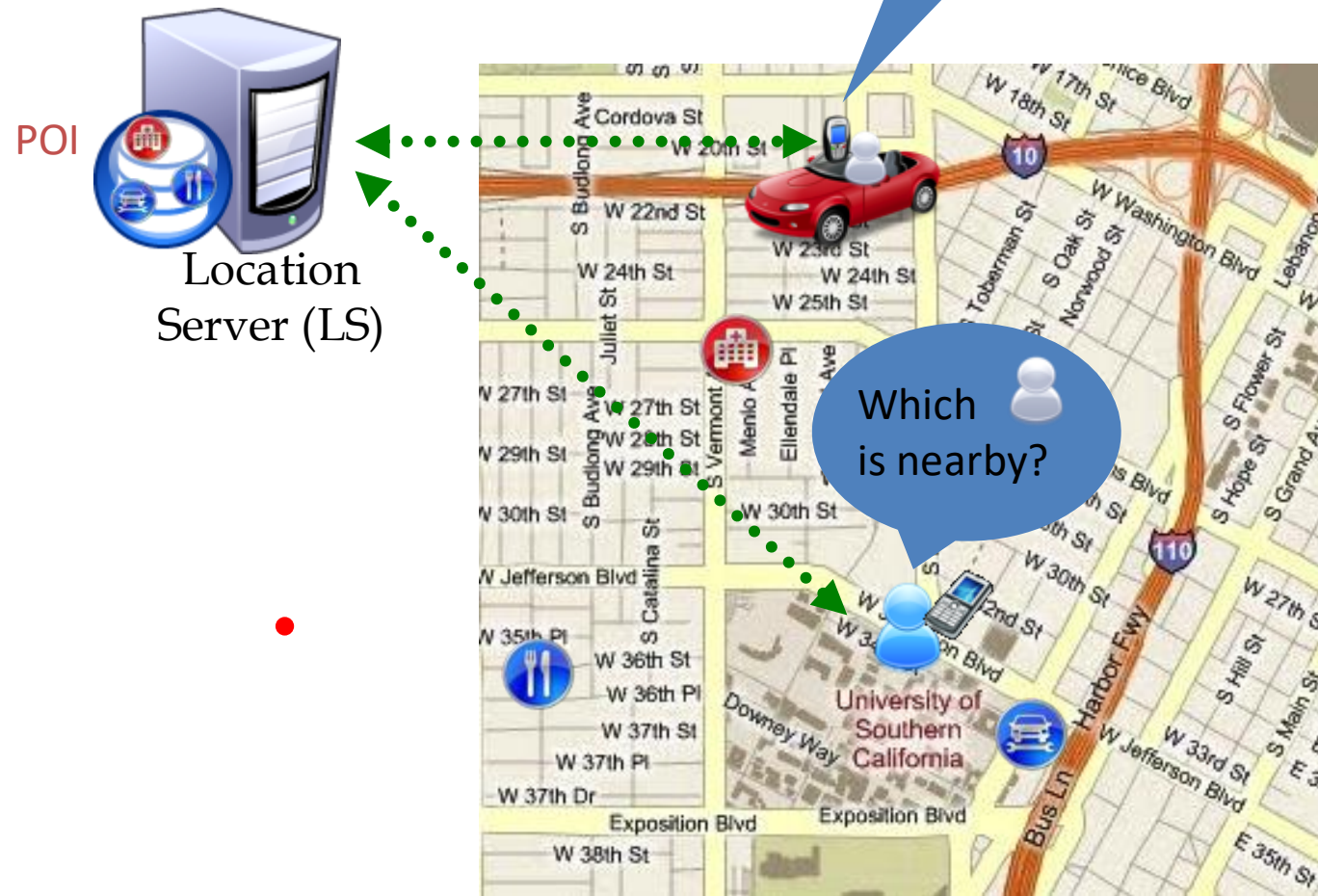




Privacy in Location Based Services



Where's the nearest ?





Privacy in Location Based Services





Privacy in Location Based Services



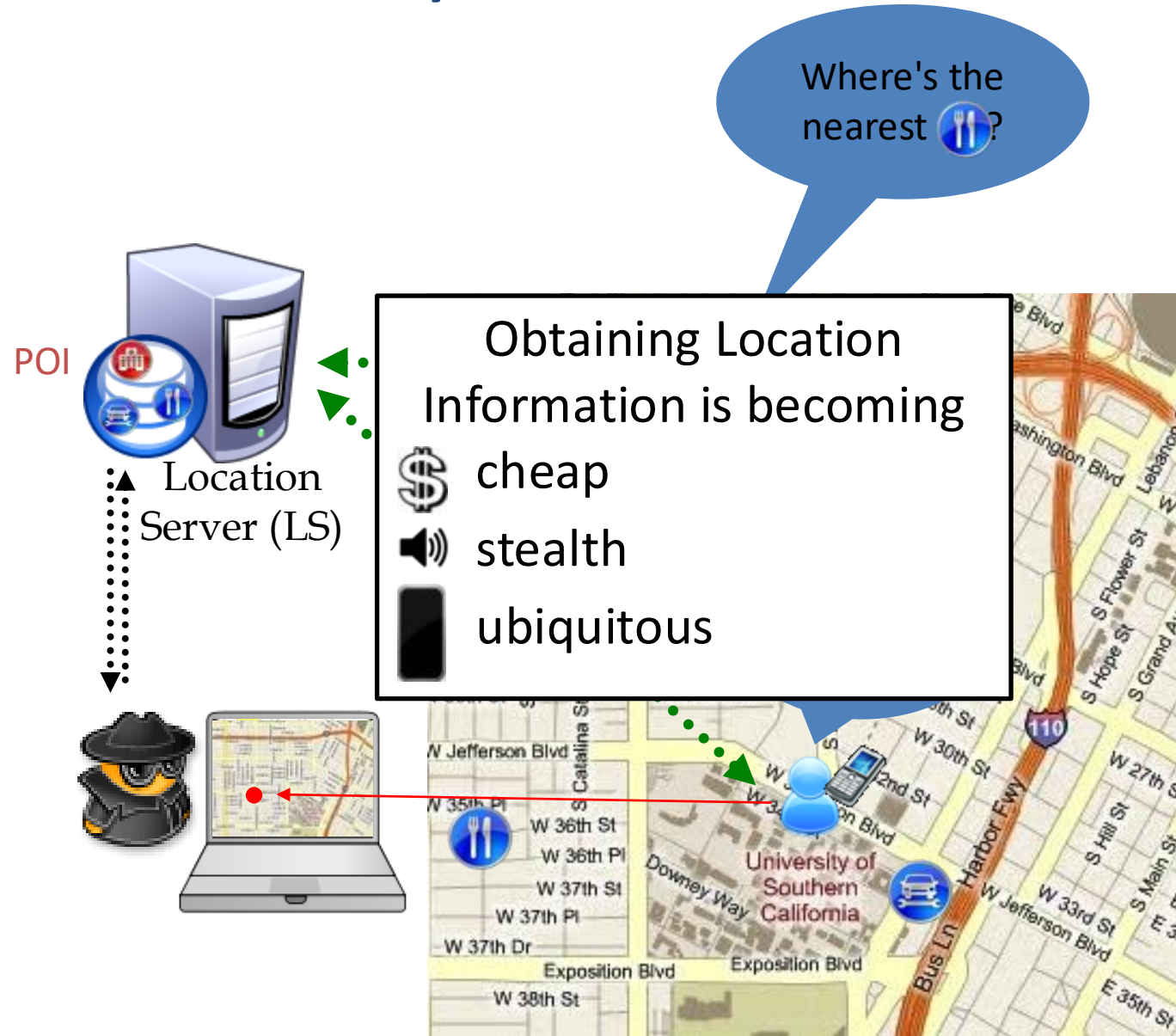


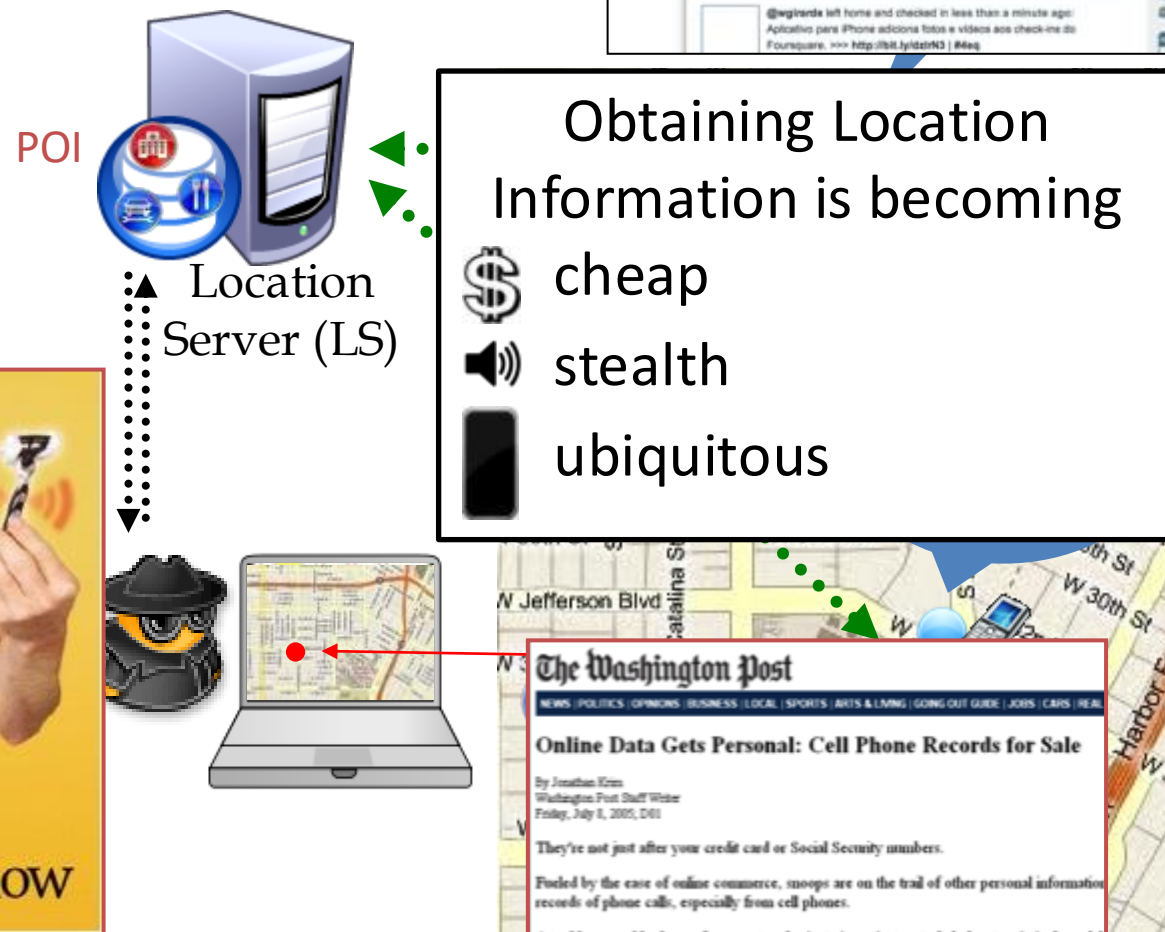
Privacy in Location Based Services





Privacy in Location Based Services





Location Privacy Threats



FOX NEWS .com **U.S. & WORLD**
Updated: 3-28-06 9:42pm ET
SEARCH GO

E-MAIL STORY PRINTER FRIENDLY FOXFAN CENTRAL

Man Accused of Stalking Ex-Girlfriend With GPS

Saturday, September 04, 2004
Associated Press

GLENDAL, Calif. — Police arrested a man they said tracked his ex-girlfriend's whereabouts by attaching a global positioning system (**search**) to her car.

Ara Gabrielyan, 32, was arrested Aug. 29 on one count of **stalking** (**search**) and three counts of making criminal threats. He was being held on \$500,000 bail and was to be arraigned Wednesday.

"This is what I would consider stalking of the 21st century," police Lt. Jon Perkins said.

<http://www.foxnews.com/story/0,2933,131487,00.html>



USA TODAY Classifieds: [cars.com](#) | [careerbuilder.com](#) | [eHarmony.com](#) |

Home News Travel Money Sports Life Tech Weather
Search

powered by **YAHOO!** GO

Tech Products
[Products home](#)
[Edward C. Baig](#)
[Kim Komando](#)
[Ask Kim](#)
Gaming
[Gaming home](#)
[Arcade](#)
[Jinny Gudmundsen](#)
[Marc Saltzman](#)
Science & Space
[Science & Space](#)
[April Holladay](#)
[Dan Vergano](#)

Tech

• [E-MAIL THIS](#) • [PRINT THIS](#) • [SAVE THIS](#) • [MOST POPULAR](#) • [SUBSC](#)

Posted 12/30/2002 7:57 PM

Authorities: GPS system used to stalk woman

KENOSHA, Wis. (AP) — A man was charged Monday with stalking his former live-in girlfriend with help from a high-tech homing device placed under the hood of her car.

Paul Seidler, 42, was arrested during the weekend. On Monday, he was charged with stalking, burglary, second-degree reckless endangerment and disorderly conduct, and ordered held on \$50,000 bail.

According to a criminal complaint, Connie Adams asked Seidler to move out of her apartment Oct. 25 after a three-year relationship. Prosecutors say he immediately began following her, including when she ran errands and went to work.

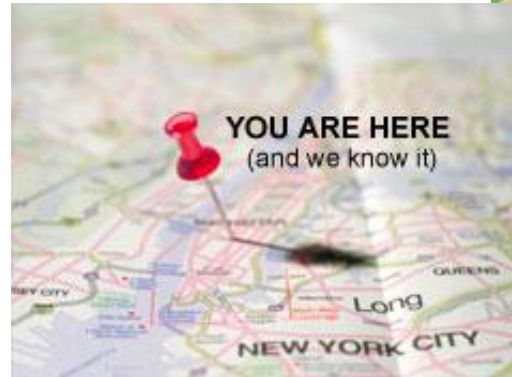
http://www.usatoday.com/tech/news/2002-12-30-gps-stalker_x.htm



Location Privacy in Industry

- ~ 26,000 persons are victims of GPS stalking annually, including by cellphone

– [Jan 2009 report by the Department of Justice]



- ~ 50% top apps for Apple iPhones and Google Android smartphones disclosed a user's location to third parties without his or her consent

– [Dec 2010 investigation by the *Wall Street Journal*]





Location Privacy in Industry

- In April 2011, consumers learned that their smartphones were automatically sending out information about their smartphone's location

[The Location Privacy Protection Act of 2011]

Google location tracking can invade privacy, hackers say
Unique IDs + router addresses = potential abuse



Got an iPhone or 3G iPad? Apple is recording your moves

A hidden file in iOS 4 is regularly recording the position of devices.

(37.7881876, -122.4043698)

Map Satellite Hybrid

Map data ©2011 Google - [Terms of Use](#)

```
"latitude": 37.7881876
"longitude": -122.4043698
"country": "United States"
"country code": "US"
"area": "San Francisco"
"city": "San Francisco"
"state": "California"
"zip": "94102"
"reg router": "192.168.1.1"
```

Mobile Apps for Kids:
Current Privacy Disclosures are Disappointing

Reg router, courtesy of Google Maps



Location Privacy Protection Act 2011

- **The Location Privacy Protection Act of 2011** requires any company that may obtain a customer's location information from his smartphone to
 - 1) Get that customer's express consent before collecting his location data
 - 2) Get that customer's express consent before sharing his or her location data with third parties

**Apple, Google, And Others Agree To
Mobile App Privacy Policy Guidelines**





Location Privacy in Industry

Location Information

- ☒ Include me in the public list of people who are currently checked in at a venue ?
- ☒ Let me earn Mayorships (I realize that Mayorship is a public office) ?
- ☒ When my friends check in with me, it's okay to include my name on their check-in tweets or Facebook wall posts ?
- ☒ Let venue managers see when I check in to their business, or when I am one of their best customers ?

- All these options in Foursquare default to on
- One is providing a sort of path that strangers can follow from a face on the street to a name, other photos, current location, and a number of other things



Location Priv

Location Infor

- ☒ Include i
- ☒ Let me e
- ☒ When m
- Facebook
- ☒ Let venu
- custome

- All these
- One is p
- from a f
- location

Tech



Where Everybody Knows Your Name. Apps tell strangers what they have in common

By Harry McCracken

SOCIAL NETWORKS FIRST persuaded millions of us to start cataloging our friends, family members and high school classmates. The networks got us to post photos, tweet our every thought and tend our virtual farms. Now the next wave wants to cross over into the real world and introduce us to nearby strangers with common interests—and perhaps a desire to make a new friend.

There are at least 15 new smart-phone apps pushing this concept, which techies call ambient social networking. Silicon Valley is rushing to fund these “people discovery” start-ups, and everybody at South by Southwest (SXSW) Interactive—the annual nerdfest in Austin that famously gave Twitter its big break in 2007—seemed to be tinkering with one of them: Highlight, an eight-week-old iPhone app, is designed to reveal real-life connections you didn’t know you had, as well as alert you to the presence of friends you might otherwise miss. Co-founder Paul

Davison calls it a “sixth sense.”

Highlight, which has yet to make public how many people are using the app, works by rummaging through your Facebook account to see whom you know and what topics you like. Then it uses your iPhone’s GPS to inform you when, say, a fellow conference attendee who’s a former co-worker’s buddy is in your immediate vicinity or when a good-looking patron who sits down at the other end of the bar.

It’s a big shift for the tech industry. Unlike Foursquare—2009’s SXSW darling, which now has 15 million members sharing their locations by “checking in” so they can earn discounts and other rewards—Highlight monitors your whereabouts continuously and automatically shares them with fellow members both in and outside your existing circle of friends. That introduces new privacy concerns and strikes some critics as enabling a form of high-tech stalking.

In its current form, Highlight is a rough draft of a powerful idea. Some problems are minor: Highlight has an odd habit of telling you who’s nearby even when you’re passing in a moving vehicle. It also drains your phone’s battery as it constantly sends location data back to its servers, a problem the company says it is addressing.

But getting Highlight’s algorithm to highlight people you actually want to meet is the biggest challenge of all. “We’re just scratching the surface,” says Davison. “If we both went to the same high school, it’s more interesting if the school is 4,000 miles away than if it’s two miles away.” At SXSW, I wasn’t moved to track down any of the individuals Highlight identified as people of interest. I did, however, keep striking up rewarding conversations with folks I encountered in hotel lobbies and at parties, no app required. Serendipity in its natural form is a wonderful thing—and manufacturing it won’t be easy.

Come Here Often? Apps that help you meet people



Highlight

Launched Jan. 24, this buzzed-about app uses GPS to let you know when a friend of one of your Facebook friends is close by



Glancee

Gaining steam, with 10,000 members, the app pinpoints others who are “steep away” and share your interests



Foursquare

One of the first location-sharing apps, it has 15 million users “checking in,” a.k.a. broadcasting where they are, to earn perks



OkCupid

Matchmaking gone mobile. This app’s Locoals feature sorts nearby singles by compatibility

—KEITH WAGSTAFF

venue ?

their check-in tweets or

in one of their best

can follow
photos, current



Isn't Confidentiality Enough?

Sensitive information obtained by
anonymous location data



Isn't Confidentiality Enough?

Sensitive information obtained by
anonymous location data

- Barabási et al., Nature'08

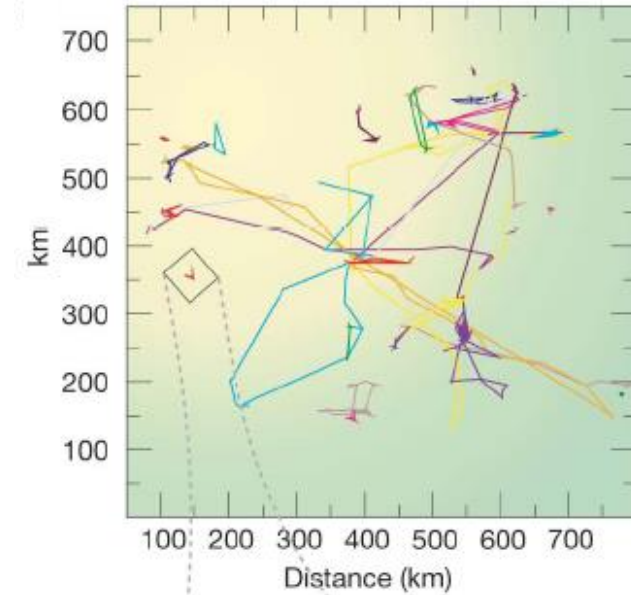


Isn't Confidentiality Enough?

Sensitive information obtained by
anonymous location data

- Barabási et al., Nature'08

Human Mobility $\Rightarrow$ Spatial Probability Distribution



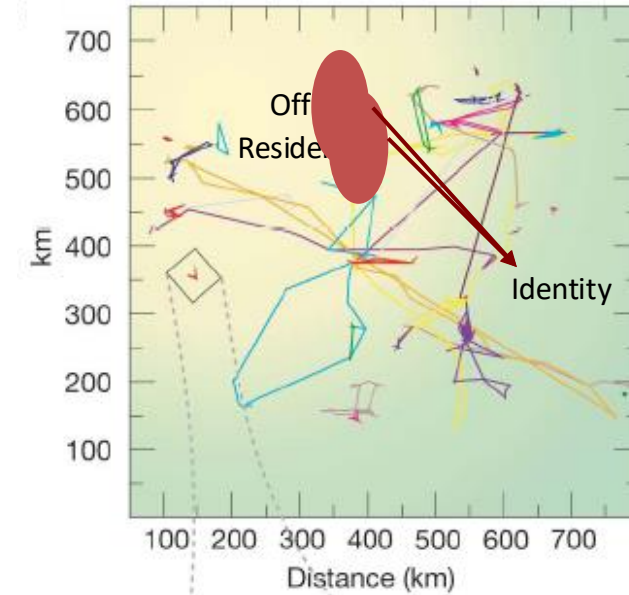


Isn't Confidentiality Enough?

Sensitive information obtained by
anonymous location data

- Barabási et al., Nature'08

Human Mobility $\Rightarrow$ Spatial Probability Distribution





Isn't Confidentiality Enough?

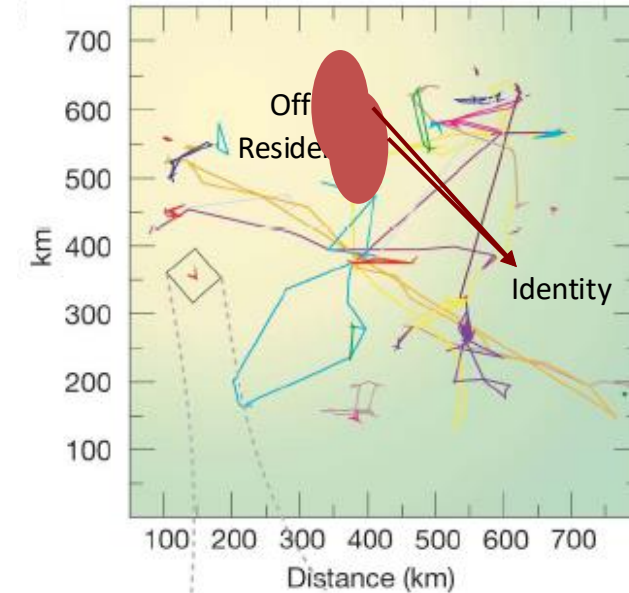


Sensitive information obtained by anonymous location data

- Barabási et al., Nature'08

Human Mobility $\Rightarrow$ Spatial Probability Distribution

- Four spatiotemporal points are enough to uniquely re-identify 90% of individuals





Isn't Confidentiality Enough?

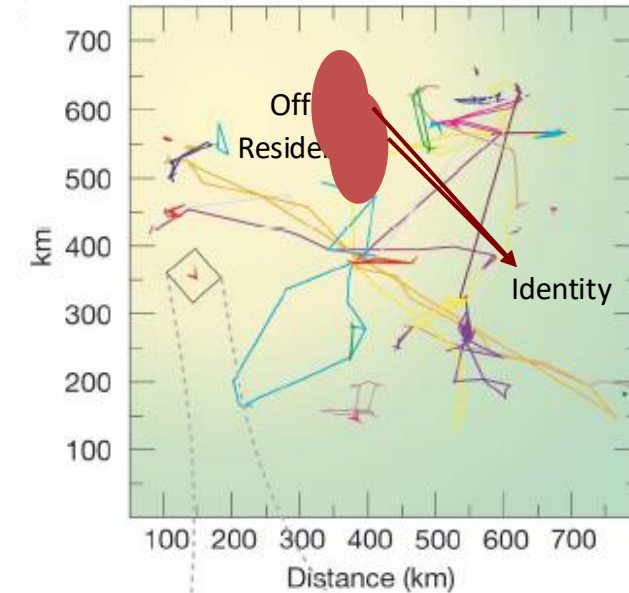


Sensitive information obtained by anonymous location data

- Barabási et al., Nature'08

Human Mobility $\Rightarrow$ Spatial Probability Distribution

- Four spatiotemporal points are enough to uniquely re-identify 90% of individuals
- Anonymous queries leak information





Isn't Confidentiality Enough?



Sensitive information obtained by anonymous location data

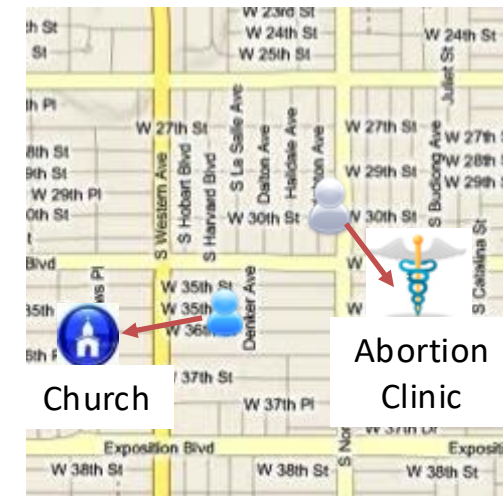
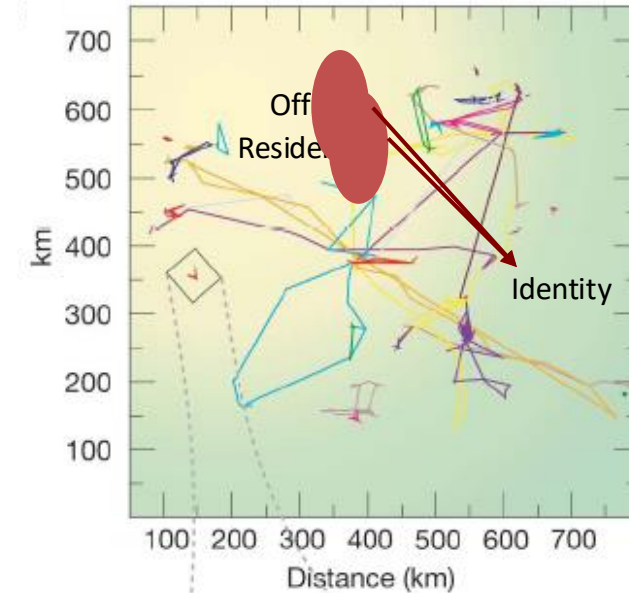
- Barabási et al., Nature'08

Human Mobility $\Rightarrow$ Spatial Probability Distribution

- Four spatiotemporal points are enough to uniquely re-identify 90% of individuals

- Anonymous queries leak information

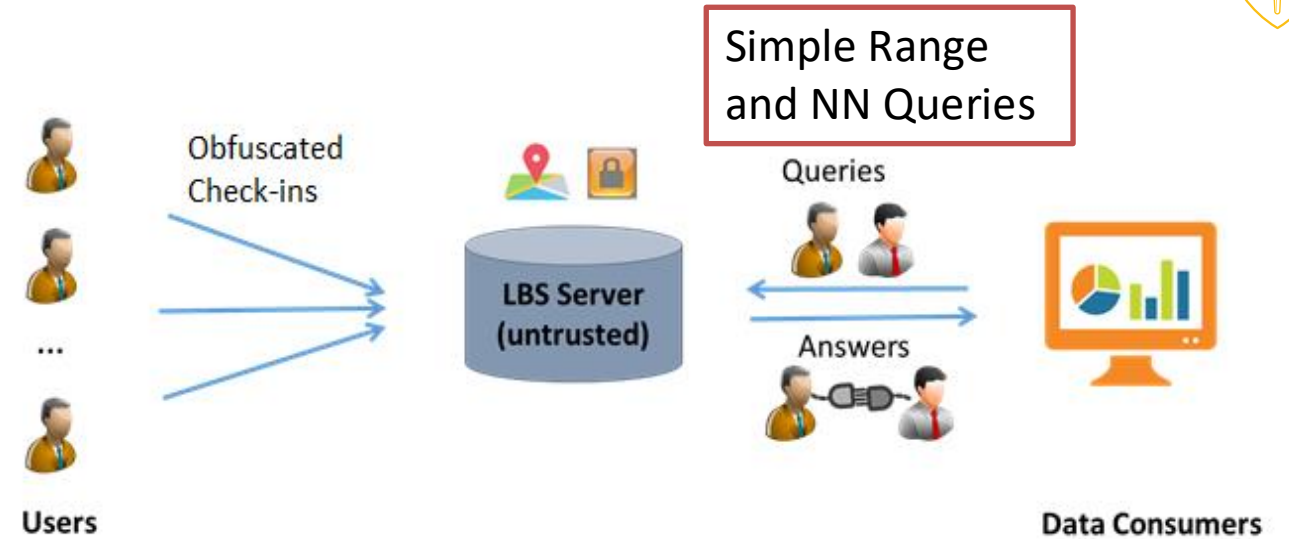
Location Queries $\Rightarrow$ Affiliations (political, religious, etc.)



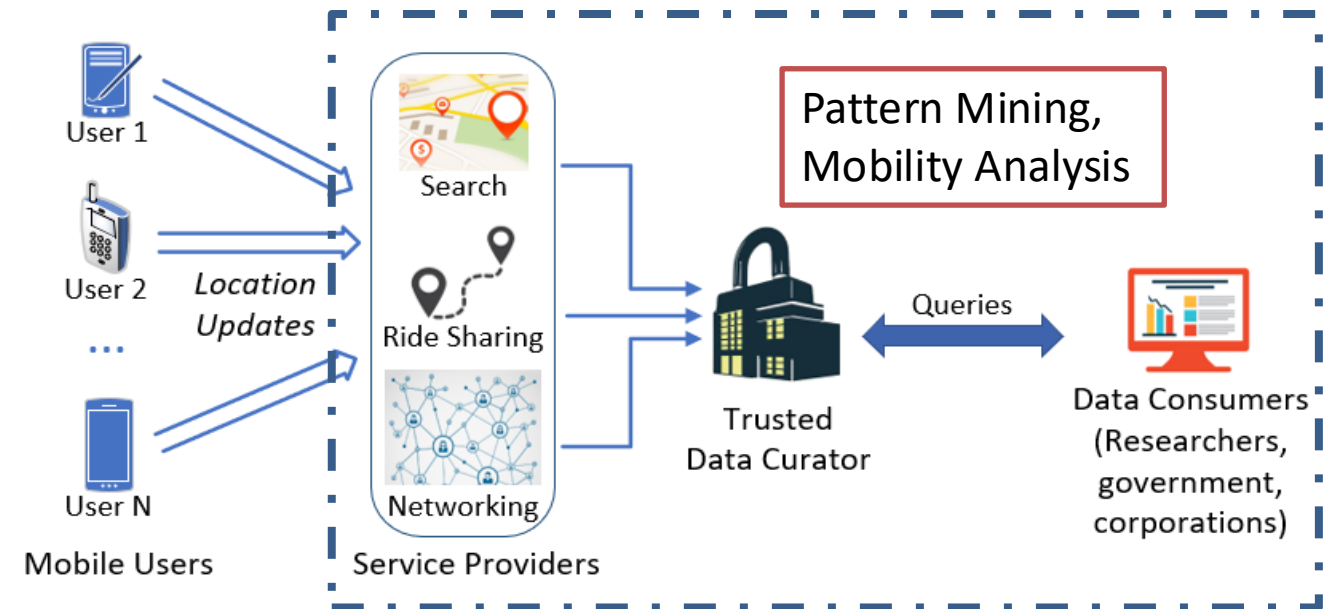


System Models

Online Setting:



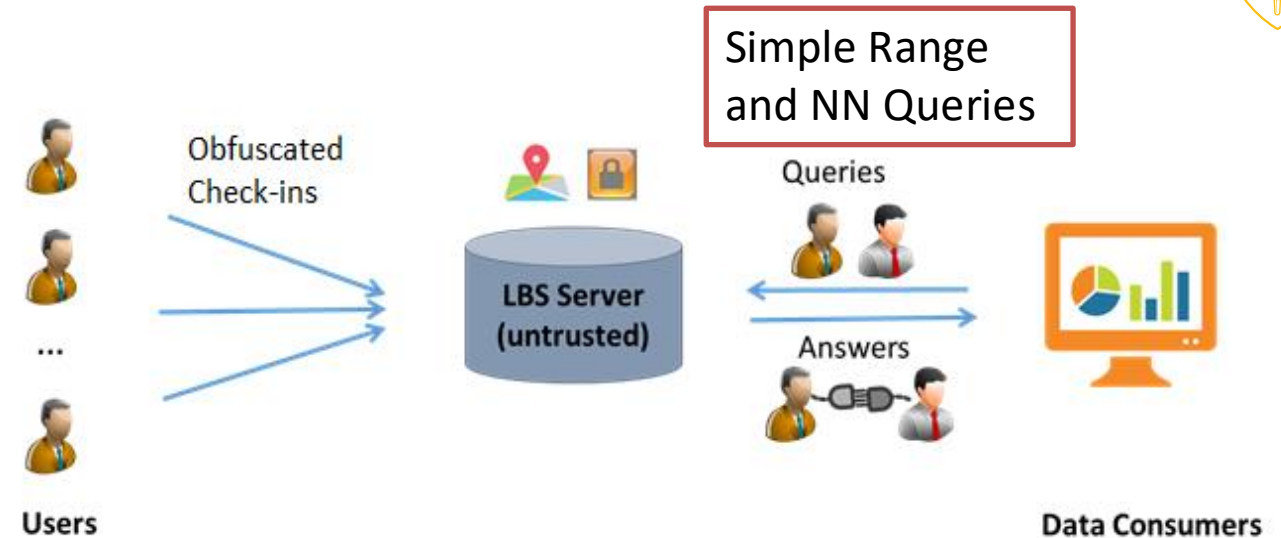
Offline Setting:



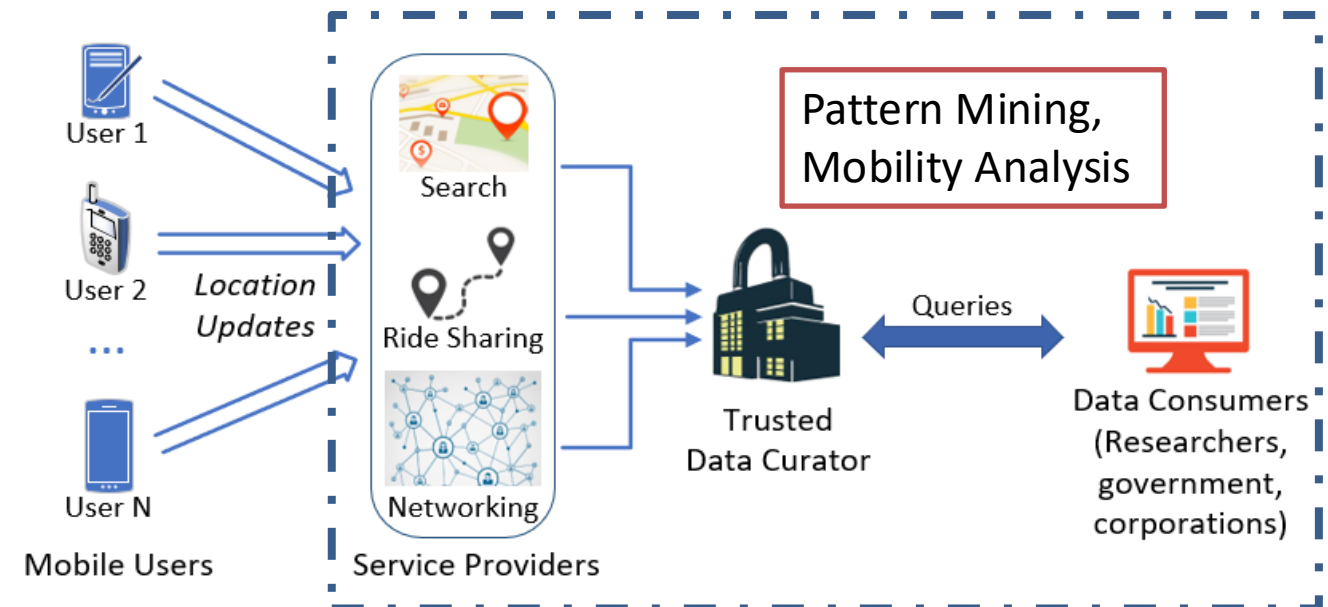


System Models

Online Setting:



Offline Setting:



System Architectures for Online Location Privacy



❖ *Third trusted party architecture*

- ❖ A centralized trusted entity is responsible for gathering information and providing the required privacy for each user
- ❖ Analogous to output perturbation

❖ *Client-Server architecture*

- ❖ Users communicate directly with the server with noisy locations.
- ❖ Analogous to input perturbation

❖ *Peer-to-Peer cooperative architecture*

- ❖ Users collaborate with each other without the interleaving of a centralized entity to provide customized privacy for each single user

System Architectures for Online Location Privacy



❖ *Third trusted party architecture*

- ❖ A centralized trusted entity is responsible for gathering information and providing the required privacy for each user
- ❖ Analogous to output perturbation

❖ *Client-Server architecture*

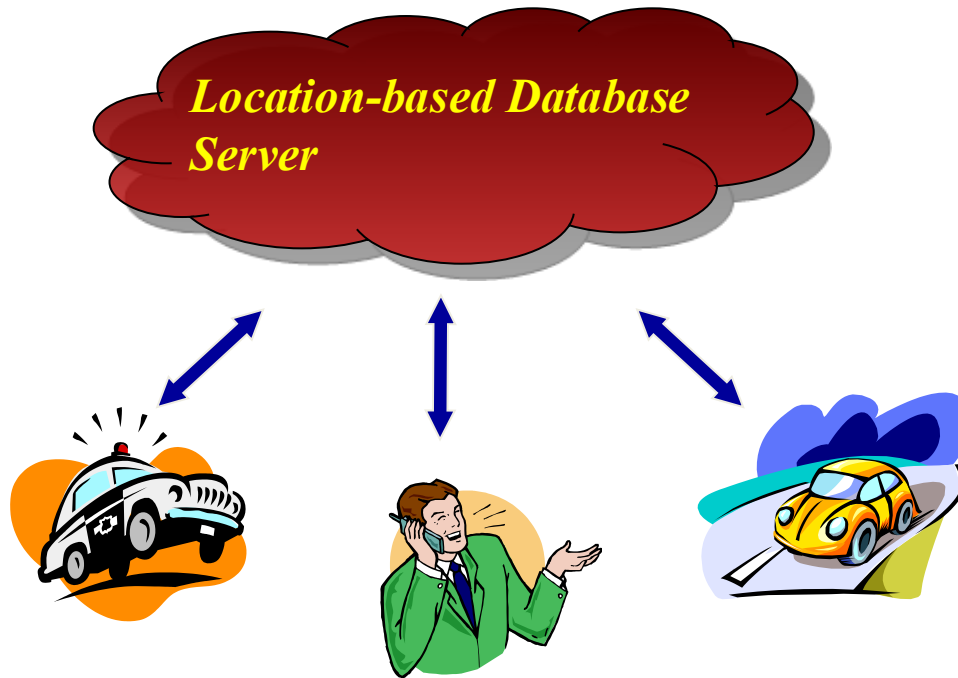
- ❖ Users communicate directly with the server with noisy locations.
- ❖ Analogous to input perturbation

❖ *Peer-to-Peer cooperative architecture*

- ❖ Users collaborate with each other without the interleaving of a centralized entity to provide customized privacy for each single user

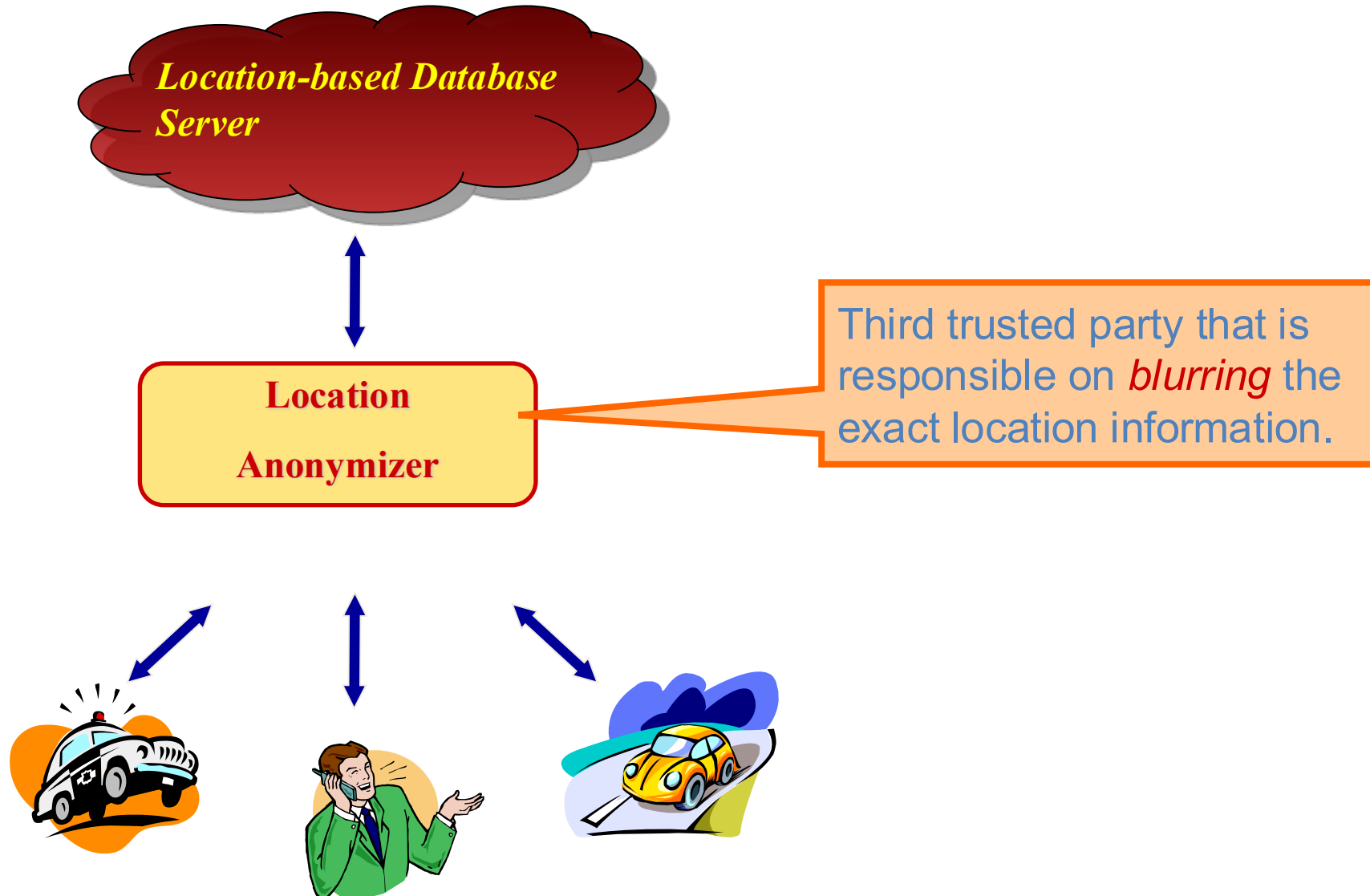


Third Trusted Party Architecture



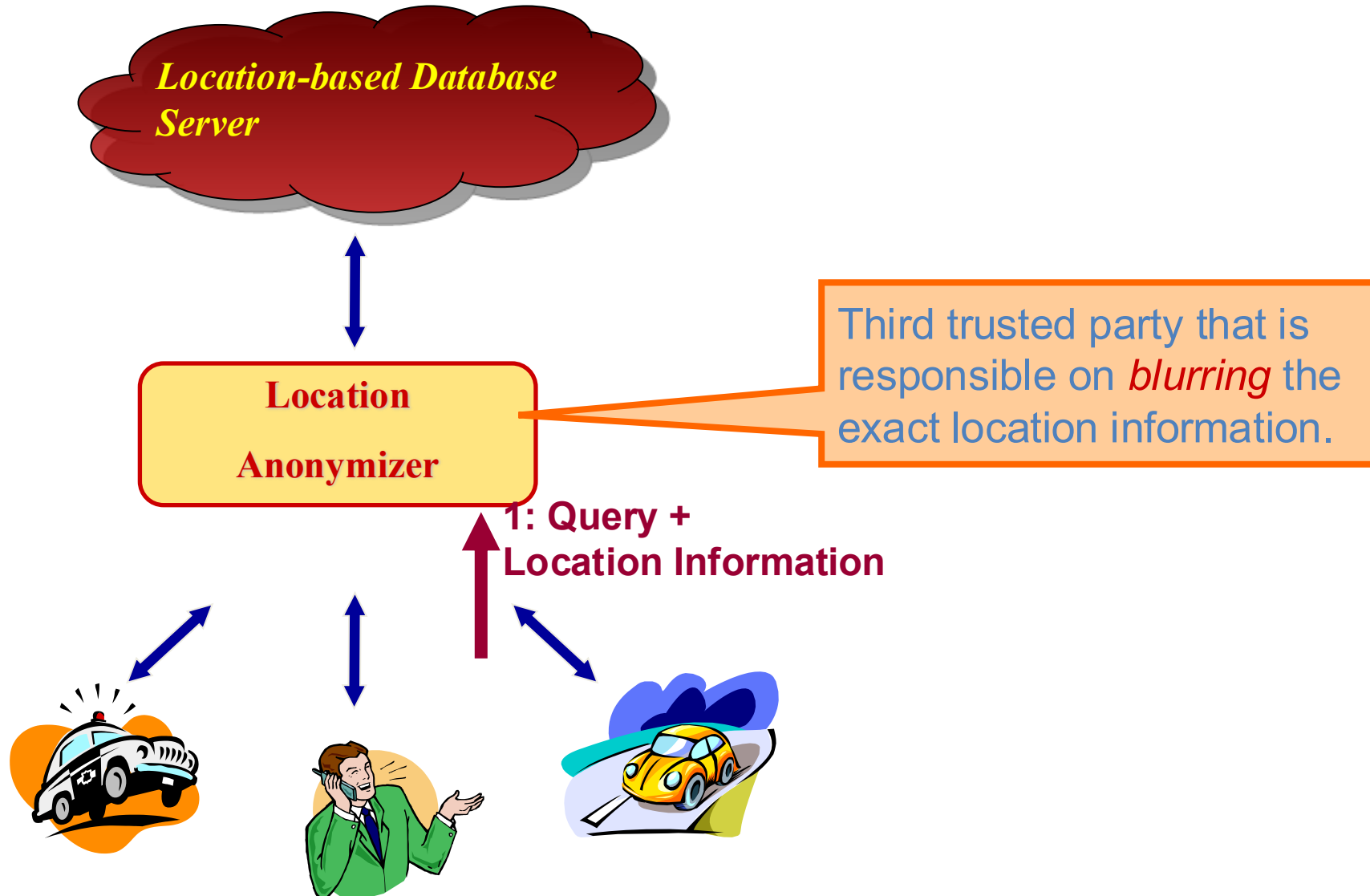


Third Trusted Party Architecture



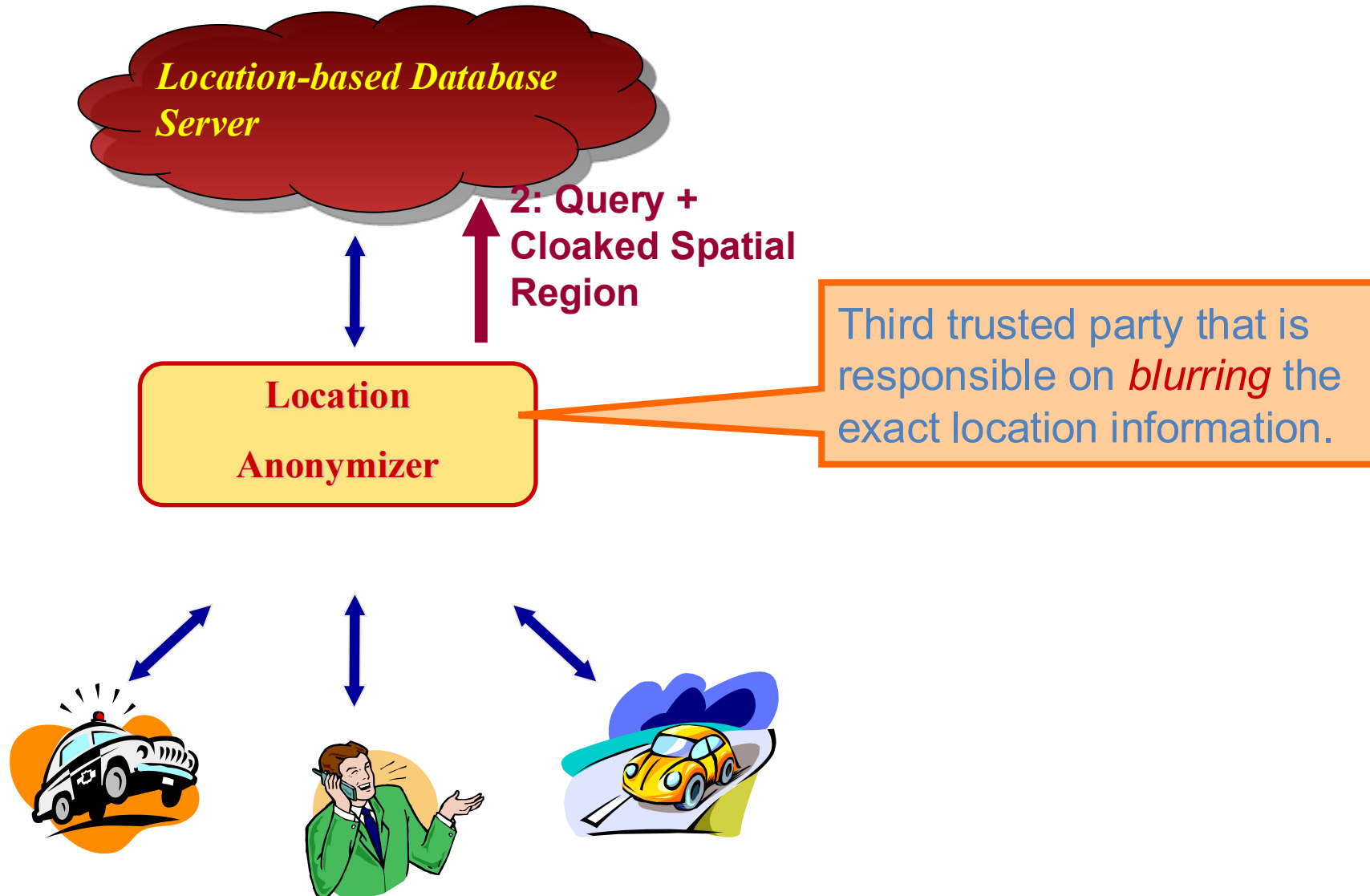


Third Trusted Party Architecture



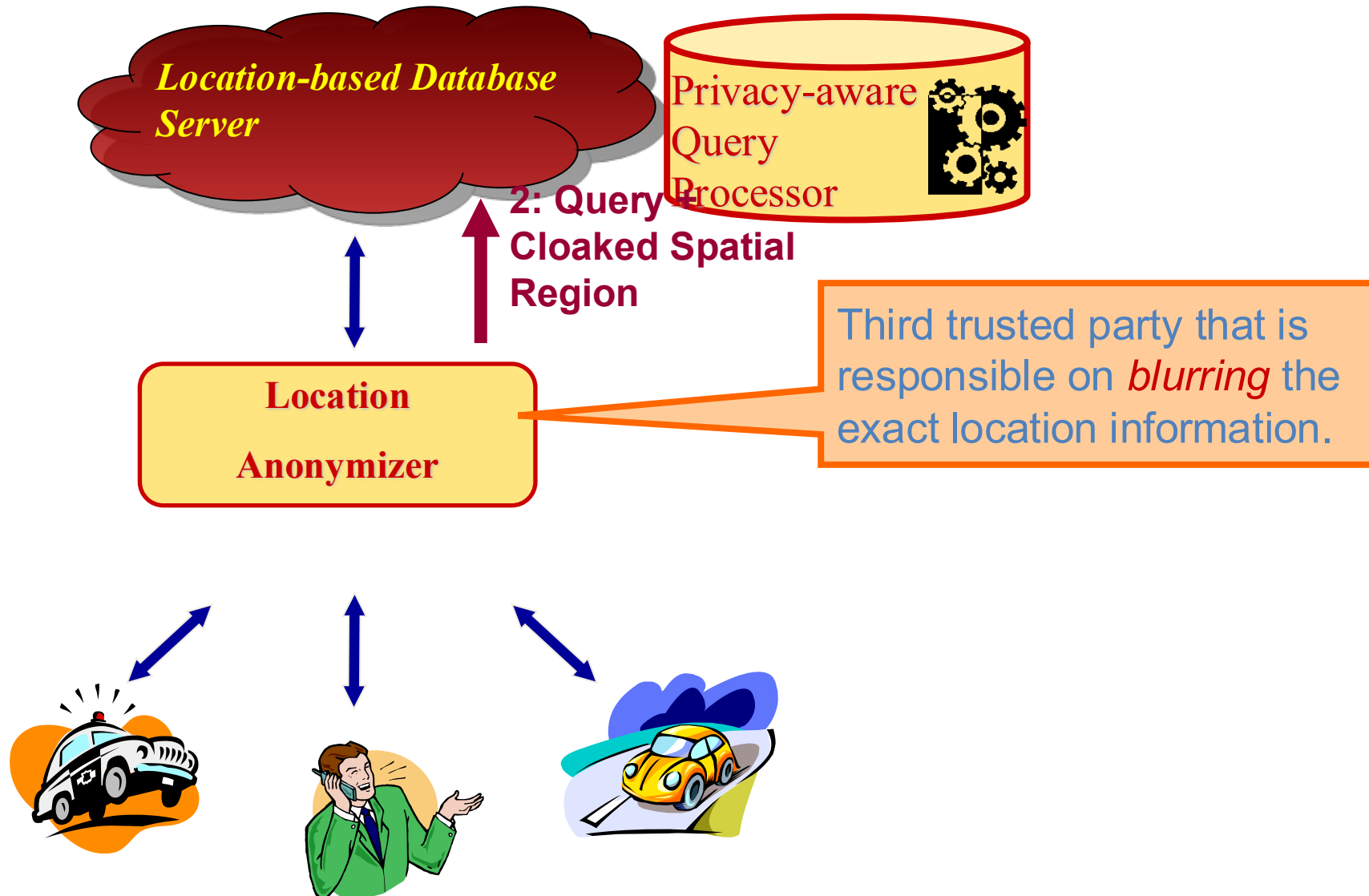


Third Trusted Party Architecture



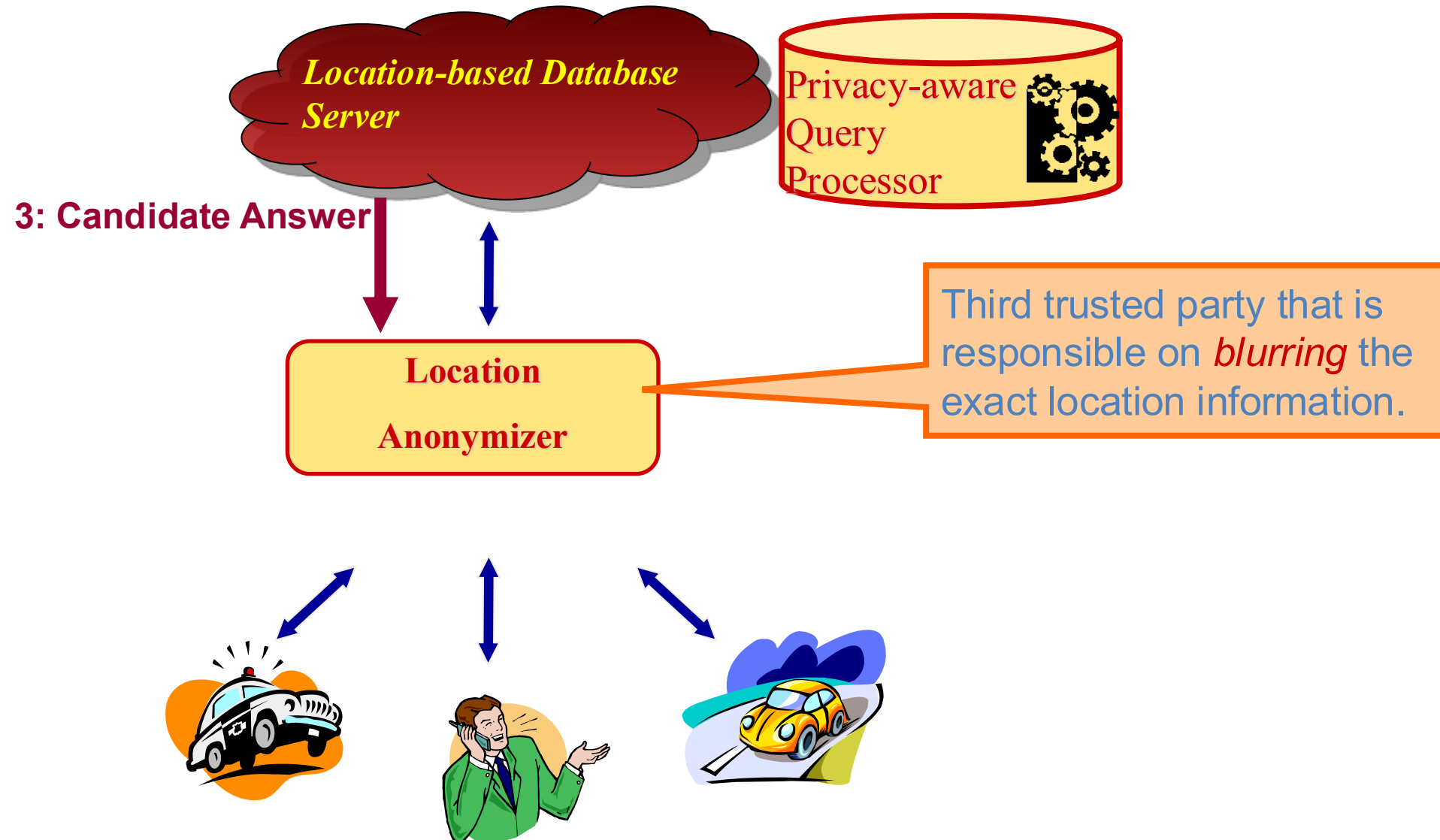


Third Trusted Party Architecture



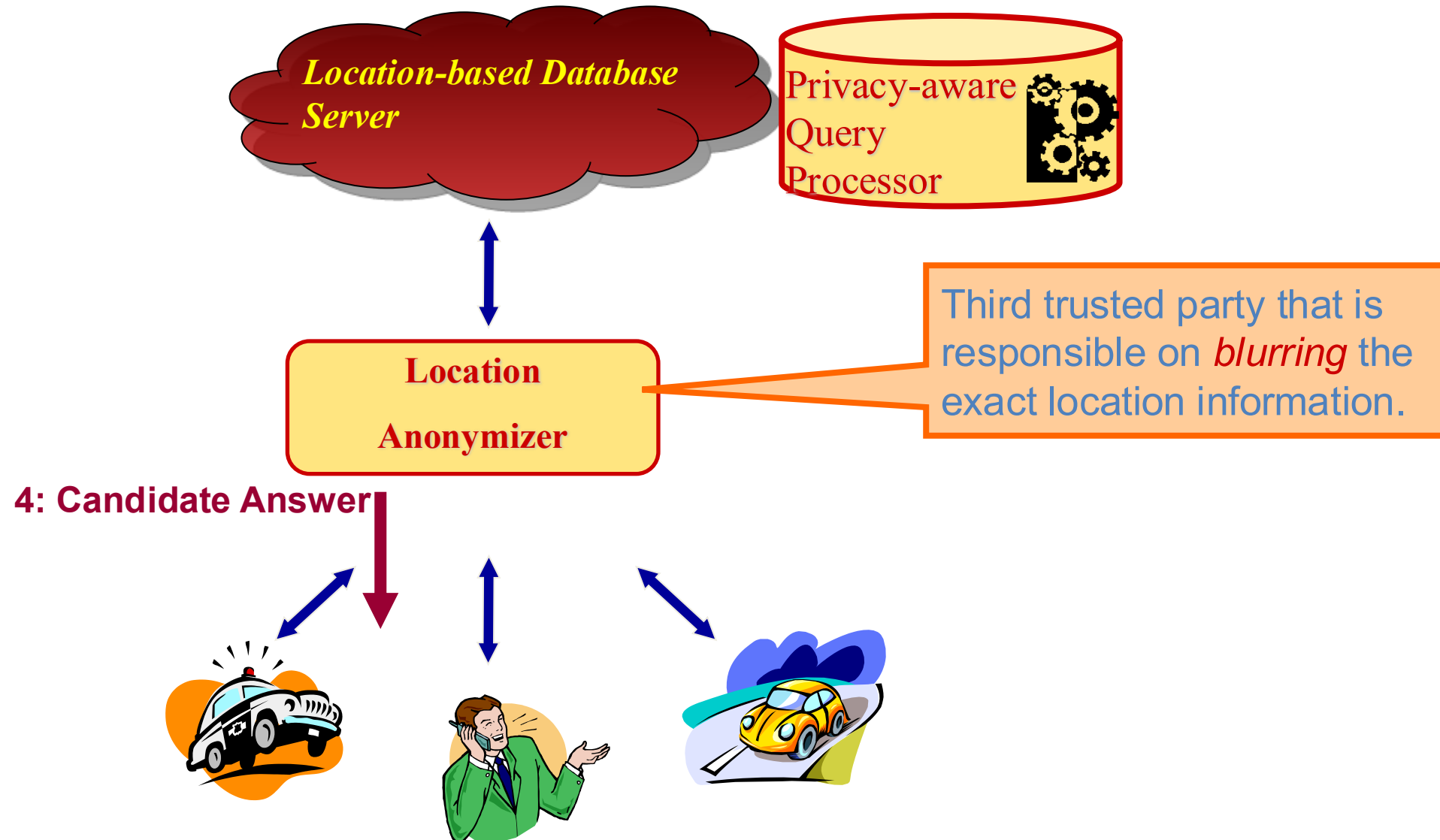


Third Trusted Party Architecture





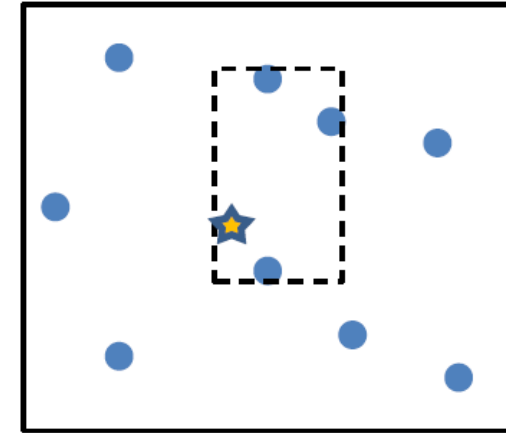
Third Trusted Party Architecture





Location k -Anonymity

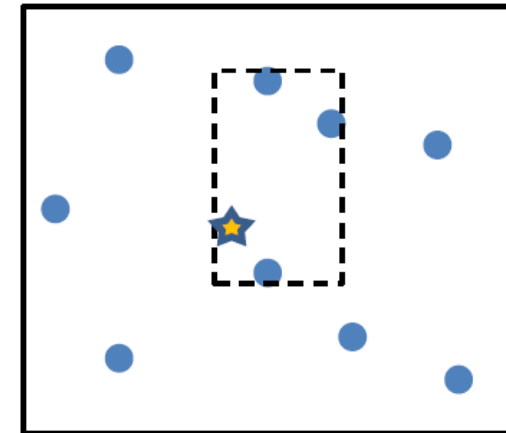
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together





Location k -Anonymity

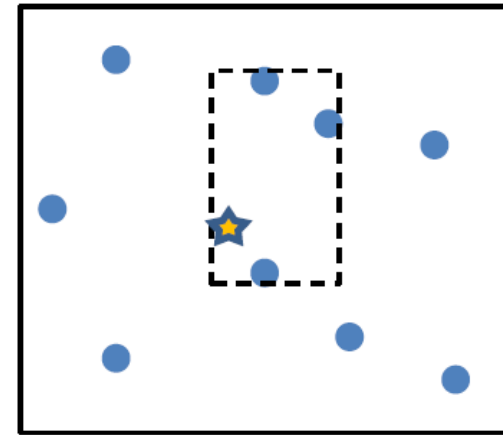
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)



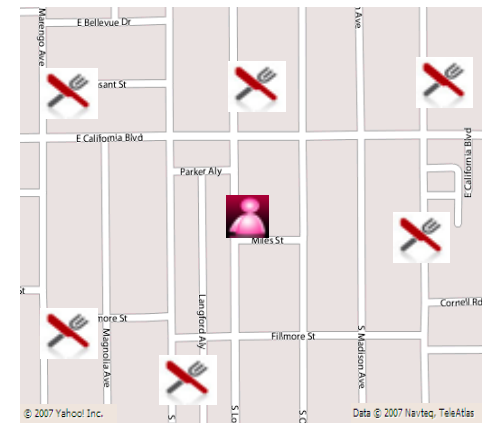


Location k -Anonymity

- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)



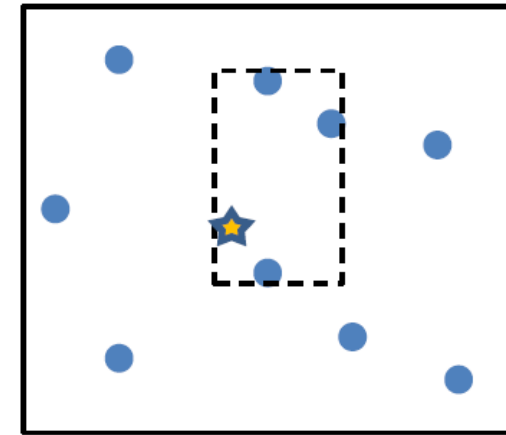
DB





Location k -Anonymity

- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)



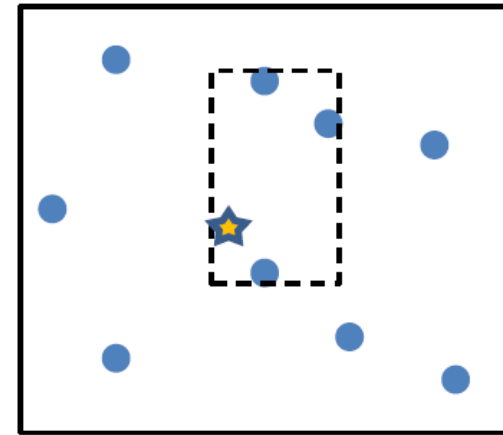
DB



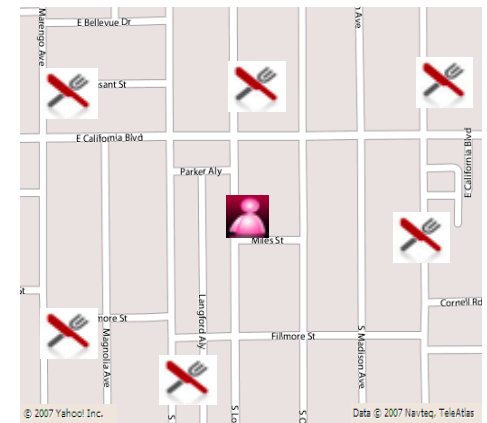


Location k -Anonymity

- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)



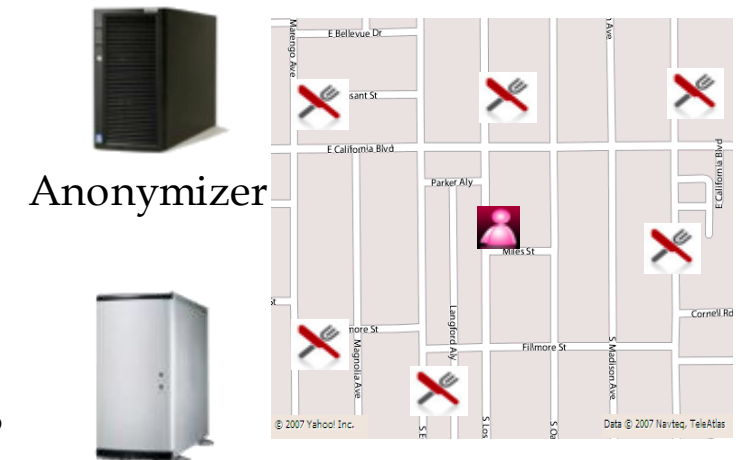
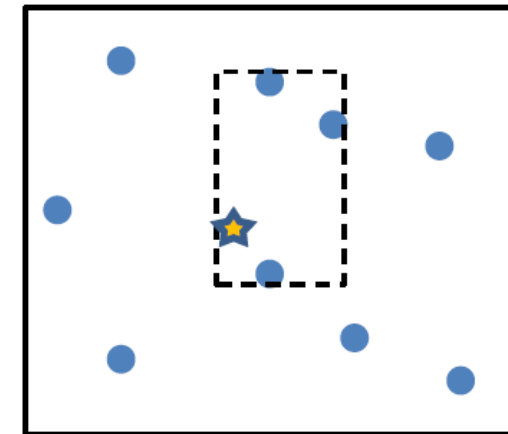
LS





Location k -Anonymity

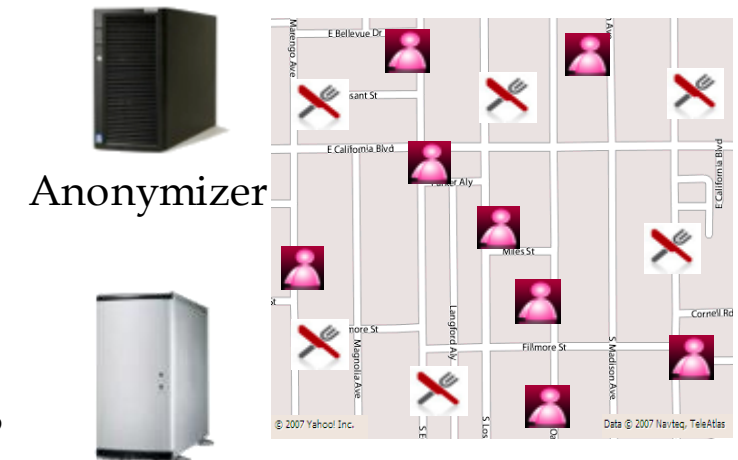
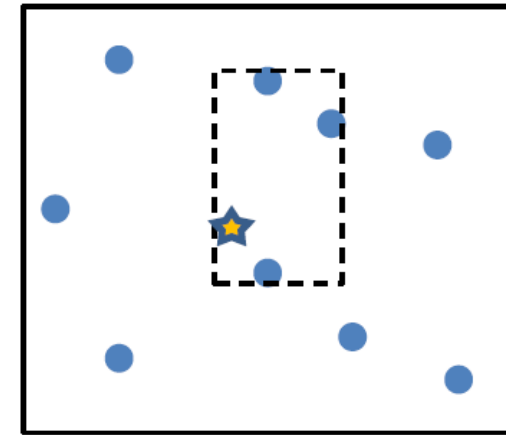
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)





Location k -Anonymity

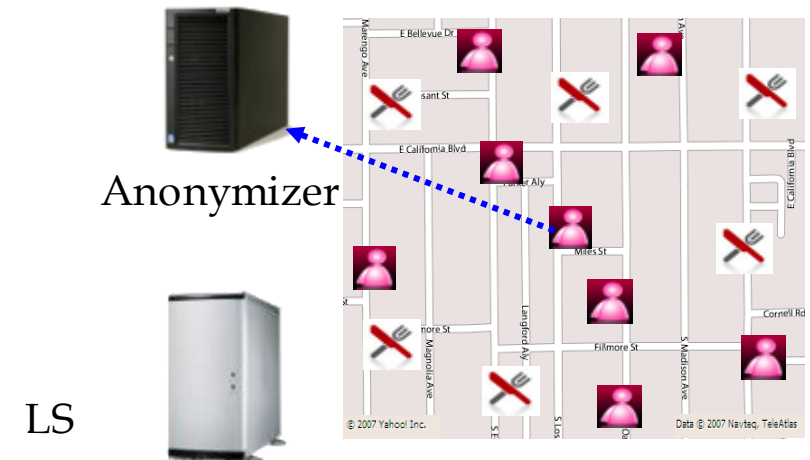
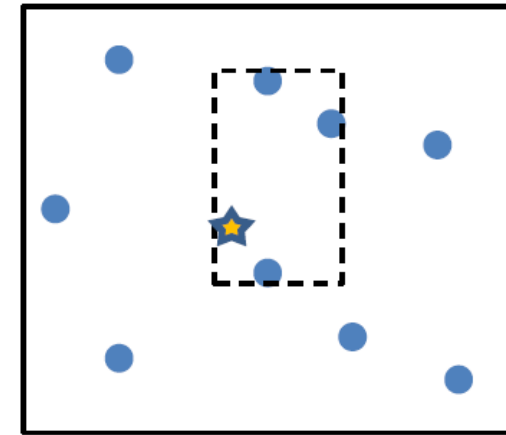
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
- If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)





Location k -Anonymity

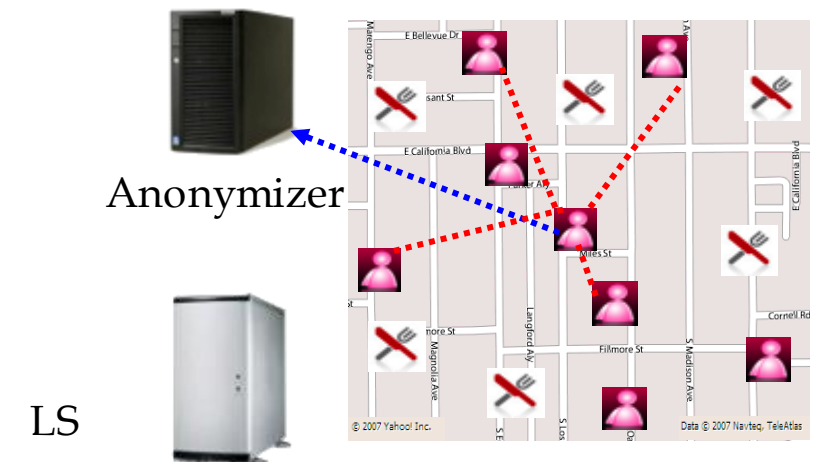
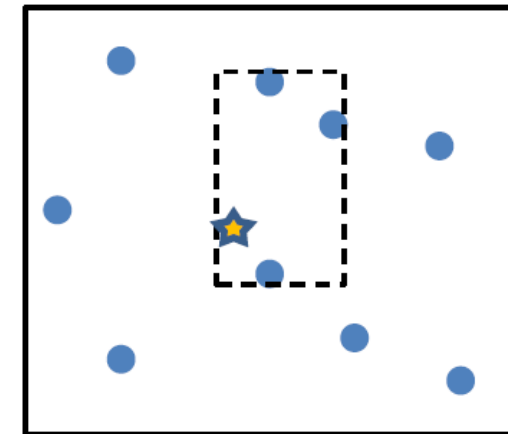
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
- If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)





Location k -Anonymity

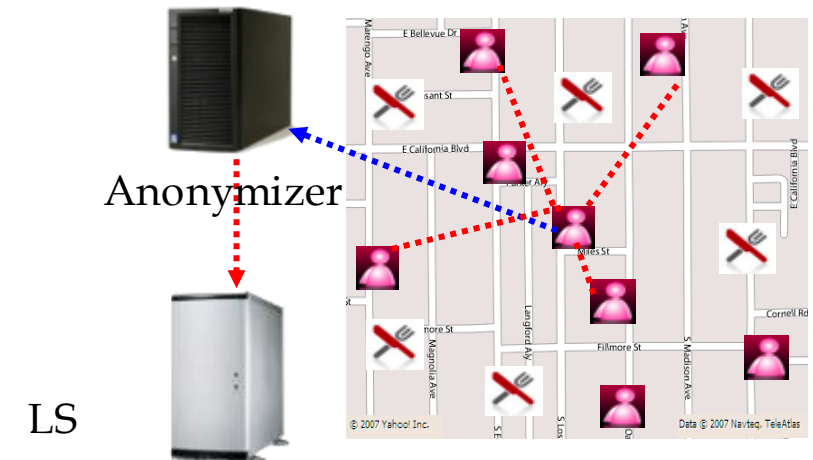
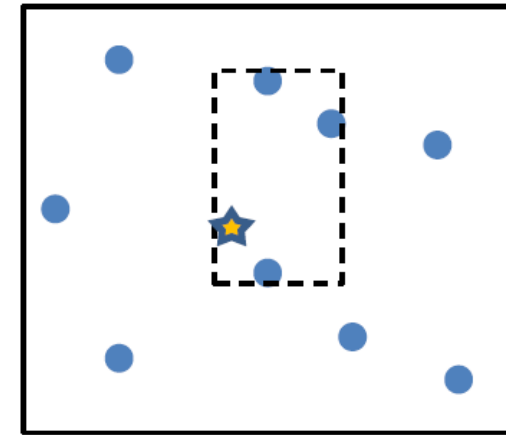
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
- If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)





Location k -Anonymity

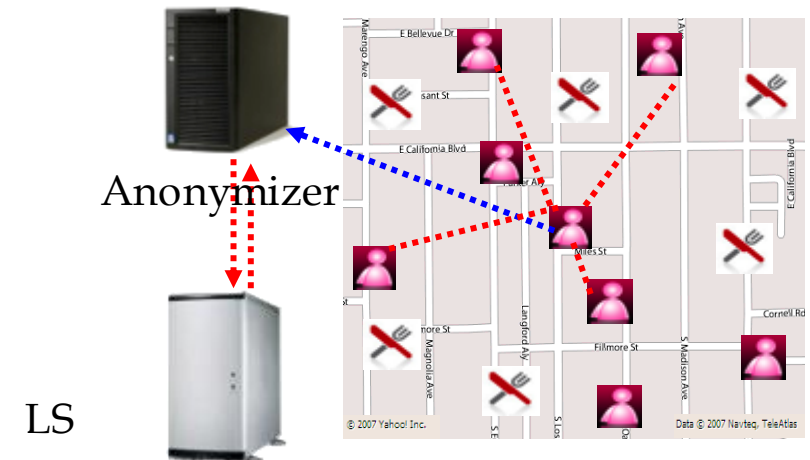
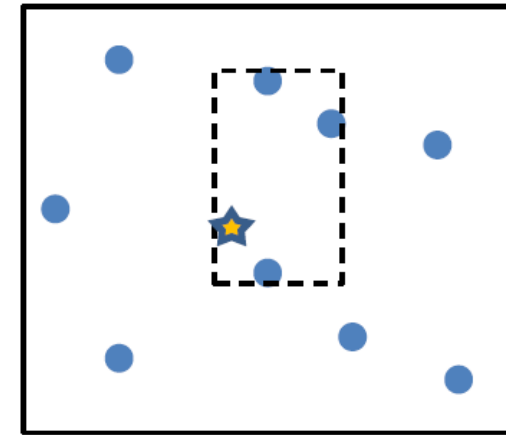
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
- If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)





Location k -Anonymity

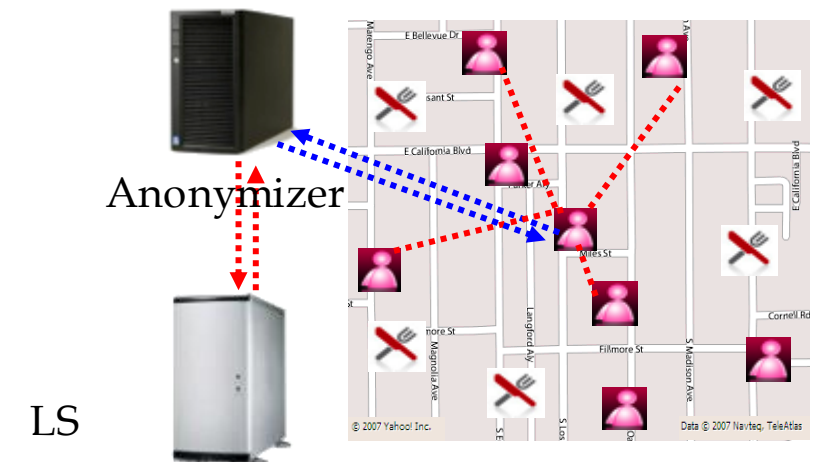
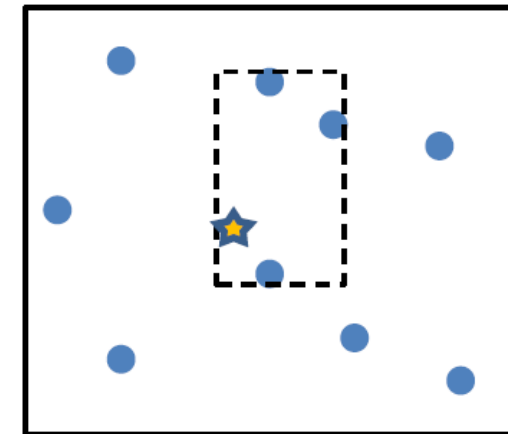
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
- If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)





Location k -Anonymity

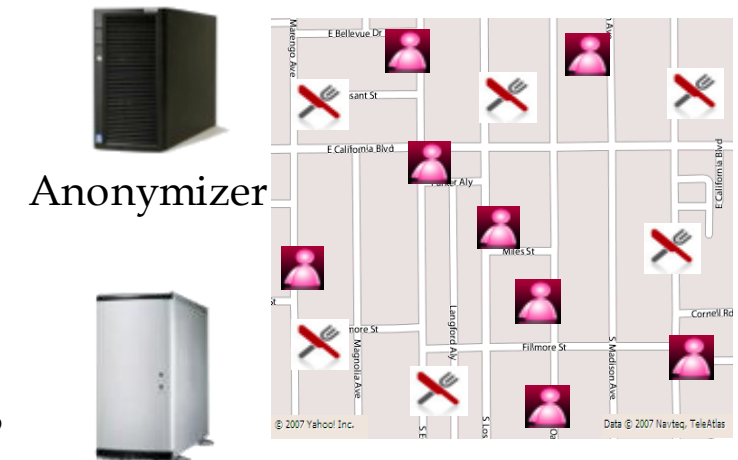
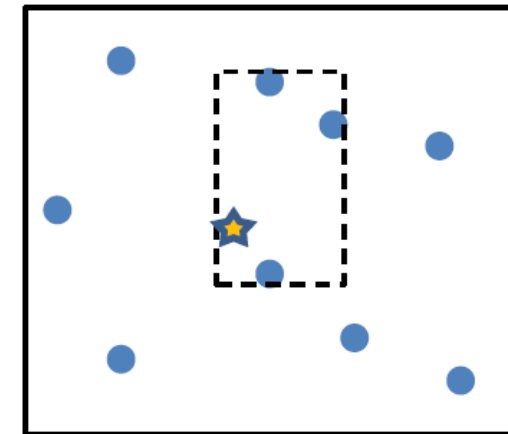
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
- If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)





Location k -Anonymity

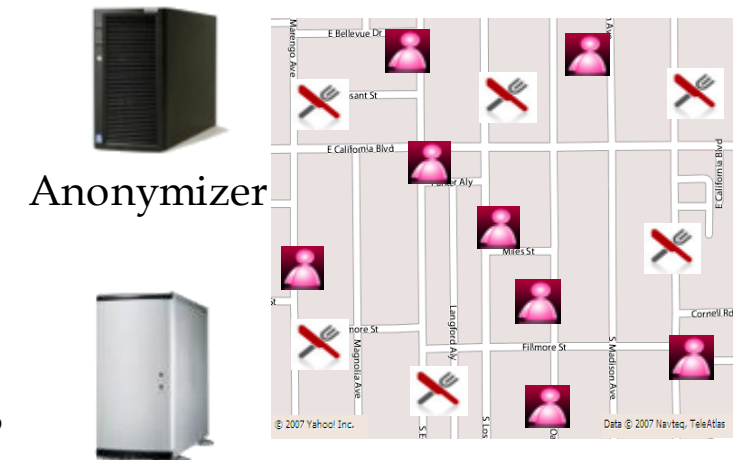
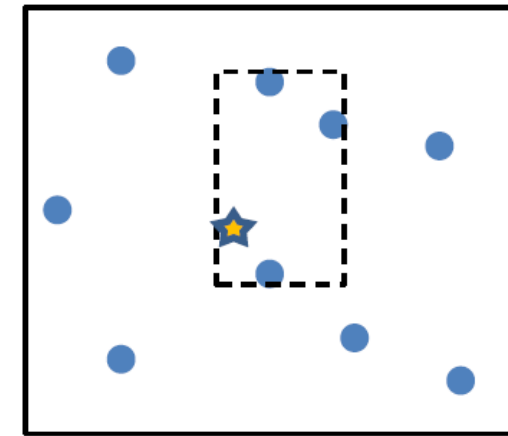
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
- If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)





Location k -Anonymity

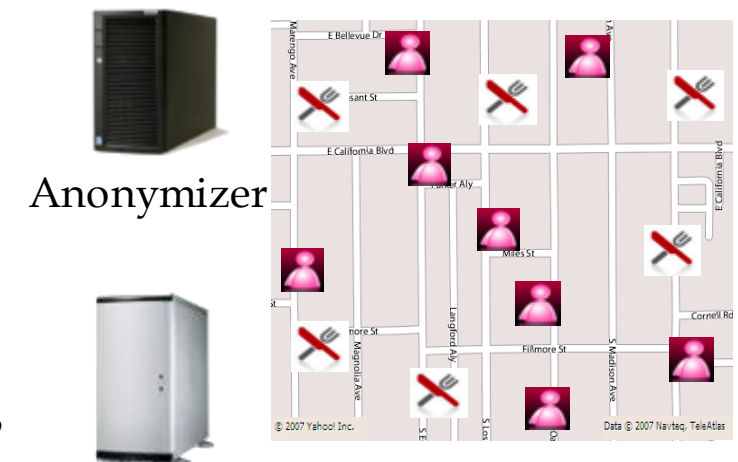
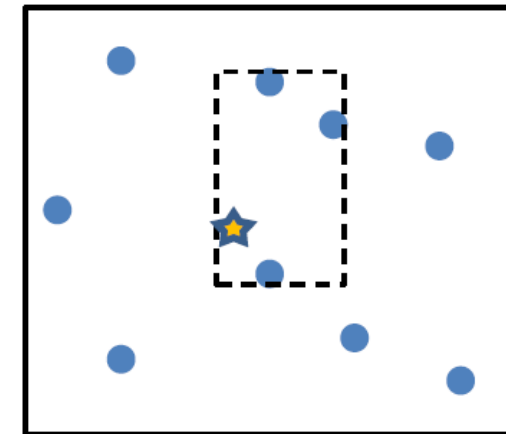
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)
- What if k other users are too close to each other?





Location k -Anonymity

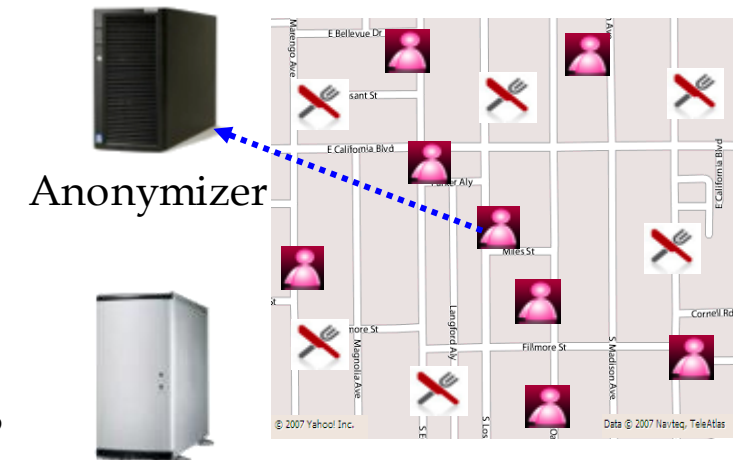
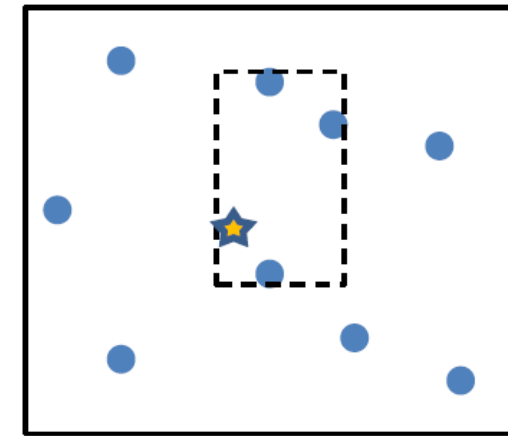
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)
- What if k other users are too close to each other?
- Cloaking





Location k -Anonymity

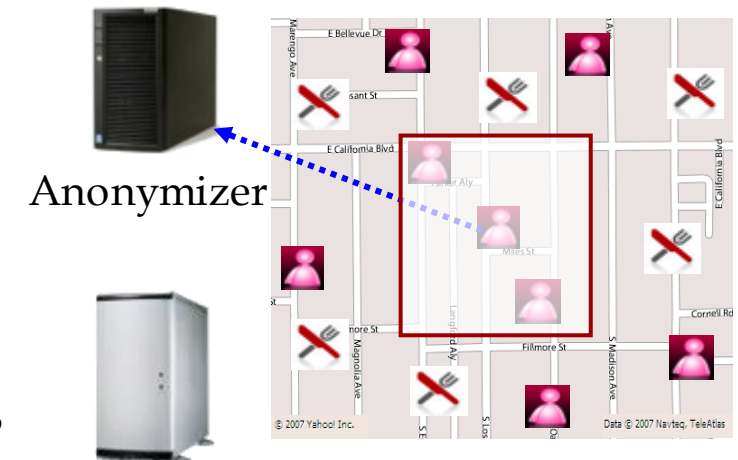
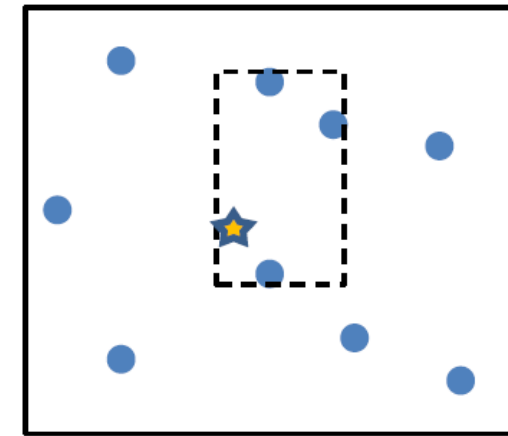
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)
- What if k other users are too close to each other?
- Cloaking





Location k -Anonymity

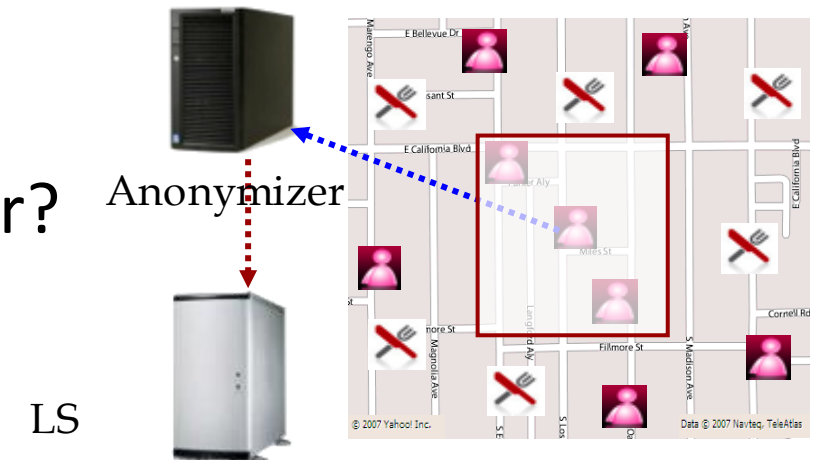
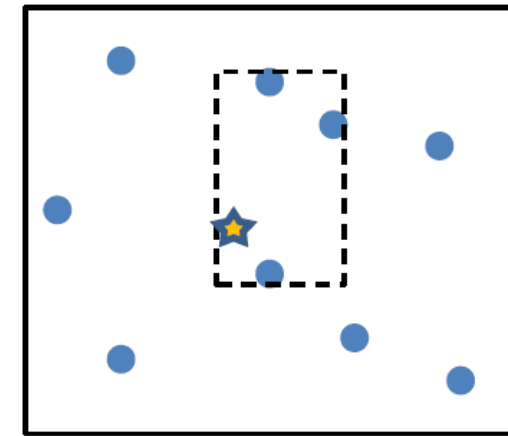
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)
- What if k other users are too close to each other?
- Cloaking





Location k -Anonymity

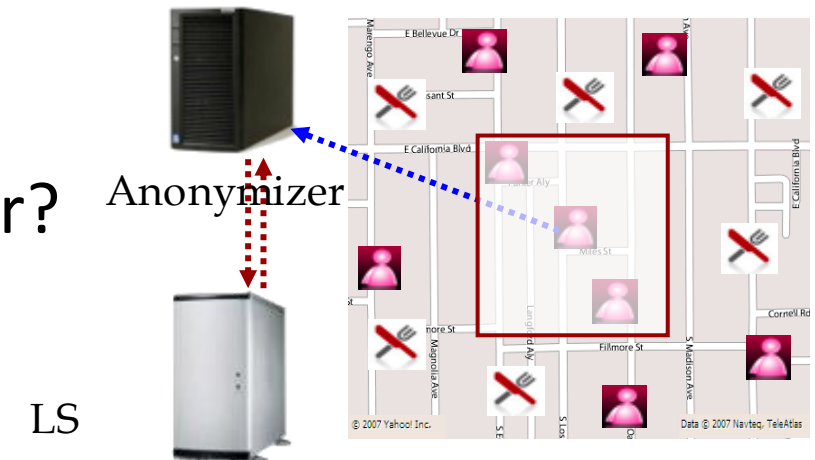
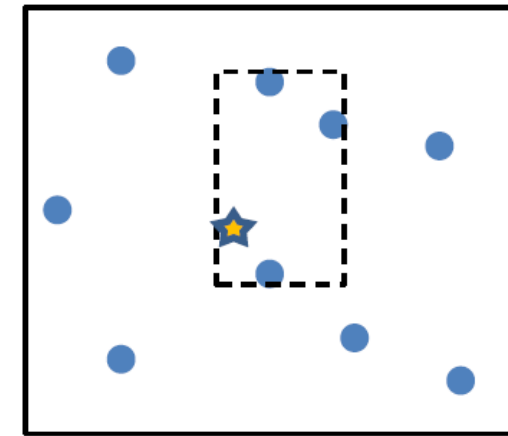
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)
- What if k other users are too close to each other?
- Cloaking





Location k -Anonymity

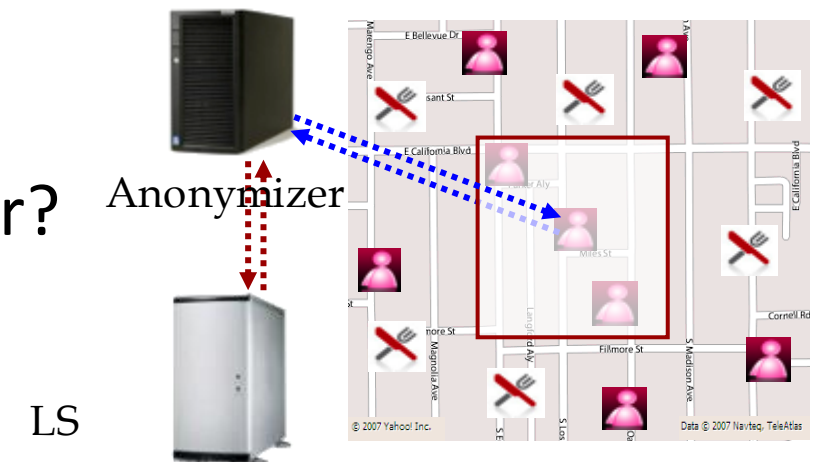
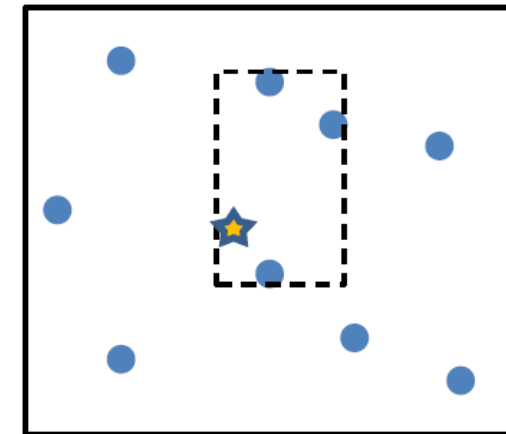
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)
- What if k other users are too close to each other?
- Cloaking





Location k -Anonymity

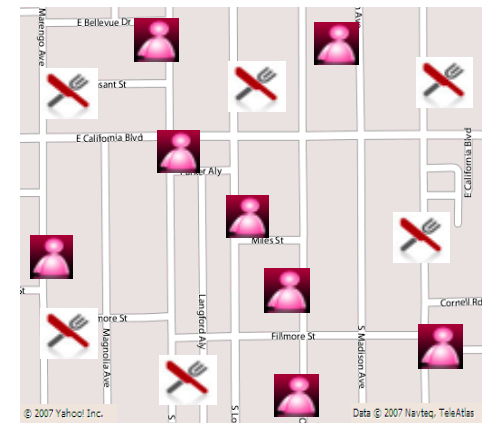
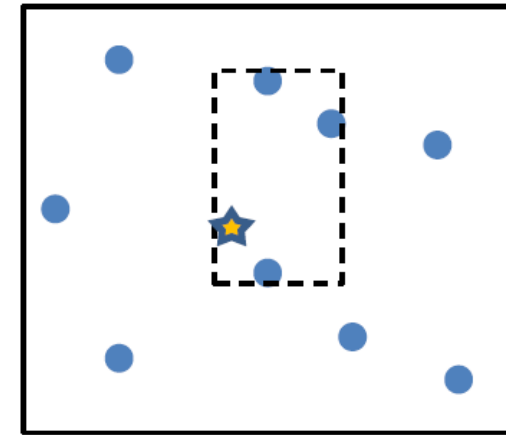
- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)
- What if k other users are too close to each other?
- Cloaking





Location k -Anonymity

- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)
- What if k other users are too close to each other?
- Cloaking



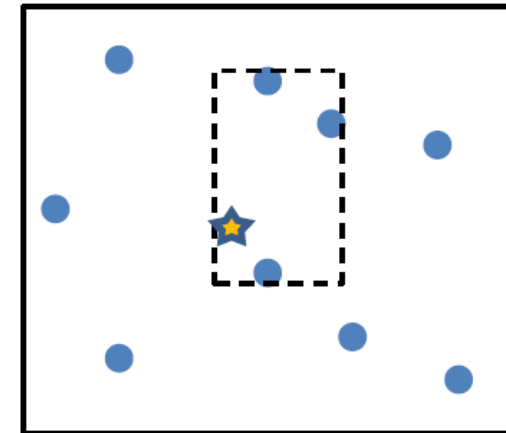
LS





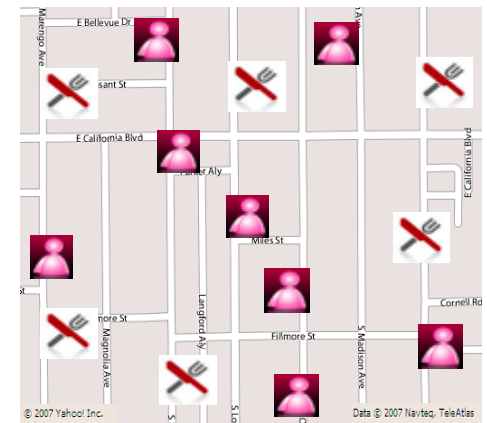
Location k -Anonymity

- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)
- What if k other users are too close to each other?
- Cloaking



What if in a sparse area?

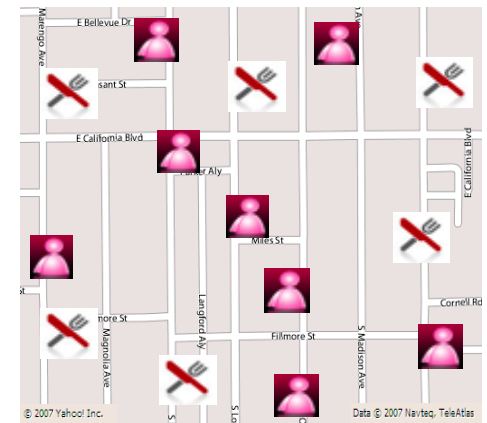
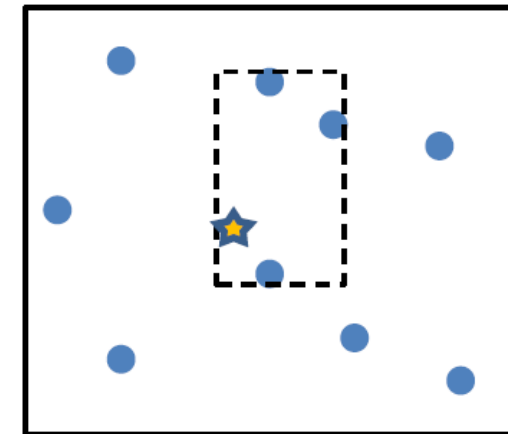
LS





Location k -Anonymity

- Submitted cloaked region must contain at least k users
 - Called the *Anonymized Spatial Region* (ASR)
 - Collect and submit k queries together
 - If not enough queries to group with
 - Drop the query (may not be acceptable)
 - Generate enough dummy (fake) queries (raises service cost)
- What if k other users are too close to each other?
- Cloaking



What if in a sparse area? Hybrid

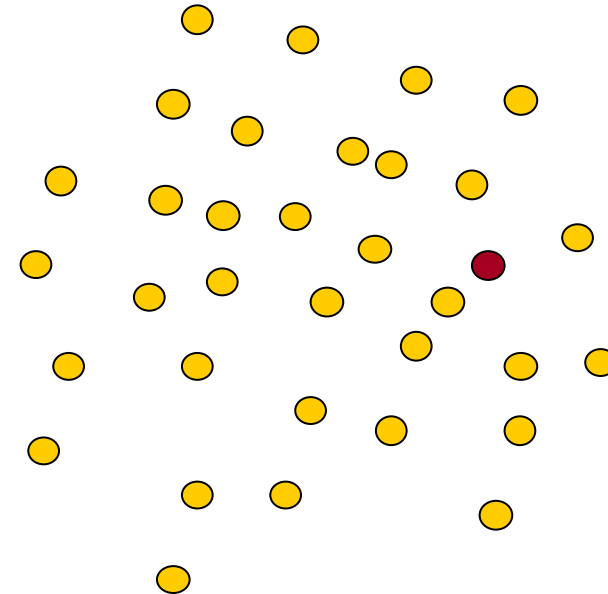
LS



Third Trusted Party Architecture: Quadtree Spatial Cloaking



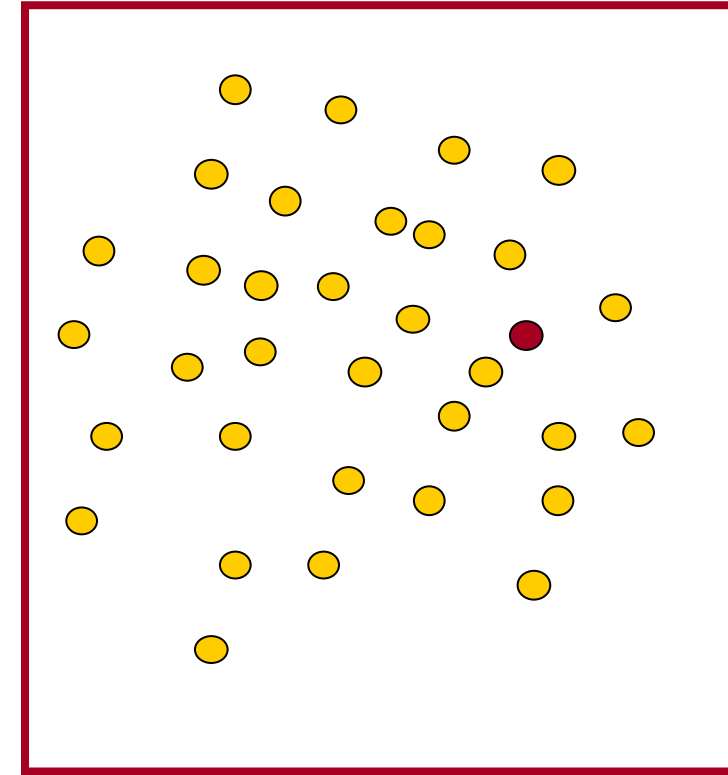
- Achieve *k-anonymity*, i.e., a user is indistinguishable from other *k-1* users
- Recursively divide the space into quadrants until a quadrant has less than *k* users.
- The previous quadrant, which still meet the *k-anonymity* constraint, is returned



Third Trusted Party Architecture: Quadtree Spatial Cloaking



- Achieve *k-anonymity*, i.e., a user is indistinguishable from other *k-1* users
- Recursively divide the space into quadrants until a quadrant has less than *k* users.
- The previous quadrant, which still meet the *k-anonymity* constraint, is returned



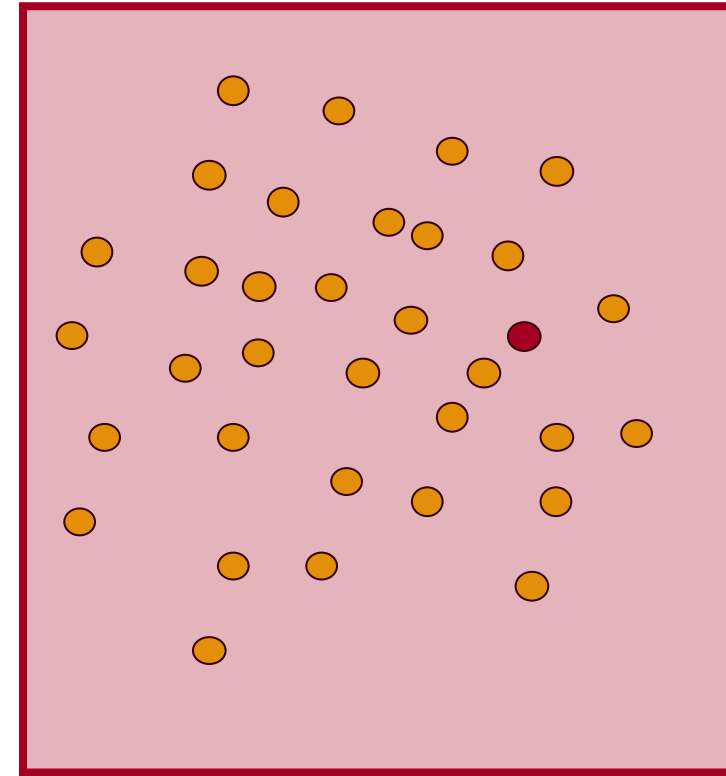
Achieve 5-anonymity for



Third Trusted Party Architecture: Quadtree Spatial Cloaking



- Achieve *k-anonymity*, i.e., a user is indistinguishable from other *k-1* users
- Recursively divide the space into quadrants until a quadrant has less than *k* users.
- The previous quadrant, which still meet the *k-anonymity* constraint, is returned



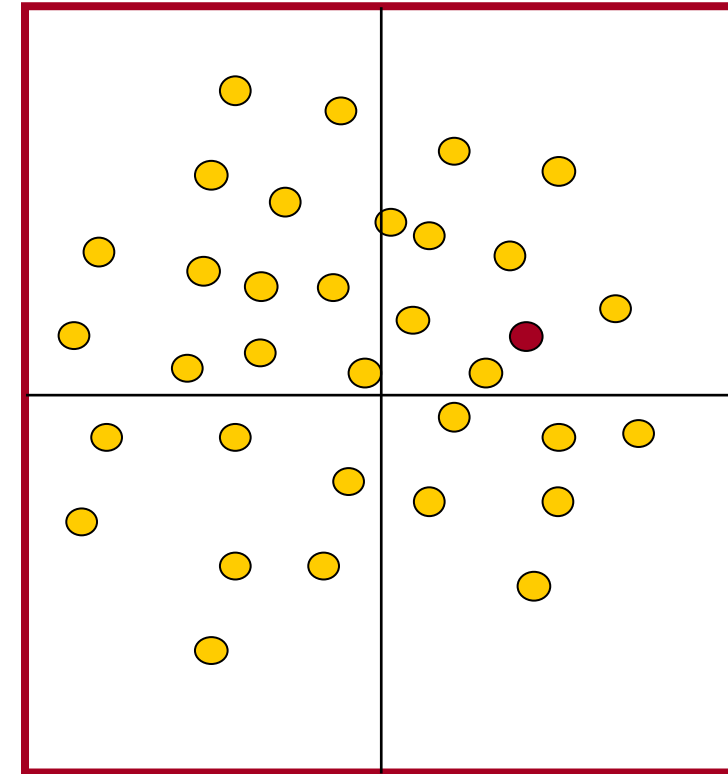
Achieve 5-anonymity for



Third Trusted Party Architecture: Quadtree Spatial Cloaking



- Achieve *k-anonymity*, i.e., a user is indistinguishable from other *k-1* users
- Recursively divide the space into quadrants until a quadrant has less than *k* users.
- The previous quadrant, which still meet the *k-anonymity* constraint, is returned



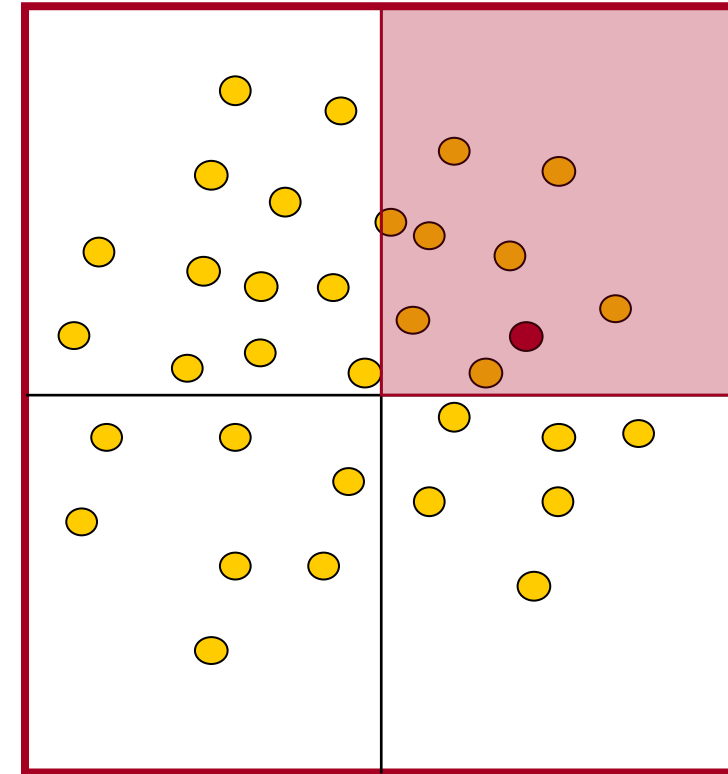
Achieve 5-anonymity for



Third Trusted Party Architecture: Quadtree Spatial Cloaking



- Achieve *k-anonymity*, i.e., a user is indistinguishable from other *k-1* users
- Recursively divide the space into quadrants until a quadrant has less than *k* users.
- The previous quadrant, which still meet the *k-anonymity* constraint, is returned



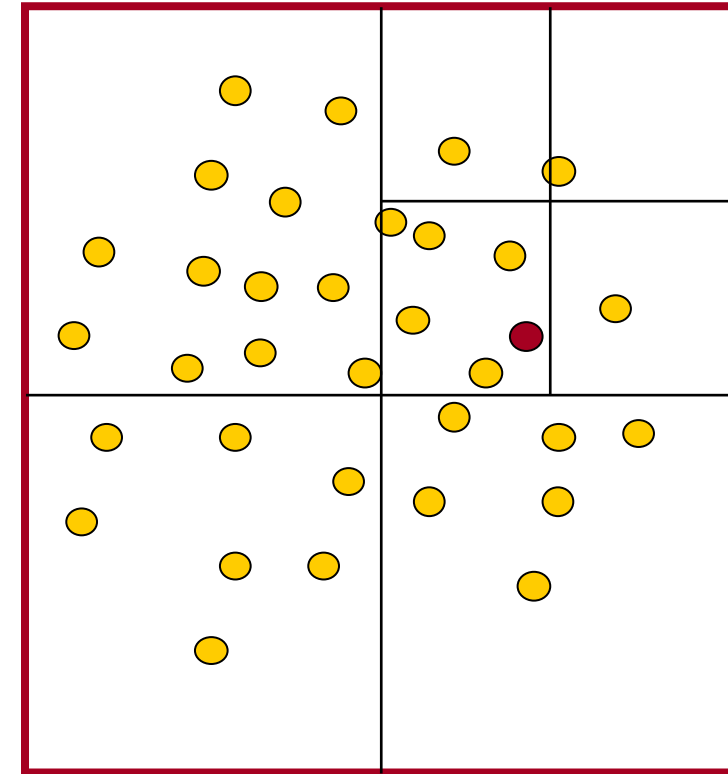
Achieve 5-anonymity for



Third Trusted Party Architecture: Quadtree Spatial Cloaking



- Achieve *k-anonymity*, i.e., a user is indistinguishable from other *k-1* users
- Recursively divide the space into quadrants until a quadrant has less than *k* users.
- The previous quadrant, which still meet the *k-anonymity* constraint, is returned



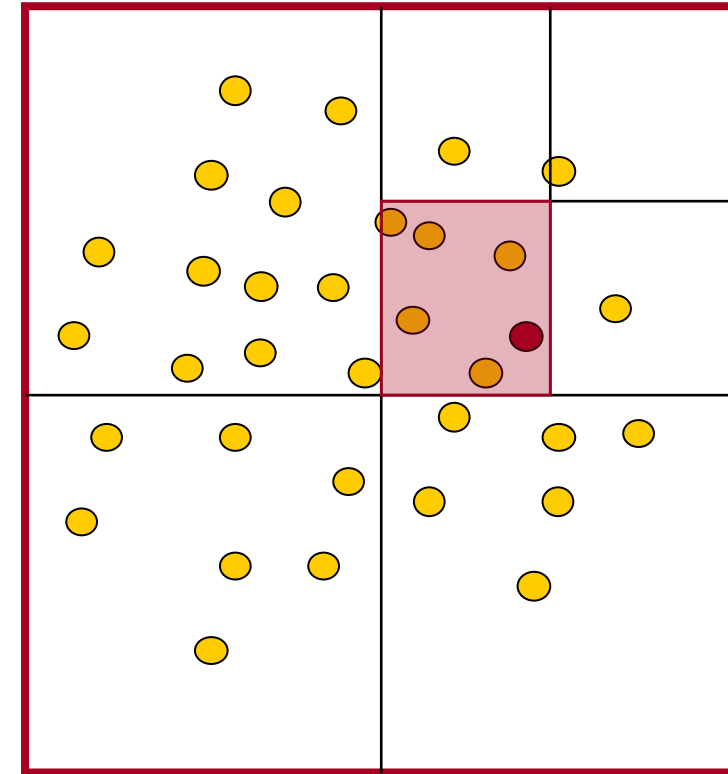
Achieve 5-anonymity for



Third Trusted Party Architecture: Quadtree Spatial Cloaking



- Achieve *k-anonymity*, i.e., a user is indistinguishable from other *k-1* users
- Recursively divide the space into quadrants until a quadrant has less than *k* users.
- The previous quadrant, which still meet the *k-anonymity* constraint, is returned



Achieve 5-anonymity for

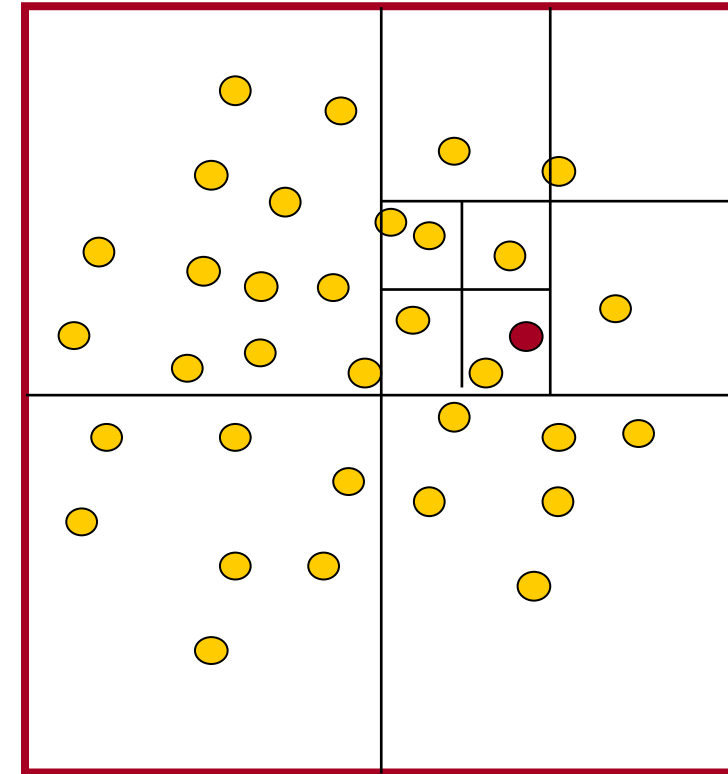


Third Trusted Party Architecture:

Quadtree Spatial Cloaking



- Achieve *k-anonymity*, i.e., a user is indistinguishable from other *k-1* users
- Recursively divide the space into quadrants until a quadrant has less than *k* users.
- The previous quadrant, which still meet the *k-anonymity* constraint, is returned



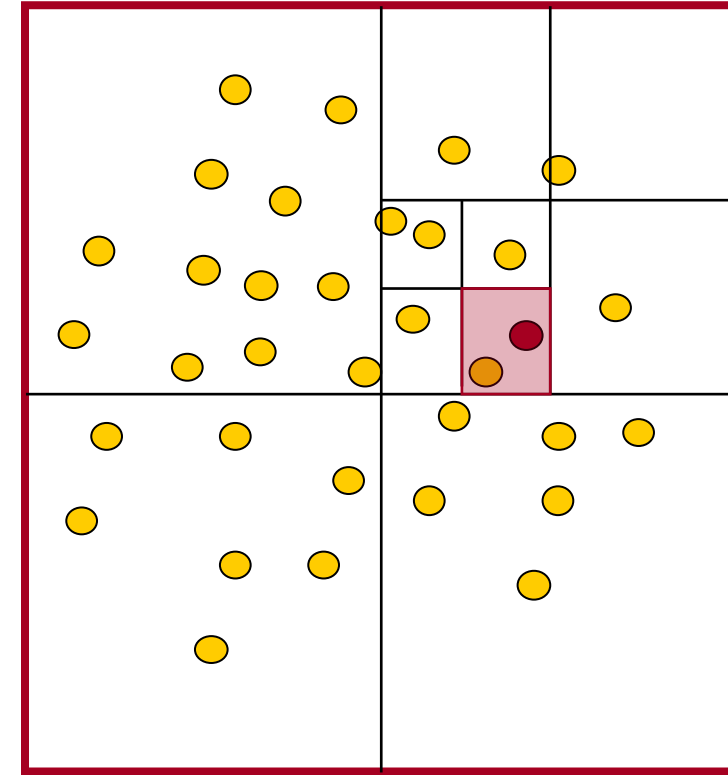
Achieve 5-anonymity for



Third Trusted Party Architecture: Quadtree Spatial Cloaking



- Achieve *k-anonymity*, i.e., a user is indistinguishable from other *k-1* users
- Recursively divide the space into quadrants until a quadrant has less than *k* users.
- The previous quadrant, which still meet the *k-anonymity* constraint, is returned



Achieve 5-anonymity for

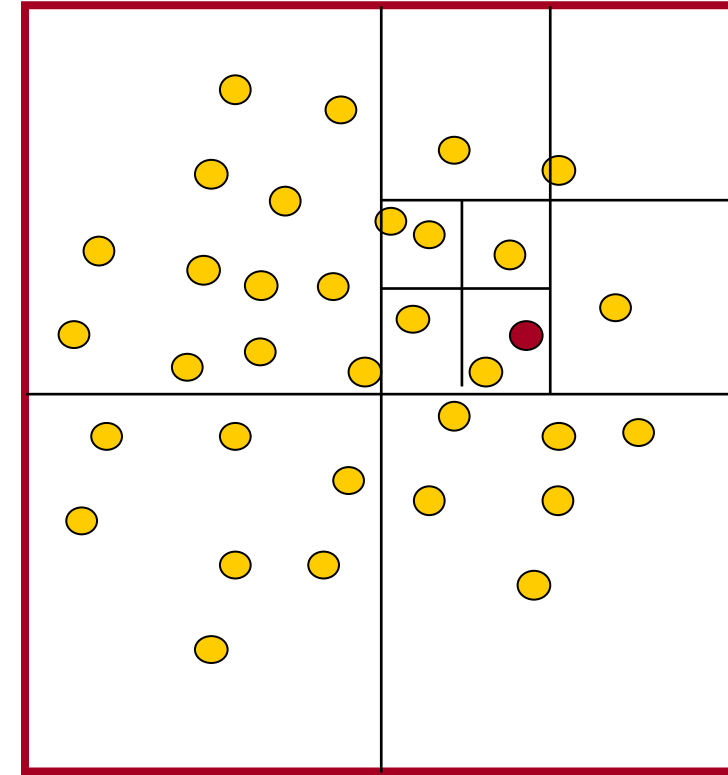


Third Trusted Party Architecture:

Quadtree Spatial Cloaking



- Achieve *k-anonymity*, i.e., a user is indistinguishable from other *k-1* users
- Recursively divide the space into quadrants until a quadrant has less than *k* users.
- The previous quadrant, which still meet the *k-anonymity* constraint, is returned



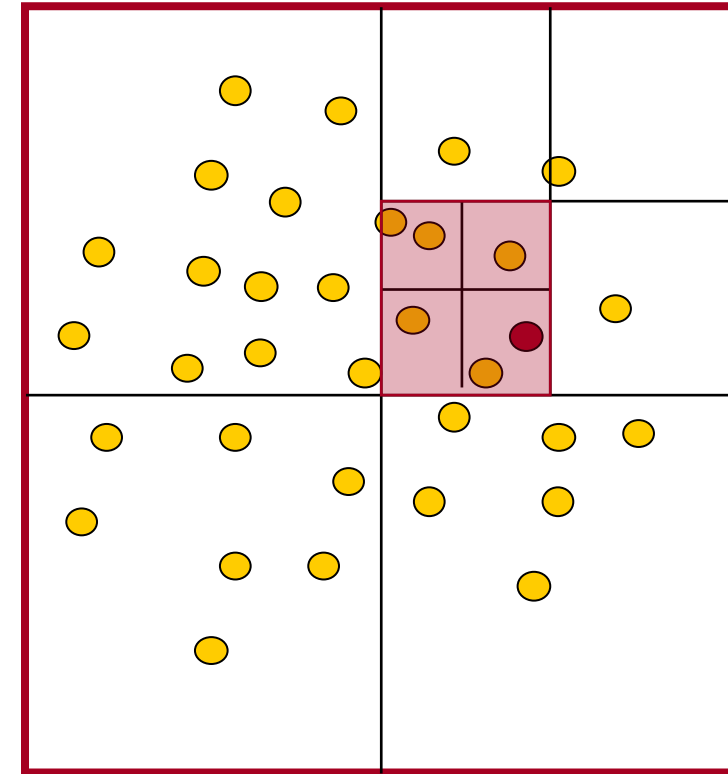
Achieve 5-anonymity for



Third Trusted Party Architecture: Quadtree Spatial Cloaking



- Achieve *k-anonymity*, i.e., a user is indistinguishable from other *k-1* users
- Recursively divide the space into quadrants until a quadrant has less than *k* users.
- The previous quadrant, which still meet the *k-anonymity* constraint, is returned

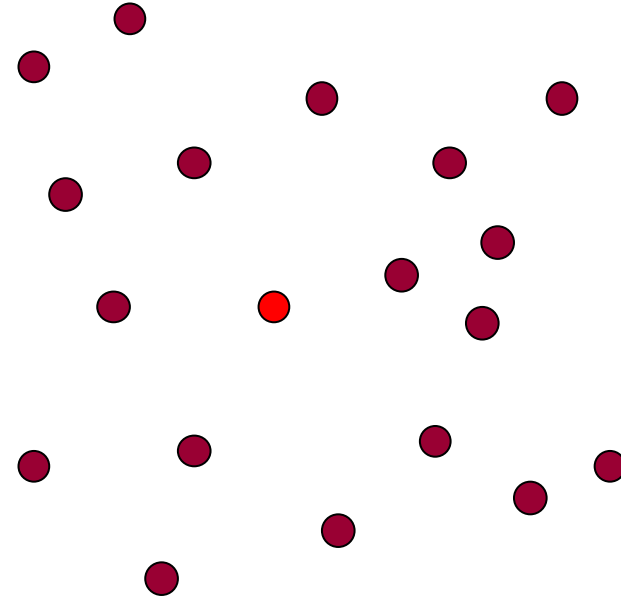


Achieve 5-anonymity for



Third Trusted Party Architecture:

Nearest-Neighbor k-Anonymizing

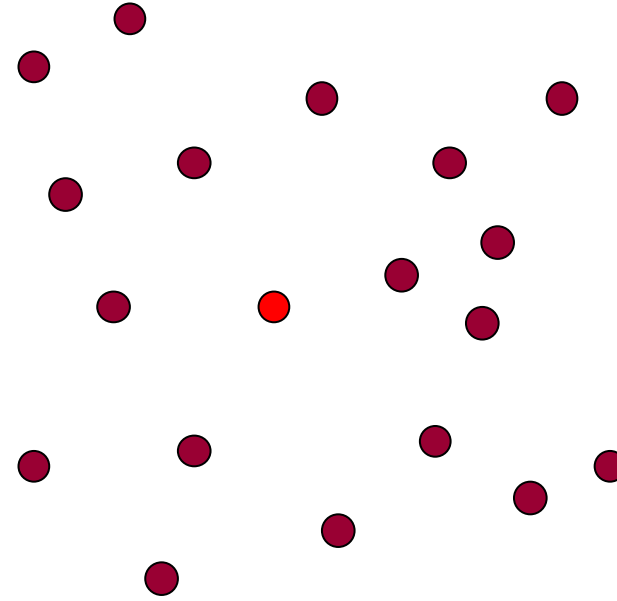


Third Trusted Party Architecture:

Nearest-Neighbor k -Anonymizing



- **STEP 1:** Determine a set S containing u and $k - 1$ u 's nearest neighbors.

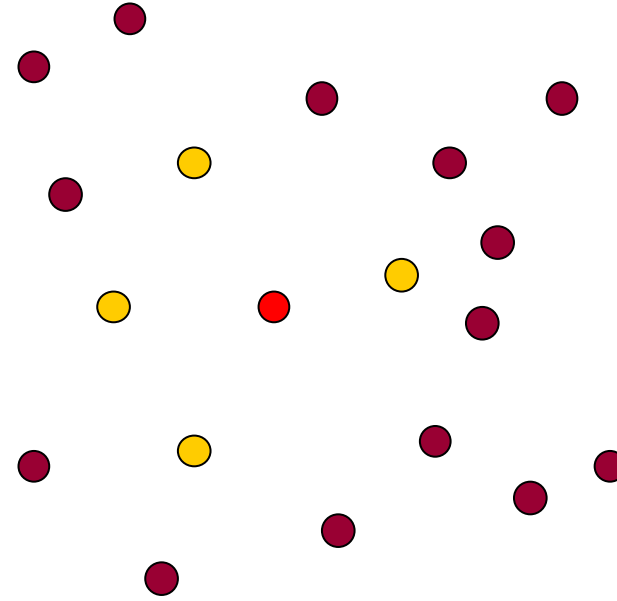


Third Trusted Party Architecture:

Nearest-Neighbor k -Anonymizing



- **STEP 1:** Determine a set S containing u and $k - 1$ u 's nearest neighbors.
- Can we return the MBR of set S as anonymity region ?

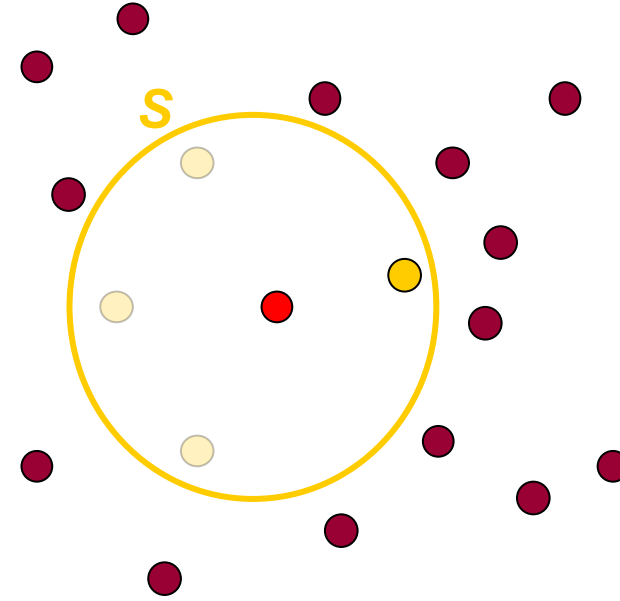


Third Trusted Party Architecture:

Nearest-Neighbor k -Anonymizing



- **STEP 1:** Determine a set S containing u and $k - 1$ u 's nearest neighbors.
- Can we return the MBR of set S as anonymity region ?
- **STEP 2:** Randomly select v from S .

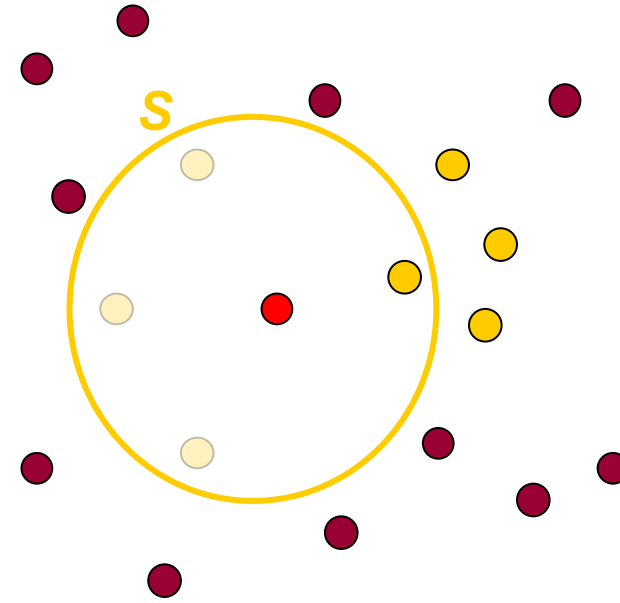


Third Trusted Party Architecture:

Nearest-Neighbor k -Anonymizing



- **STEP 1:** Determine a set S containing u and $k - 1$ u 's nearest neighbors.
- Can we return the MBR of set S as anonymity region ?
- **STEP 2:** Randomly select v from S .
- **STEP 3:** Determine a set S' containing v and v 's $k - 1$ nearest neighbors.

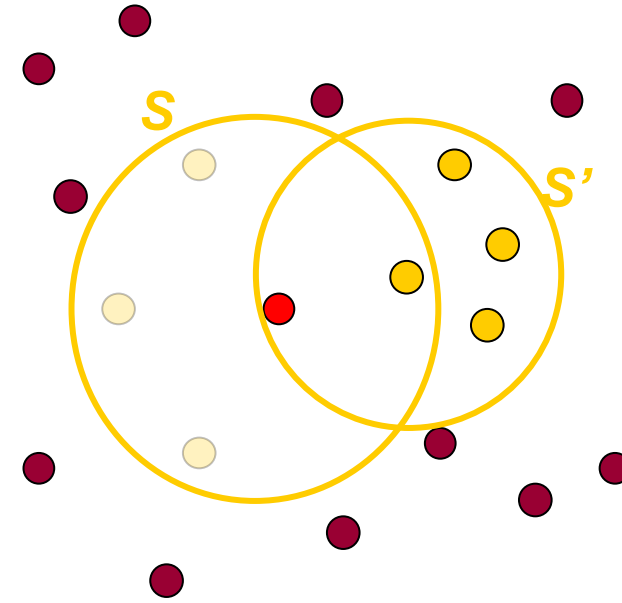


Third Trusted Party Architecture:

Nearest-Neighbor k -Anonymizing



- **STEP 1:** Determine a set S containing u and $k - 1$ u 's nearest neighbors.
- Can we return the MBR of set S as anonymity region ?
- **STEP 2:** Randomly select v from S .
- **STEP 3:** Determine a set S' containing v and v 's $k - 1$ nearest neighbors.

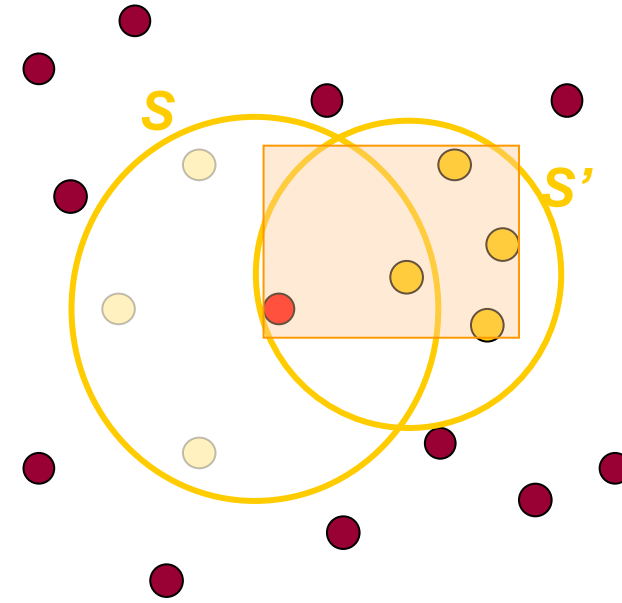


Third Trusted Party Architecture:

Nearest-Neighbor k -Anonymizing



- **STEP 1:** Determine a set S containing u and $k - 1$ u 's nearest neighbors.
- Can we return the MBR of set S as anonymity region ?
- **STEP 2:** Randomly select v from S .
- **STEP 3:** Determine a set S' containing v and v 's $k - 1$ nearest neighbors.
- **STEP 4:** A cloaked spatial region is an MBR of all users in S' and u .

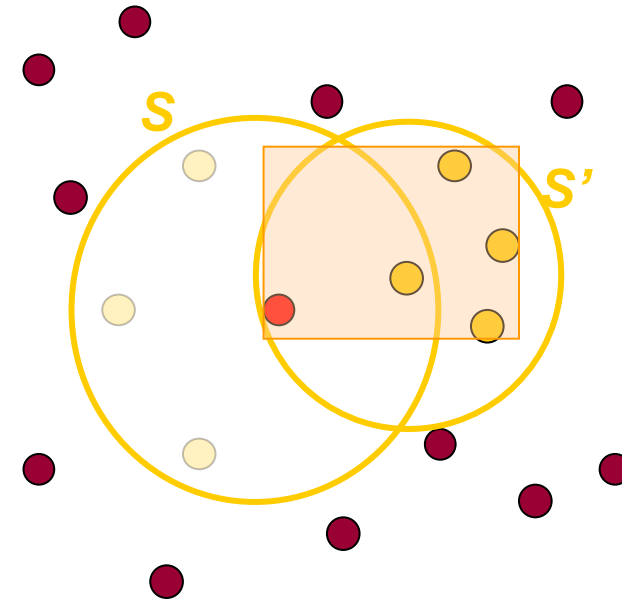


Third Trusted Party Architecture:

Nearest-Neighbor k -Anonymizing



- **STEP 1:** Determine a set S containing u and $k - 1$ u 's nearest neighbors.
- Can we return the MBR of set S as anonymity region ?
- **STEP 2:** Randomly select v from S .
- **STEP 3:** Determine a set S' containing v and v 's $k - 1$ nearest neighbors.
- **STEP 4:** A cloaked spatial region is an MBR of all users in S' and u .
- The main idea is that randomly selecting one of the k nearest neighbors achieves the k -anonymity

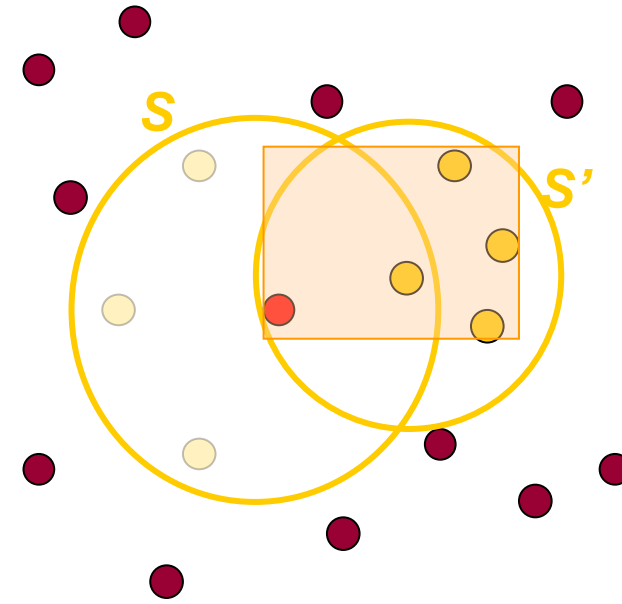


Third Trusted Party Architecture:

Nearest-Neighbor k -Anonymizing



- **STEP 1:** Determine a set S containing u and $k - 1$ u 's nearest neighbors.
- Can we return the MBR of set S as anonymity region ?
- **STEP 2:** Randomly select v from S .
- **STEP 3:** Determine a set S' containing v and v 's $k - 1$ nearest neighbors.
- **STEP 4:** A cloaked spatial region is an MBR of all users in S' and u .
- The main idea is that randomly selecting one of the k nearest neighbors achieves the k -anonymity



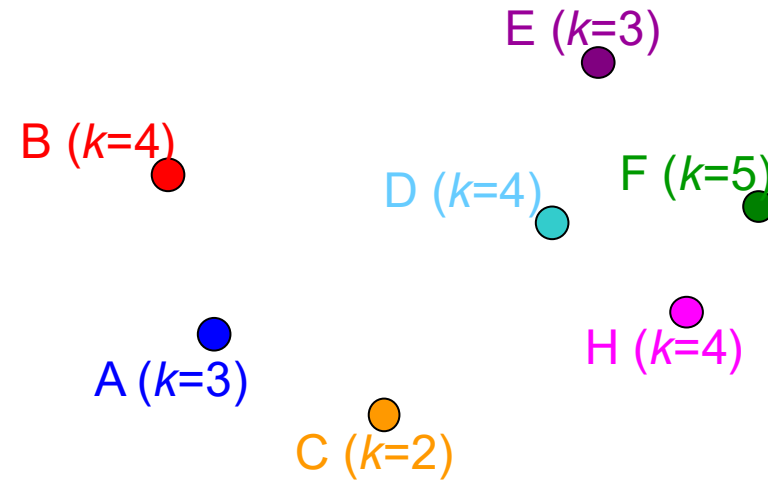
What if different users have different privacy requirements, service level needs

Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity

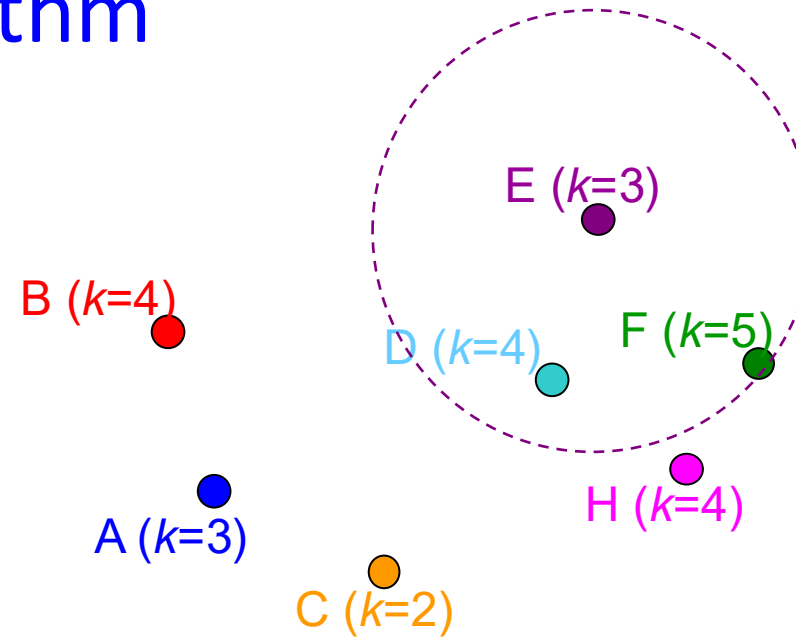


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area

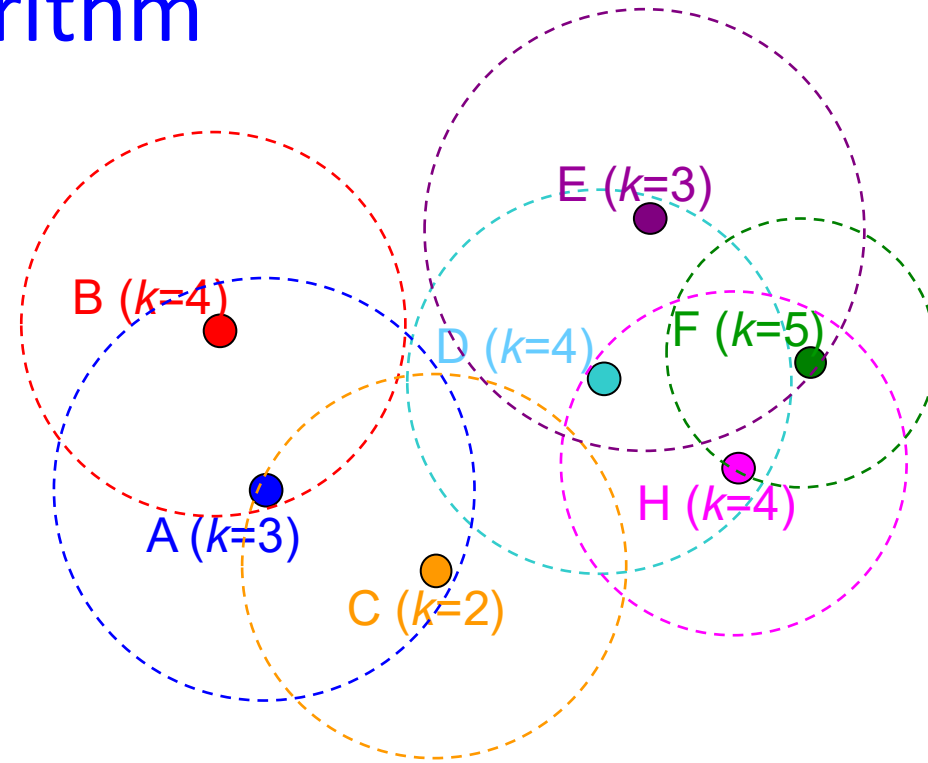


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area

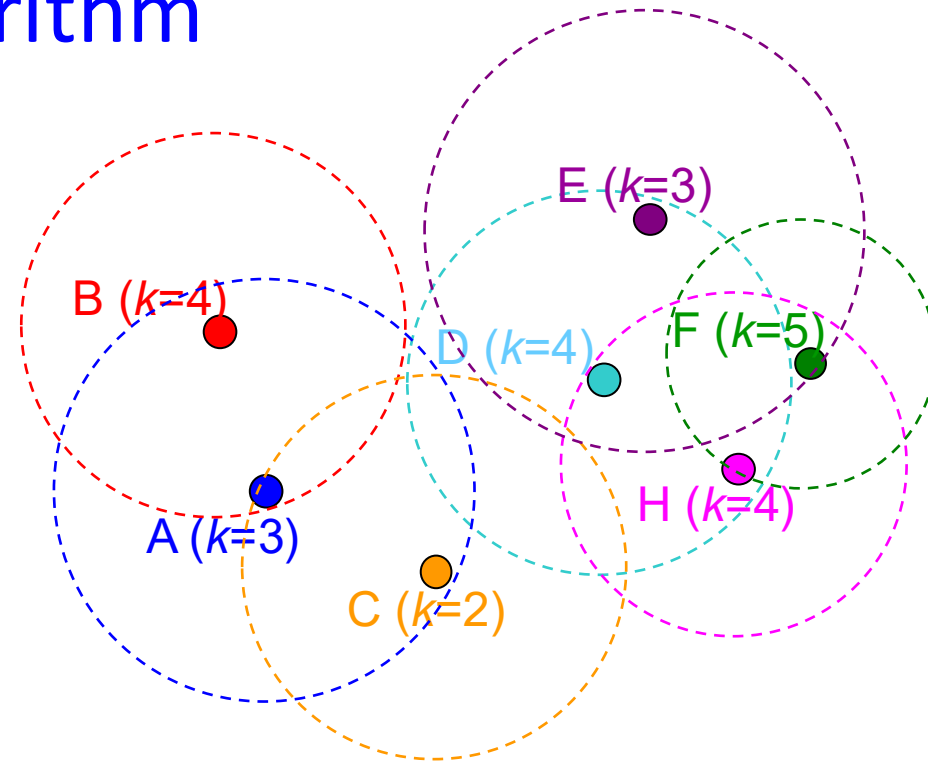


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

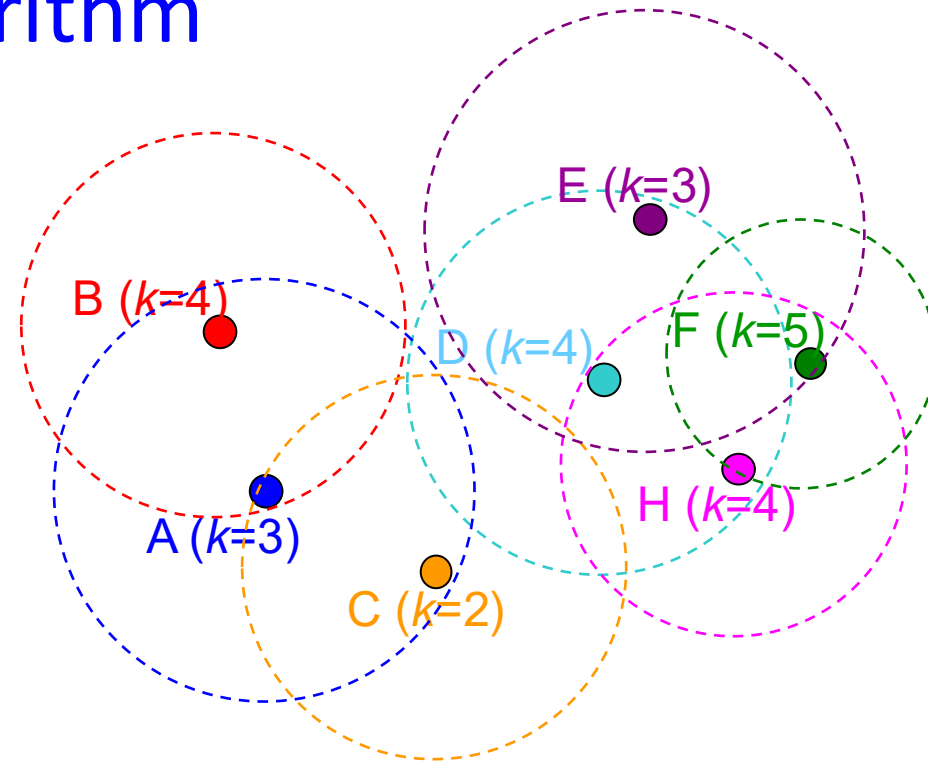


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

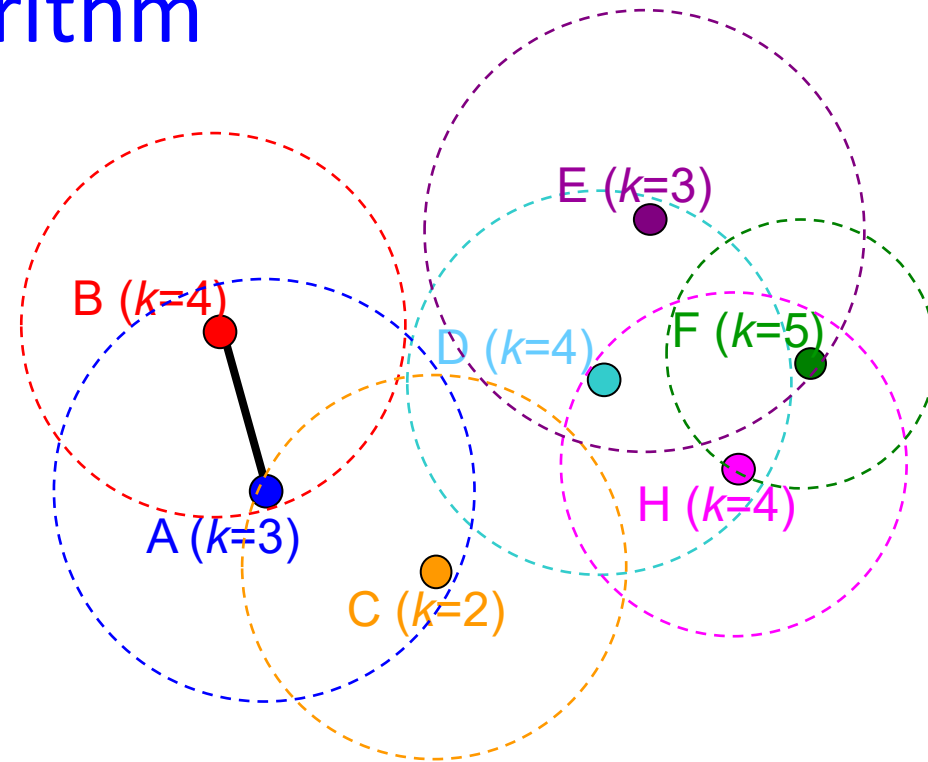


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

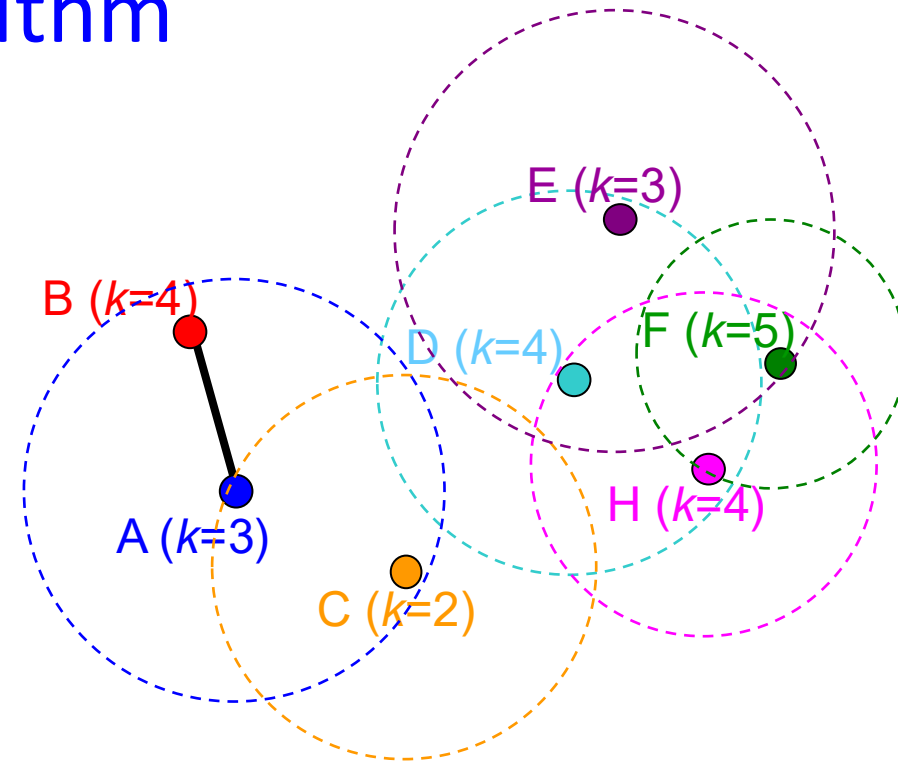


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

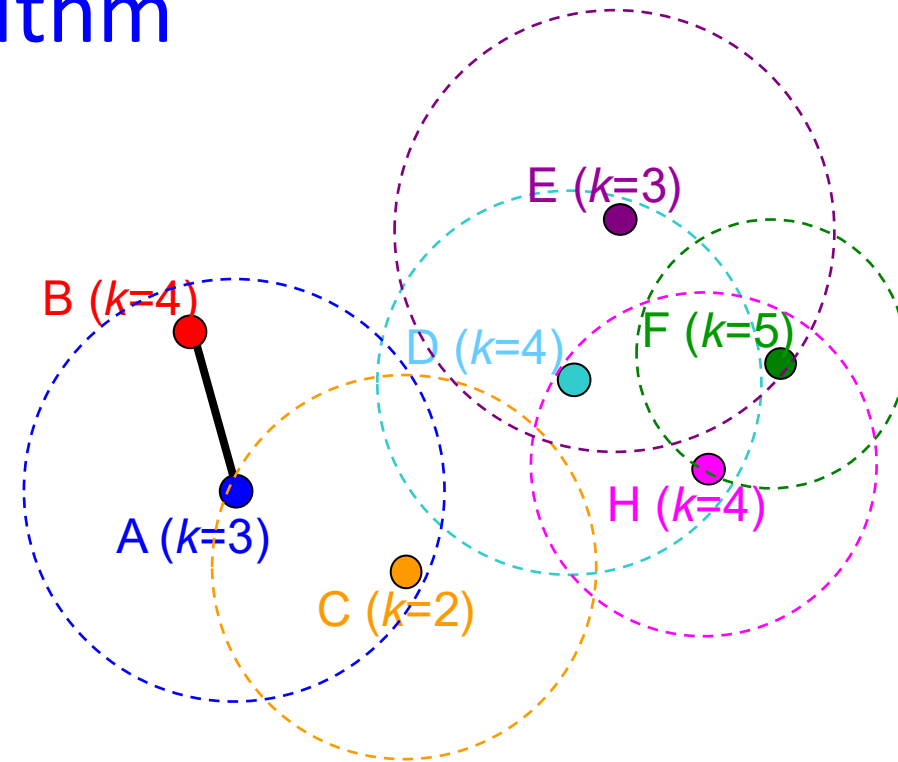


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

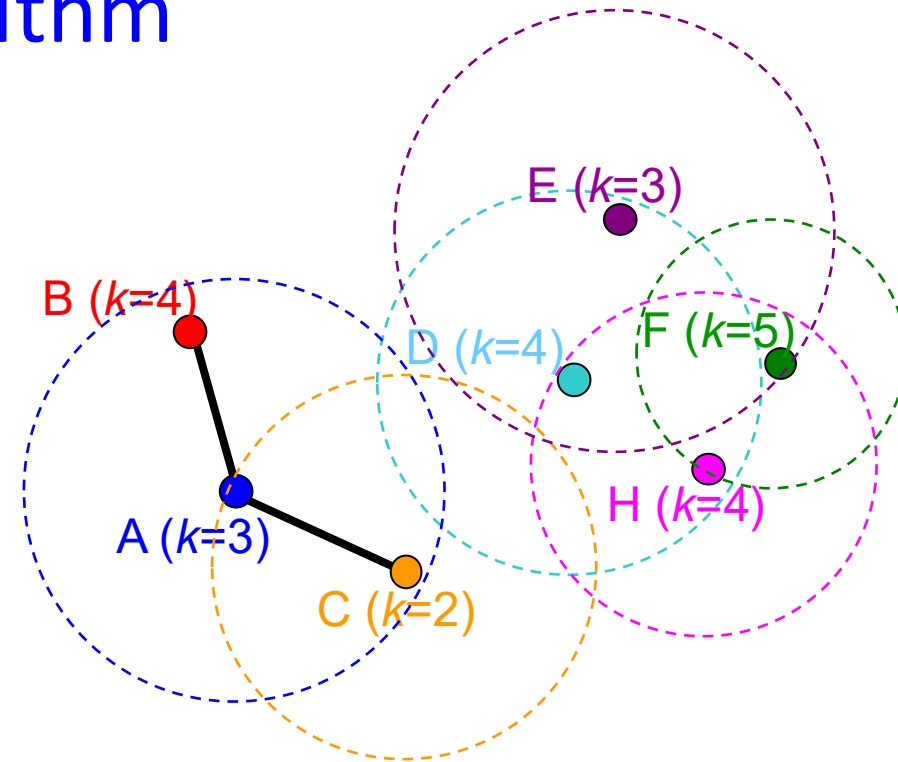


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

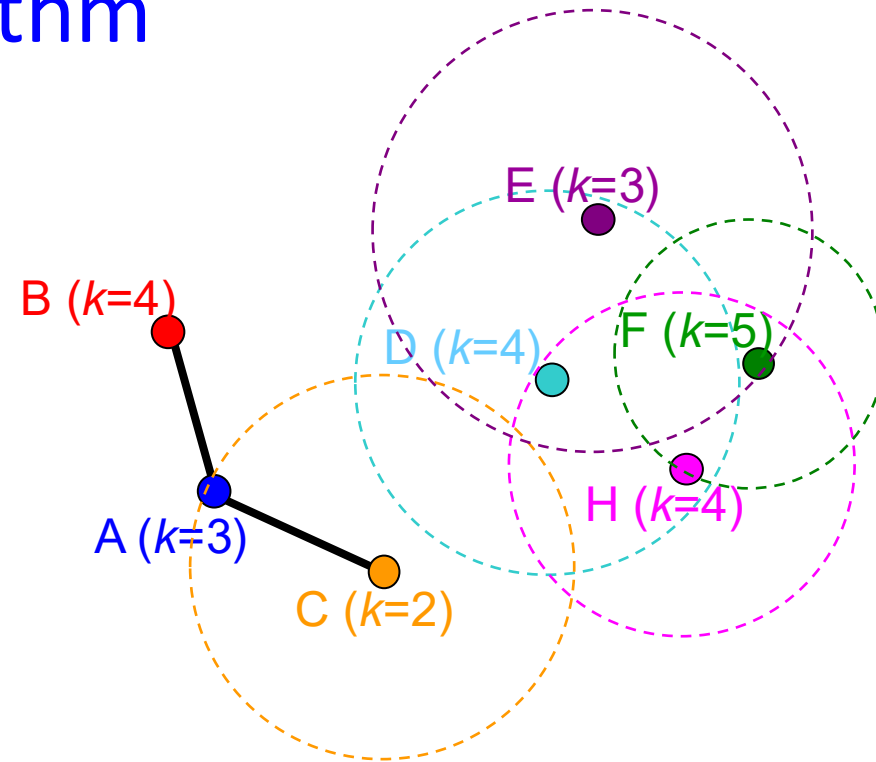


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

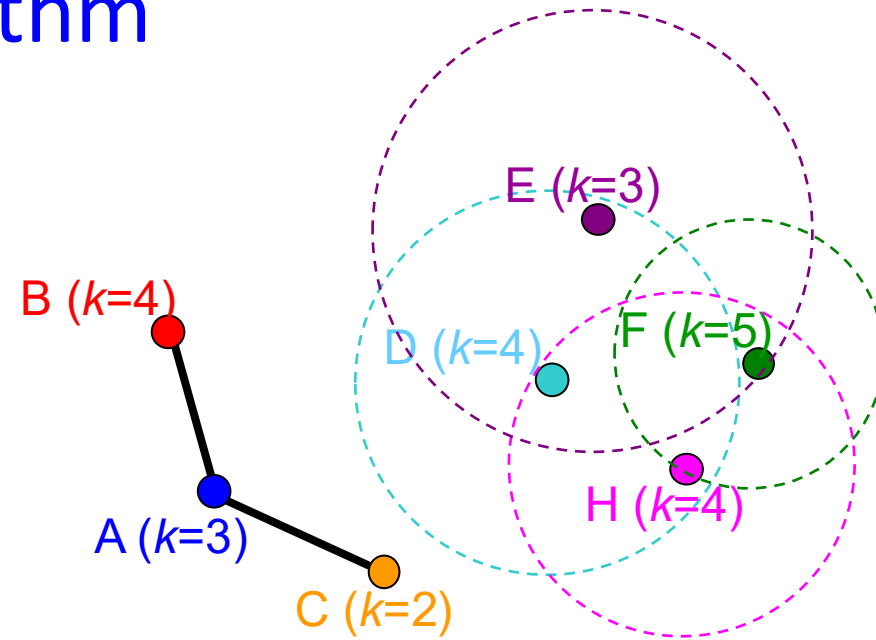


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

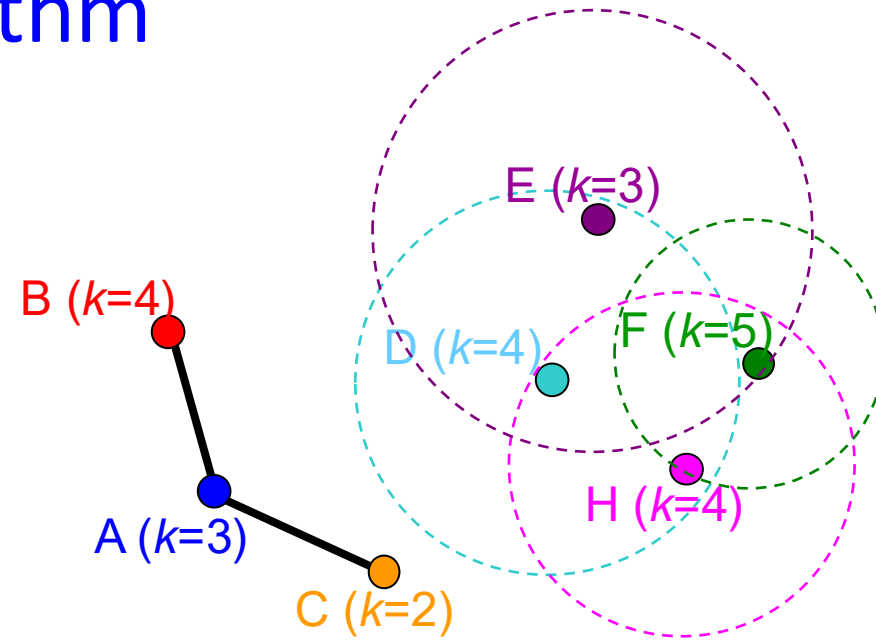


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

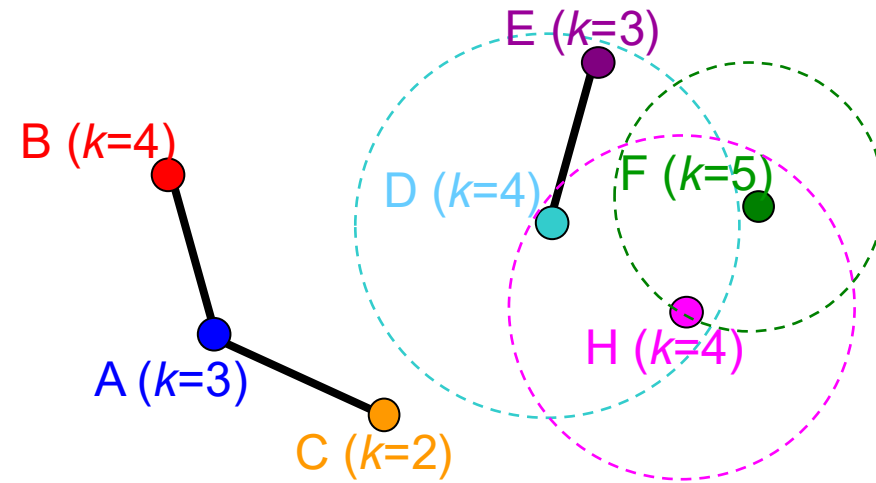


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

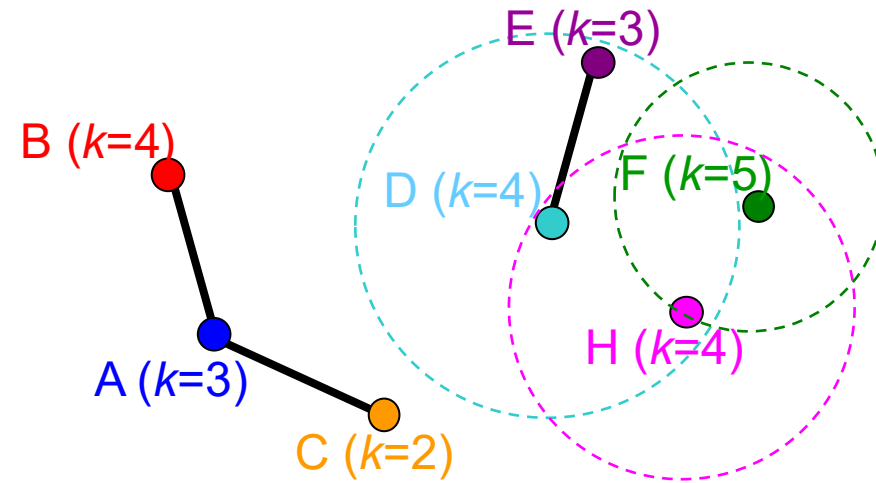


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

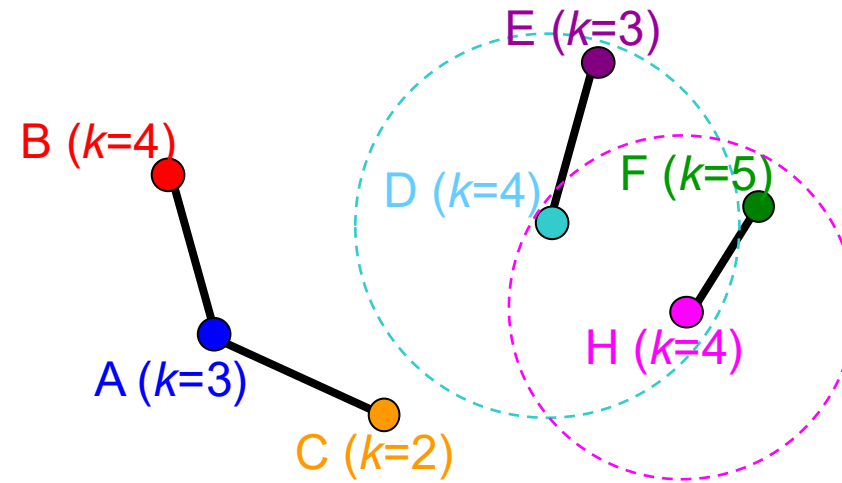


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

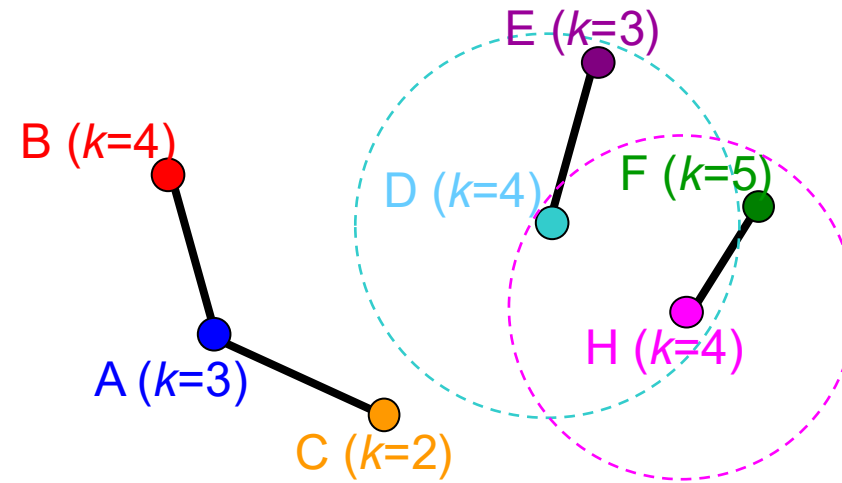


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

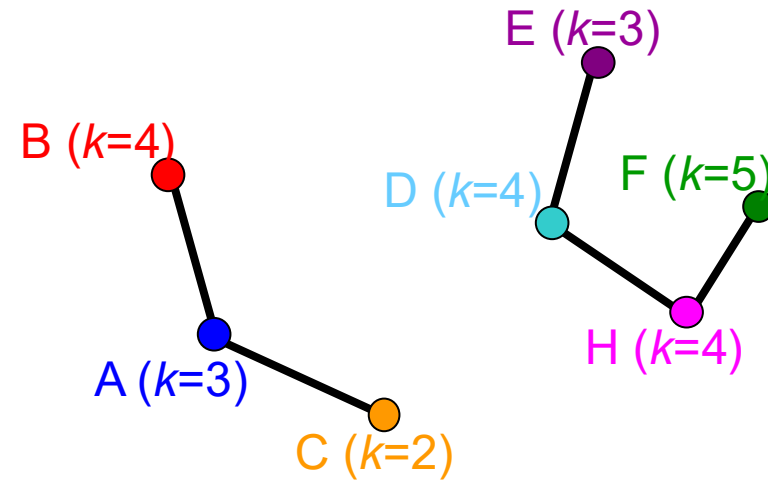


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.

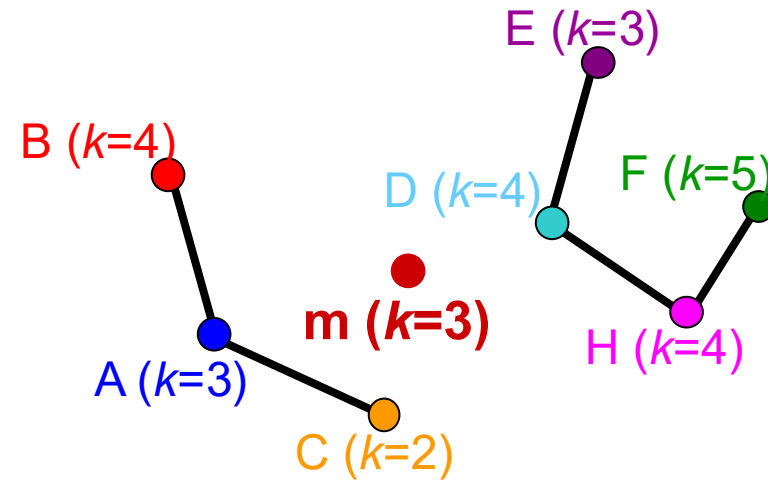


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.



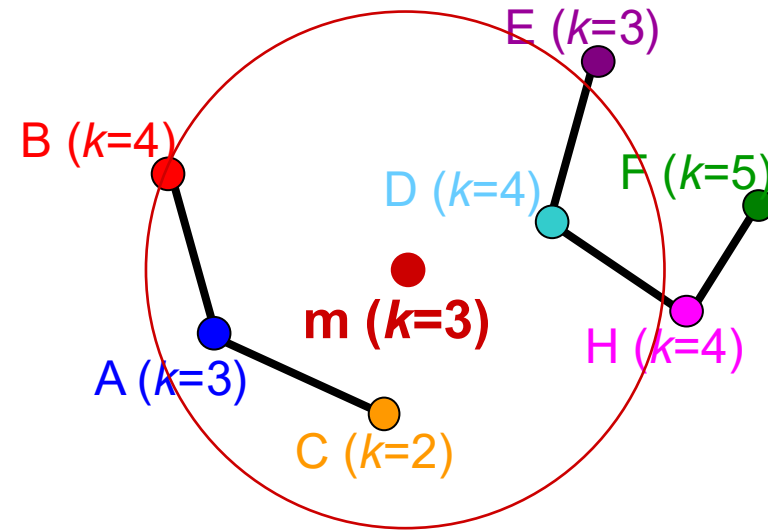
- For a new user m , add m to the graph. Find the set of nodes that are neighbors to m in the graph and has level of anonymity $\leq k$

Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.



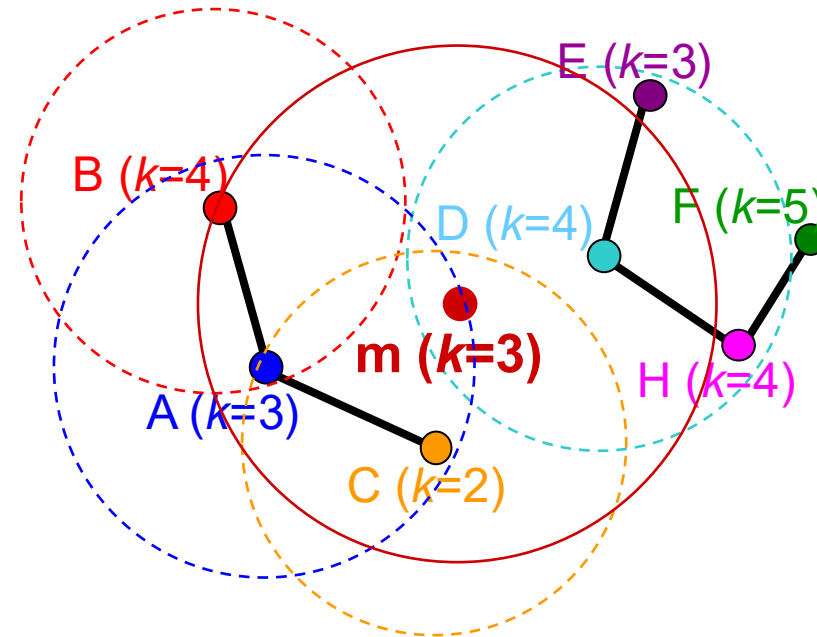
- For a new user m , add m to the graph. Find the set of nodes that are neighbors to m in the graph and has level of anonymity $\leq k$

Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.
- For a new user m , add m to the graph. Find the set of nodes that are neighbors to m in the graph and has level of anonymity $\leq k$

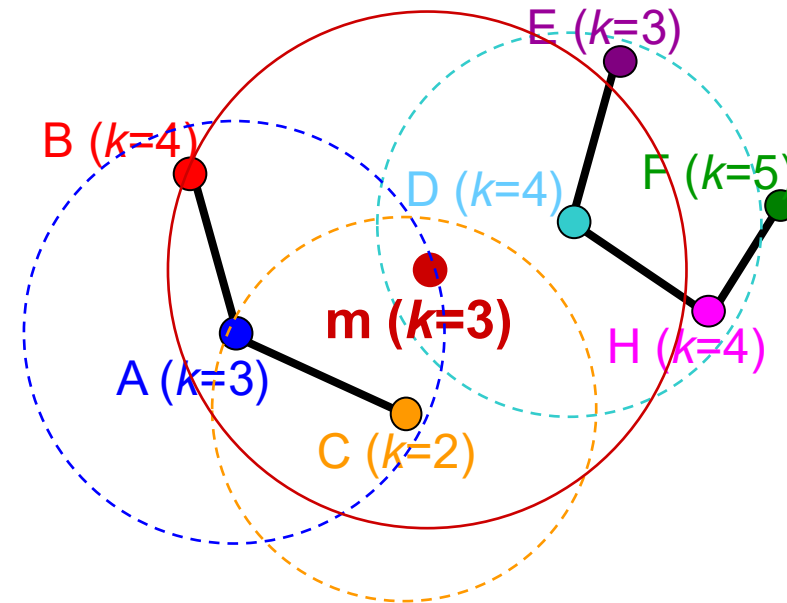


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.
- For a new user m , add m to the graph. Find the set of nodes that are neighbors to m in the graph and has level of anonymity $\leq k$

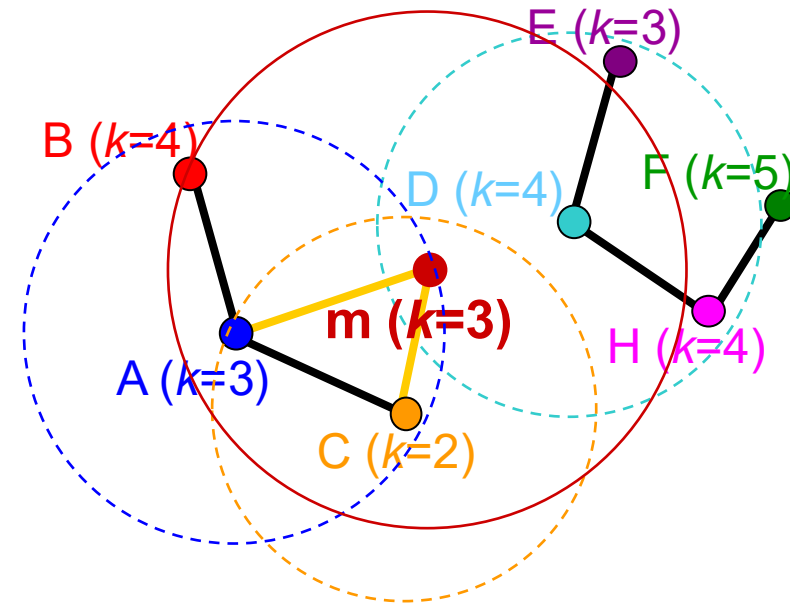


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.
- For a new user m , add m to the graph. Find the set of nodes that are neighbors to m in the graph and has level of anonymity $\leq k$

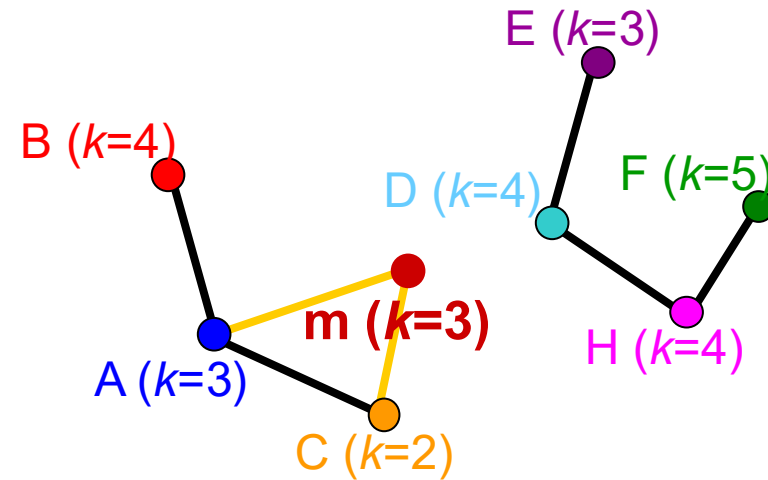


Third Trusted Party Architecture:

CliqueCloak Algorithm



- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.



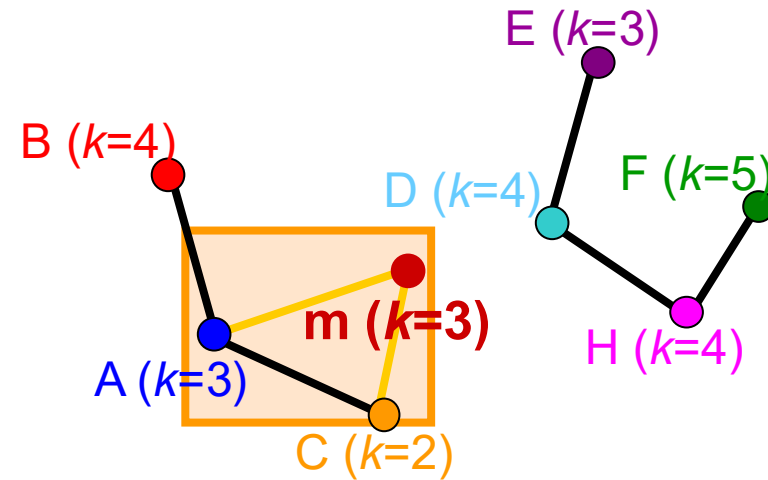
- For a new user m , add m to the graph. Find the set of nodes that are neighbors to m in the graph and has level of anonymity $\leq k$

Third Trusted Party Architecture:

CliqueCloak Algorithm



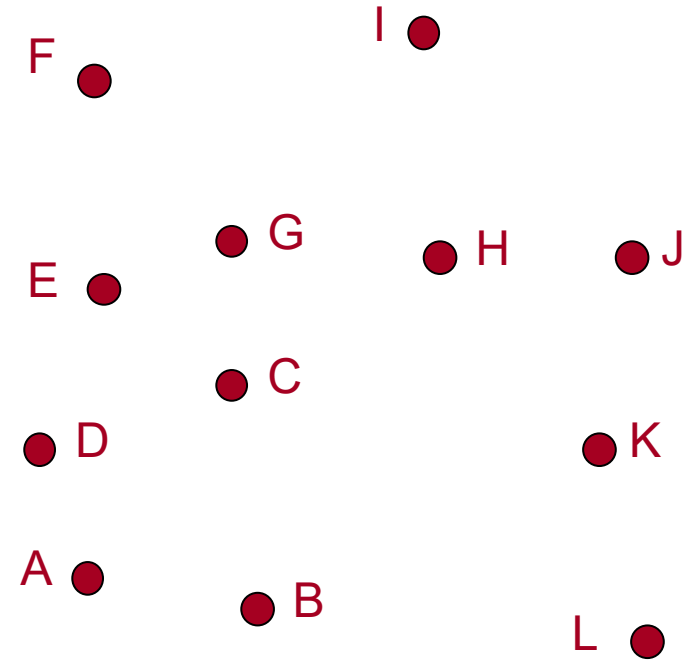
- Each user requests:
 - A level of k anonymity
 - A maximum cloaked area
- Build an undirected constraint graph. Two nodes are neighbors, if their maximum areas contain each other.



- For a new user m , add m to the graph. Find the set of nodes that are neighbors to m in the graph and has level of anonymity $\leq k$
- The cloaked region is the MBR that includes the user and neighboring nodes. All users within an MBR use that MBR as their cloaked region

Third Trusted Party Architecture:

Hilbert k-Anonymizing



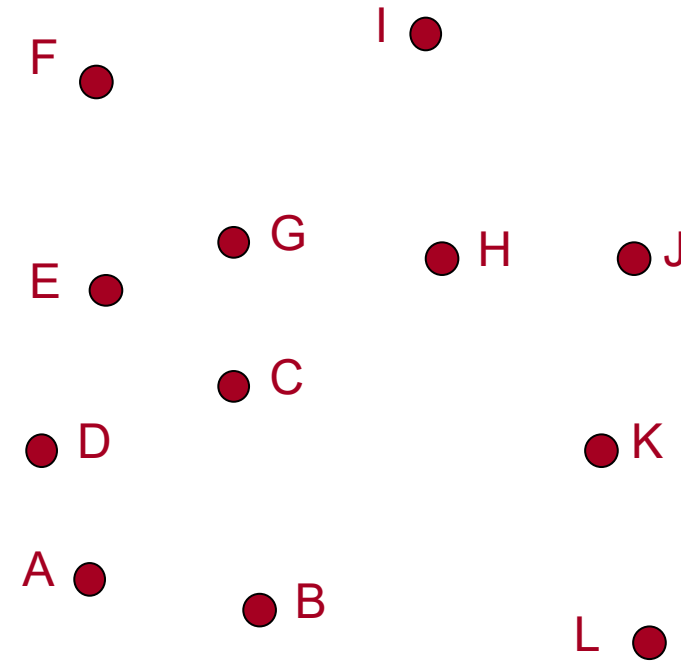
	A	B	C	D	E	F	G	H	I	J	K	L
k_u	6	5	4	5	4	5	6	5	7	4	5	4

Third Trusted Party Architecture:

Hilbert k-Anonymizing



- All user locations are sorted based on their Hilbert order



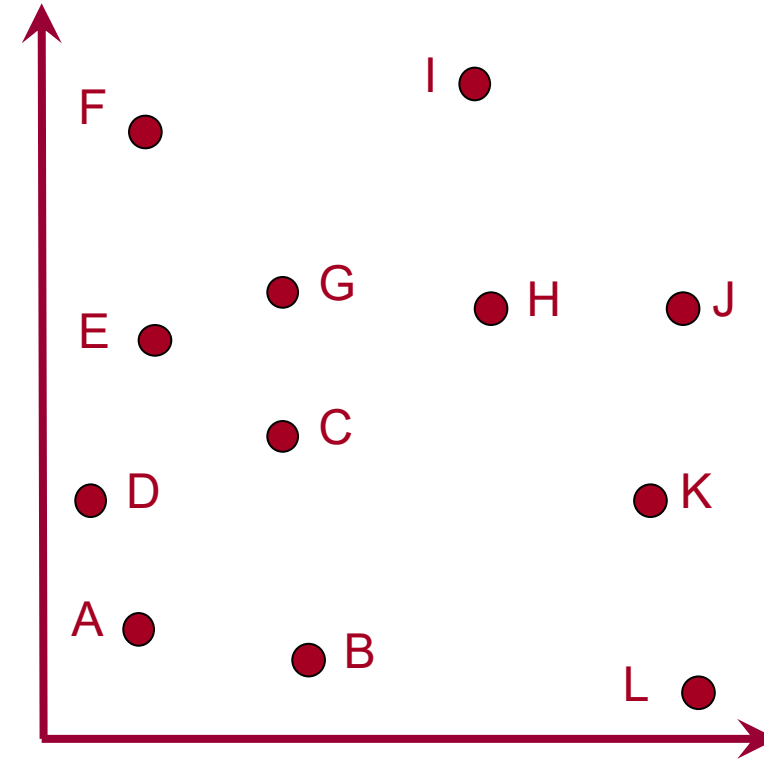
	A	B	C	D	E	F	G	H	I	J	K	L
k_u	6	5	4	5	4	5	6	5	7	4	5	4

Third Trusted Party Architecture:

Hilbert k-Anonymizing



- All user locations are sorted based on their Hilbert order



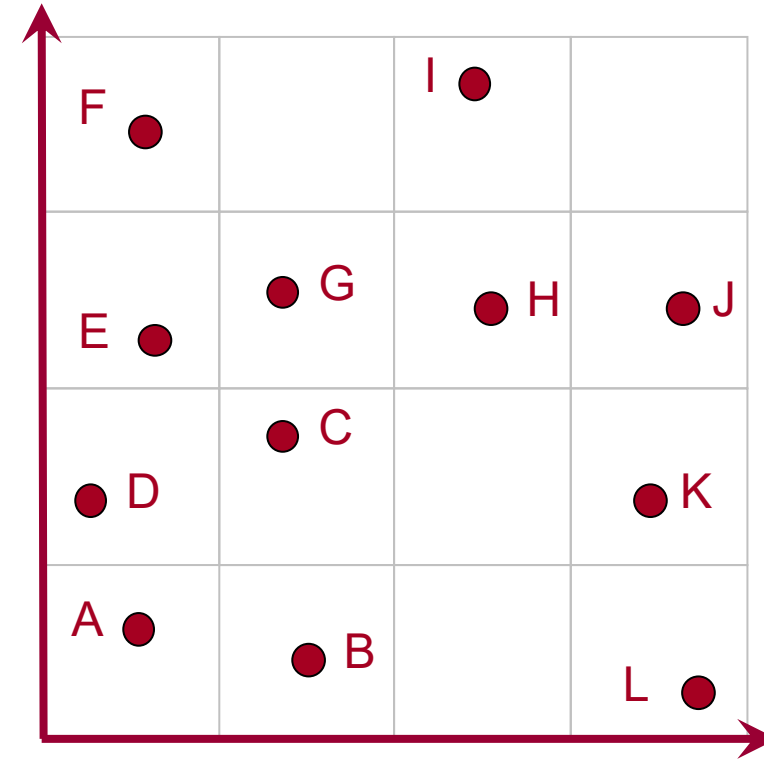
	A	B	C	D	E	F	G	H	I	J	K	L
k_u	6	5	4	5	4	5	6	5	7	4	5	4

Third Trusted Party Architecture:

Hilbert k-Anonymizing



- All user locations are sorted based on their Hilbert order



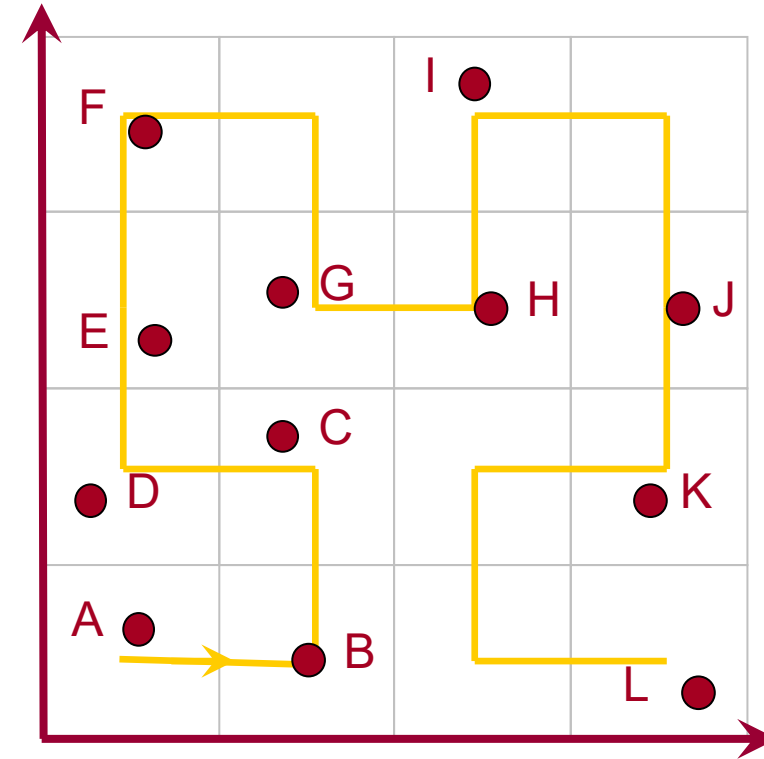
	A	B	C	D	E	F	G	H	I	J	K	L
k_u	6	5	4	5	4	5	6	5	7	4	5	4

Third Trusted Party Architecture:

Hilbert k-Anonymizing



- All user locations are sorted based on their Hilbert order



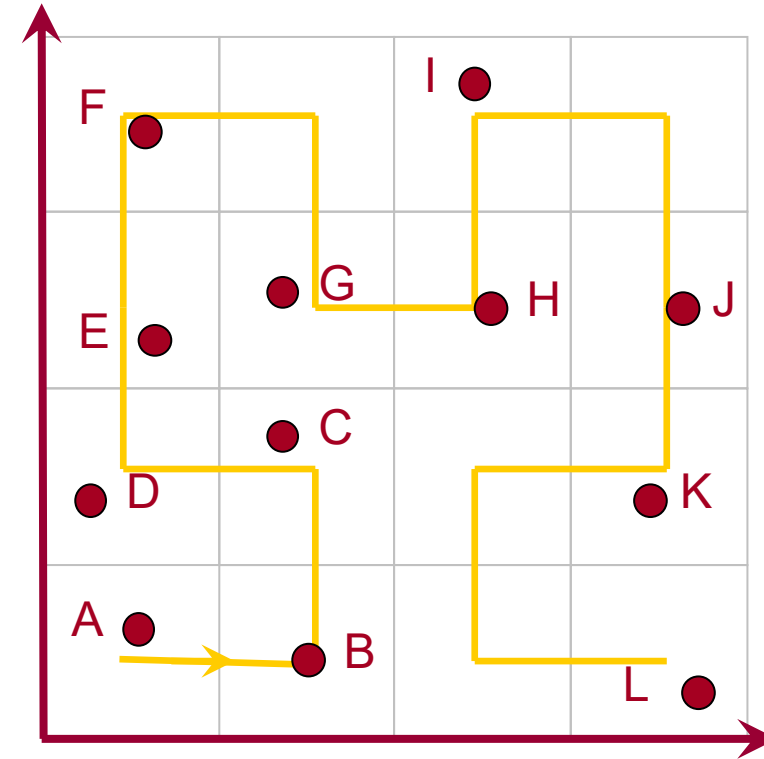
	A	B	C	D	E	F	G	H	I	J	K	L
k_u	6	5	4	5	4	5	6	5	7	4	5	4

Third Trusted Party Architecture:

Hilbert k-Anonymizing



- All user locations are sorted based on their Hilbert order



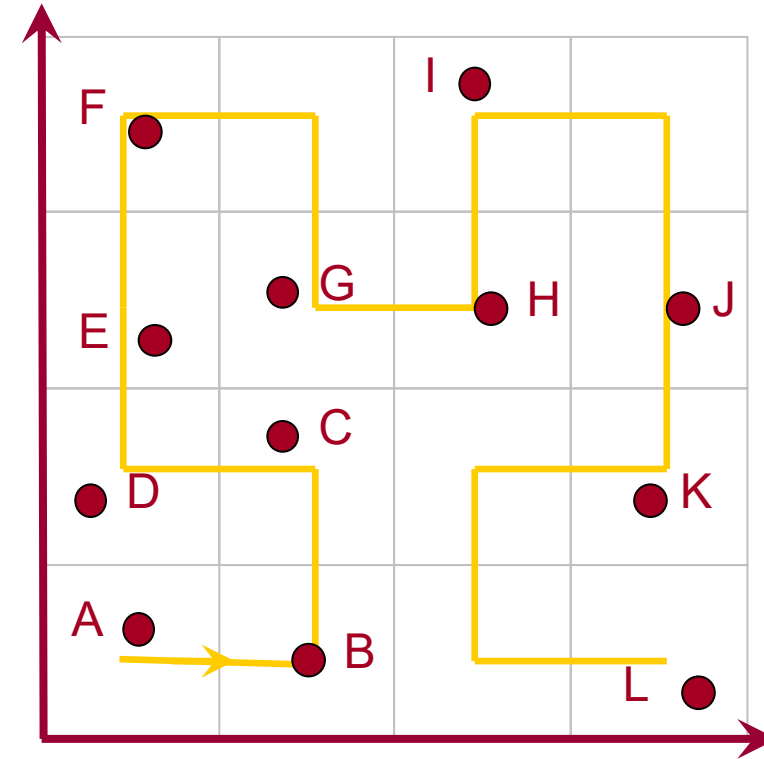
	A	B	C	D	E	F	G	H	I	J	K	L
k_u	6	5	4	5	4	5	6	5	7	4	5	4
$Rank_u$	0	1	2	3	4	5	6	7	8	9	10	11

Third Trusted Party Architecture:

Hilbert k-Anonymizing



- All user locations are sorted based on their Hilbert order
- To anonymize a user, we compute *start* and *end* values as:
 - $start = rank_u - (rank_u \bmod k_u)$
 - $end = start + k_u - 1$
- A cloaked spatial region is an MBR of all users within the range (from *start* to *end*).



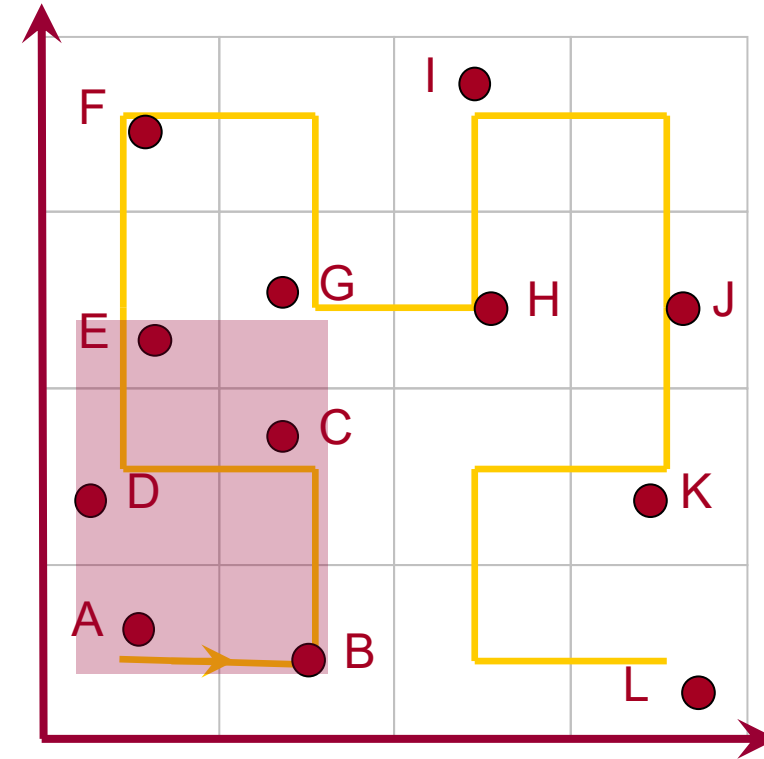
	A	B	C	D	E	F	G	H	I	J	K	L
k_u	6	5	4	5	4	5	6	5	7	4	5	4
$Rank_u$	0	1	2	3	4	5	6	7	8	9	10	11

Third Trusted Party Architecture:

Hilbert k-Anonymizing



- All user locations are sorted based on their Hilbert order
- To anonymize a user, we compute *start* and *end* values as:
 - $start = rank_u - (rank_u \bmod k_u)$
 - $end = start + k_u - 1$
- A cloaked spatial region is an MBR of all users within the range (from *start* to *end*).



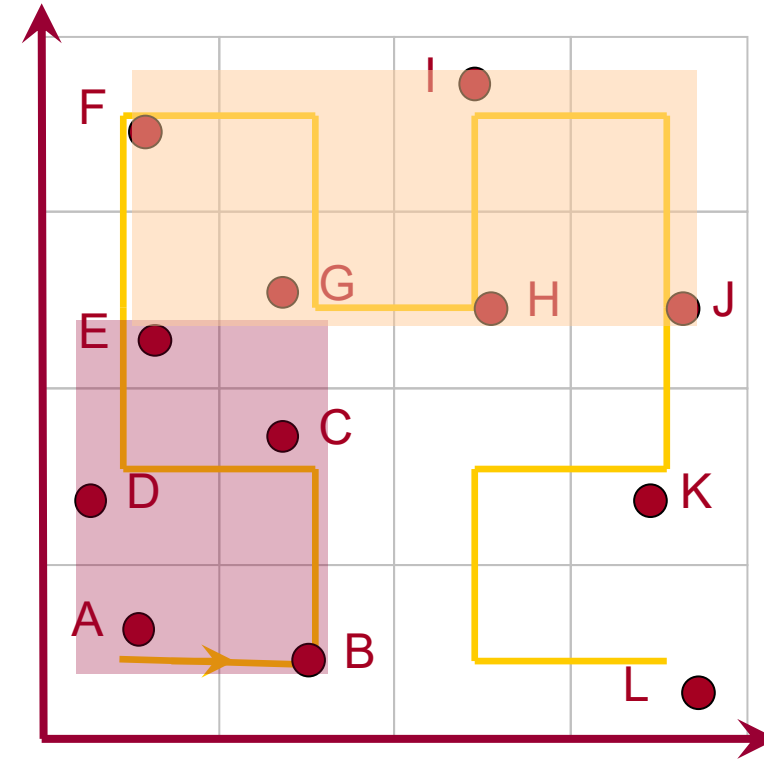
	A	B	C	D	E	F	G	H	I	J	K	L
k_u	6	5	4	5	4	5	6	5	7	4	5	4
$Rank_u$	0	1	2	3	4	5	6	7	8	9	10	11

Third Trusted Party Architecture:

Hilbert k-Anonymizing



- All user locations are sorted based on their Hilbert order
- To anonymize a user, we compute *start* and *end* values as:
 - $start = rank_u - (rank_u \bmod k_u)$
 - $end = start + k_u - 1$
- A cloaked spatial region is an MBR of all users within the range (from *start* to *end*).



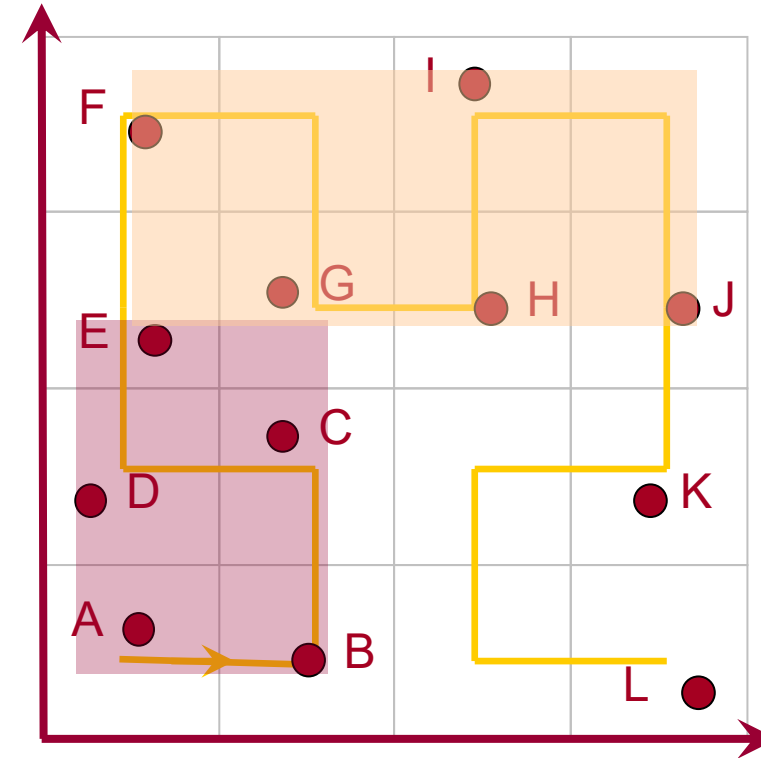
	A	B	C	D	E	F	G	H	I	J	K	L
k_u	6	5	4	5	4	5	6	5	7	4	5	4
$Rank_u$	0	1	2	3	4	5	6	7	8	9	10	11

Third Trusted Party Architecture:

Hilbert k-Anonymizing

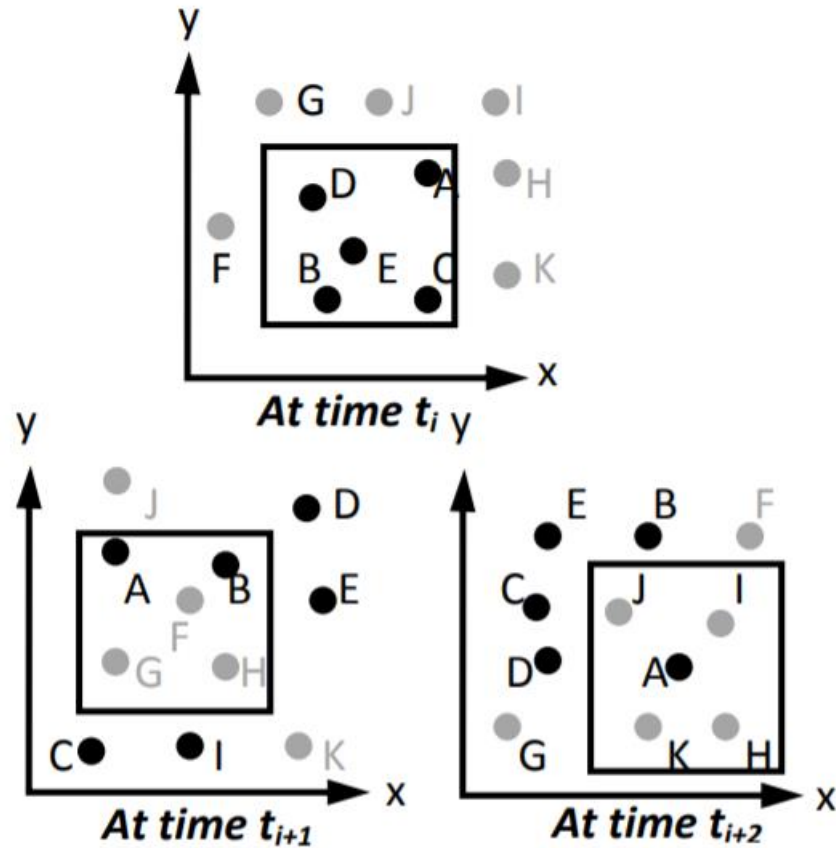


- All user locations are sorted based on their Hilbert order
- To anonymize a user, we compute *start* and *end* values as:
 - $start = rank_u - (rank_u \bmod k_u)$
 - $end = start + k_u - 1$
- A cloaked spatial region is an MBR of all users within the range (from *start* to *end*).
- The main idea is that it is always the case that k_u users would have the same $[start, end]$ interval



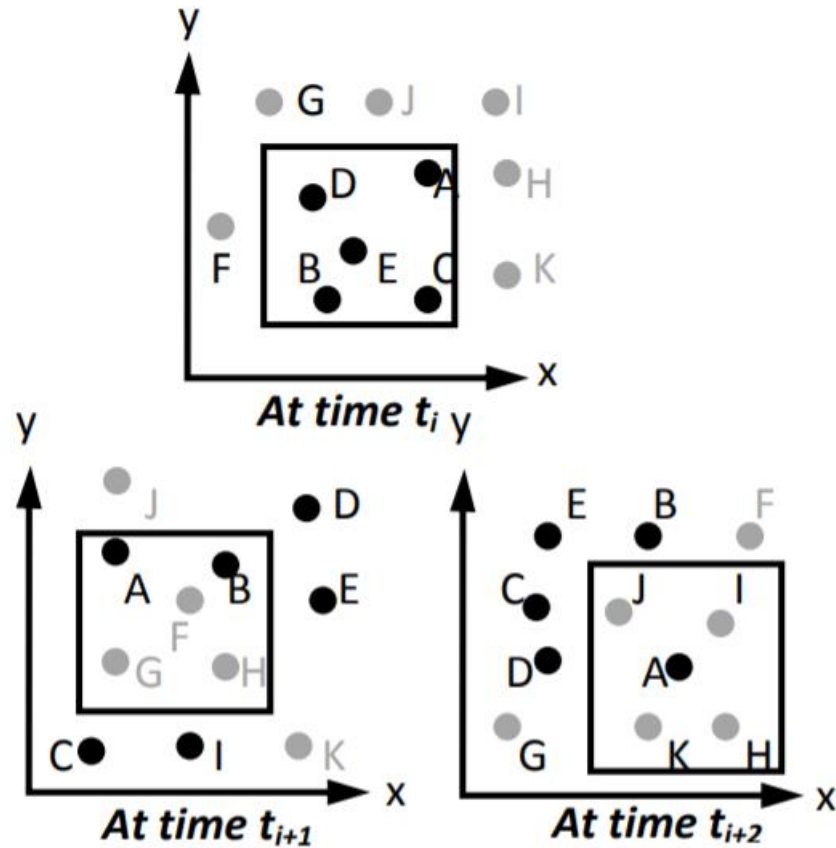
	A	B	C	D	E	F	G	H	I	J	K	L
k_u	6	5	4	5	4	5	6	5	7	4	5	4
$Rank_u$	0	1	2	3	4	5	6	7	8	9	10	11

Anonymizing Trajectories ?

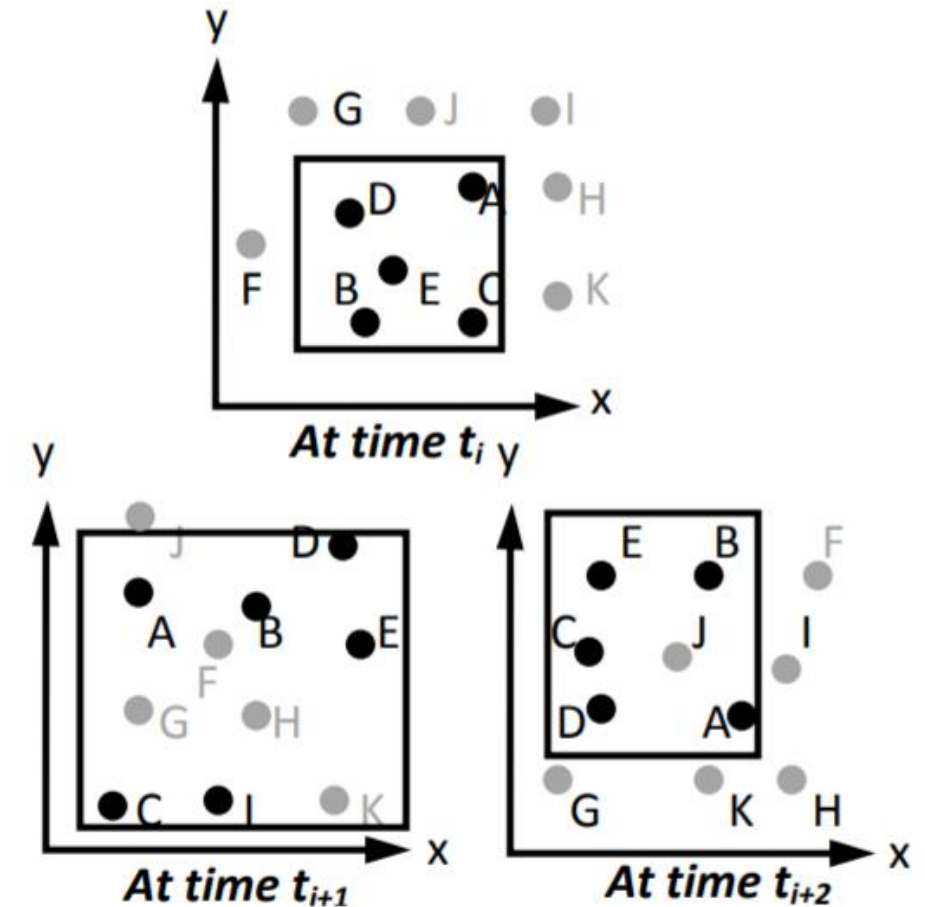


- Correlation Attack
 - User A submits query at time i for $k = 5$
 - At time $i + 1$, his anonymity reduces to $\frac{1}{2}$
 - At time $i + 2$, his identity is revealed.

Anonymizing Trajectories ?



- Correlation Attack
 - User A submits query at time i for $k = 5$
 - At time $i + 1$, his anonymity reduces to $\frac{1}{2}$
 - At time $i + 2$, his identity is revealed.



Possible Solution.
But need a lot of noise.

System Architectures for Online Location Privacy



❖ *Third trusted party architecture*

- ❖ A centralized trusted entity is responsible for gathering information and providing the required privacy for each user
- ❖ Analogous to output perturbation

❖ *Client-Server architecture*

- ❖ Users communicate directly with the server with noisy locations.
- ❖ Analogous to input perturbation

❖ *Peer-to-Peer cooperative architecture*

- ❖ Users collaborate with each other without the interleaving of a centralized entity to provide customized privacy for each single user

System Architectures for Online Location Privacy



❖ *Third trusted party architecture*

- ❖ A centralized trusted entity is responsible for gathering information and providing the required privacy for each user
- ❖ Analogous to output perturbation

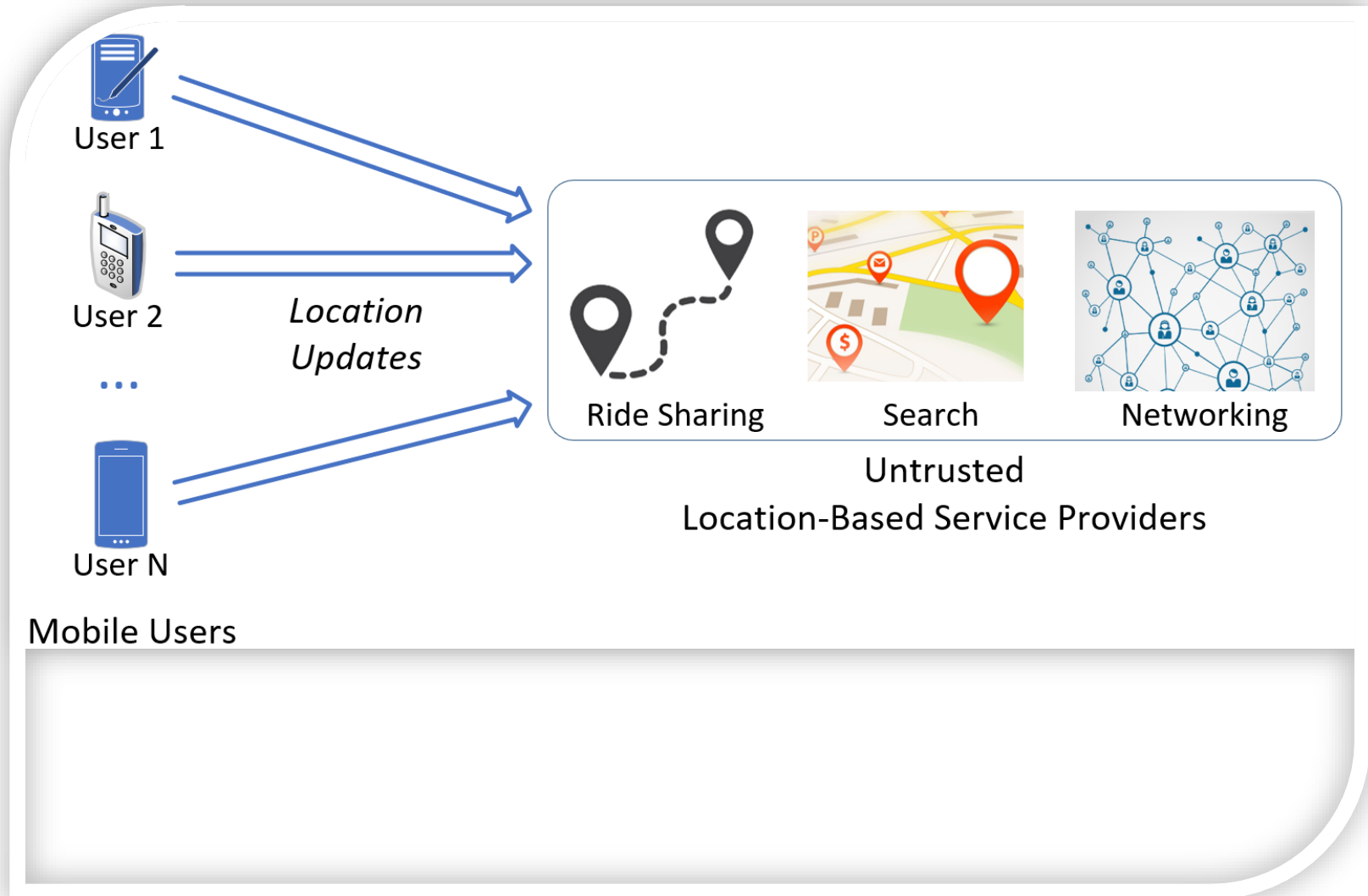
❖ *Client-Server architecture*

- ❖ Users communicate directly with the server with noisy locations.
- ❖ Analogous to input perturbation

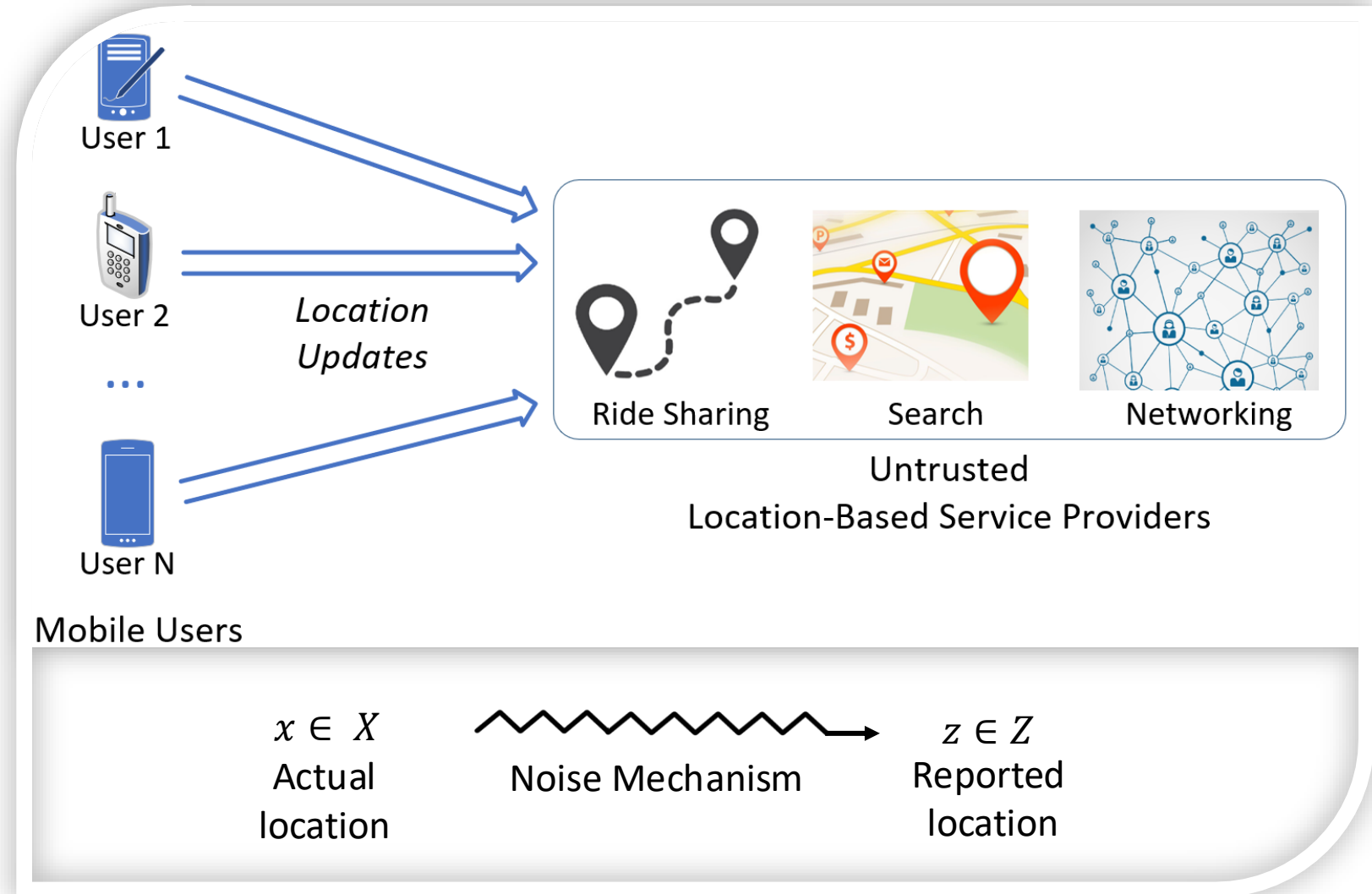
❖ *Peer-to-Peer cooperative architecture*

- ❖ Users collaborate with each other without the interleaving of a centralized entity to provide customized privacy for each single user

Client-Server Architecture



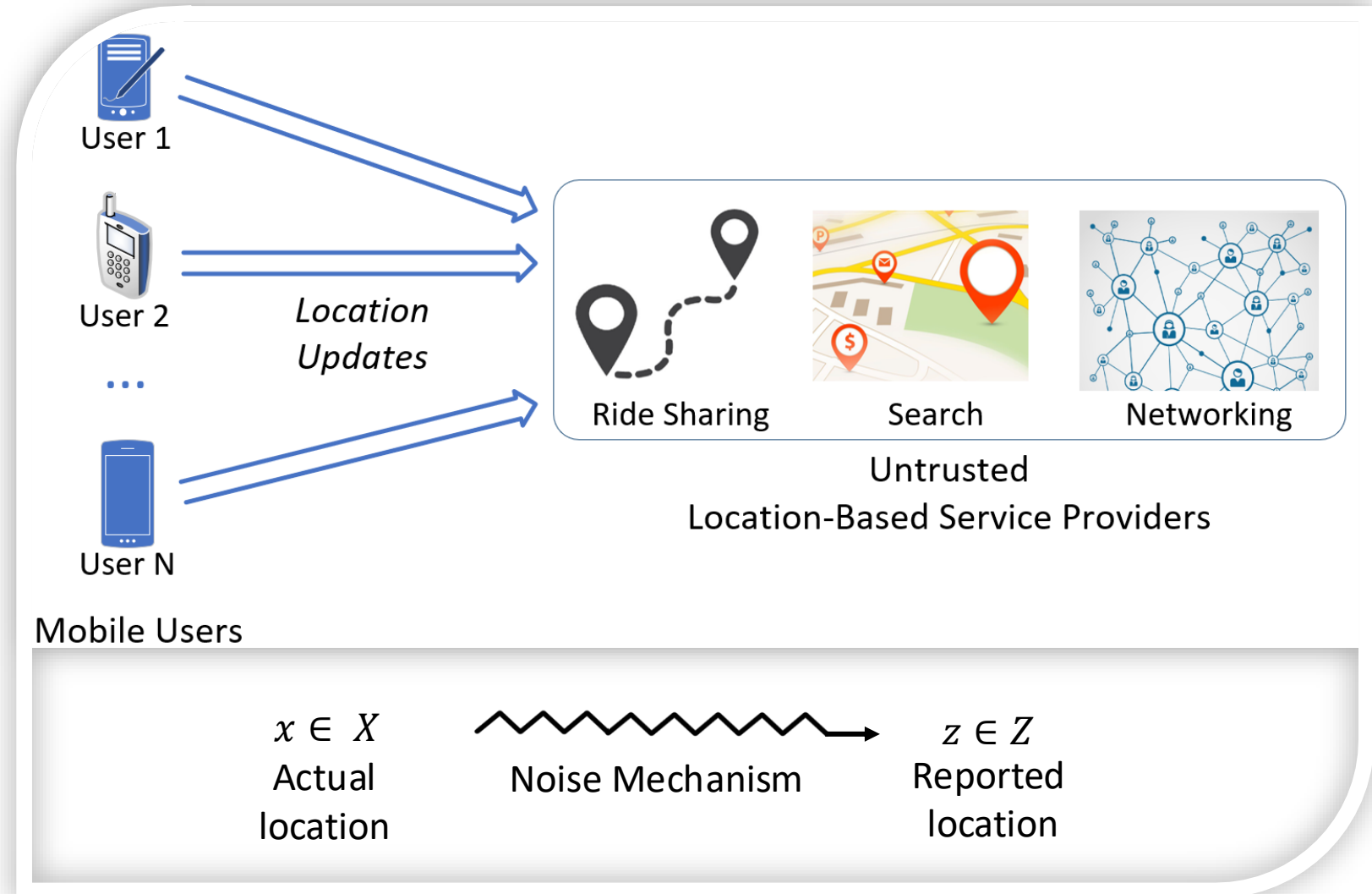
Client-Server Architecture





Client-Server Architecture

- Users randomly perturb their inputs.
- No need for a trusted centralized party.
- More obfuscation means Better Privacy $\Leftrightarrow$ Utility Loss
e.g. requesting Uber.





Client-Server Architecture

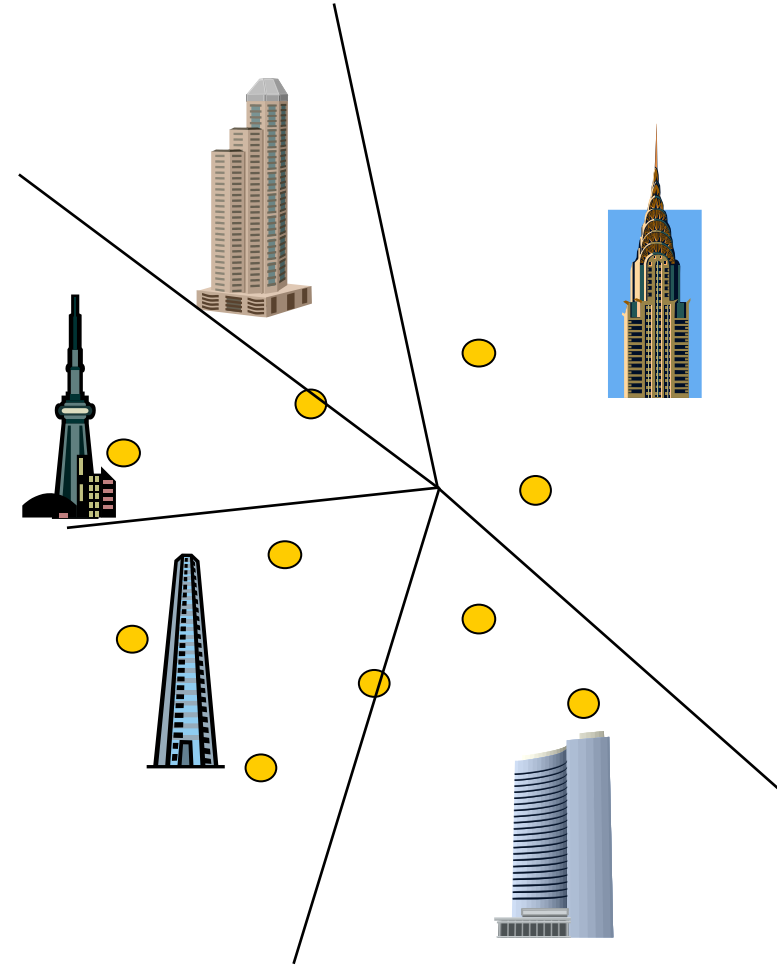
- Clients try to *cheat* the server using either fake locations or fake space
- Simple to implement, easy to integrate with existing technologies
- Lower quality of service
- *Examples*: Landmark objects, false dummies, and space transformation

Client-Server Architecture:

Landmark objects



- Instead of reporting the exact location, report the location of a closest landmark
- The query answer will be based on the landmark
- Voronoi diagrams can be used to identify the closest landmark





Moving to a better privacy definition

- Early efforts
 - Location Generalization.
 - Location Cloaking, k -anonymity models.



Moving to a better privacy definition

- Early efforts
 - Location Generalization.
 - Location Cloaking, k -anonymity models.

Lack of a formal privacy guarantee



Moving to a better privacy definition

- Early efforts
 - Location Generalization.
 - Location Cloaking, k -anonymity models.
- Lack of a formal privacy guarantee
- Geo-Indistinguishability [Andres et. al., CCS 2013]
 - A powerful model that mimics traditional Differential Privacy.
 - Broadens the scope, over distance metric.
 - prevents an adversary from inferring with high probability the user's whereabouts.



ε -Geo-Indistinguishability (GeoInd)

Let X, Z be the set of all possible user locations.

A randomized mechanism $K(X)(Z)$

satisfies ε -GeoInd iff for all $x, x', z \subseteq Z$:

$$\frac{\Pr(K(x) = z)}{\Pr(K(x') = z)} \leq e^{\varepsilon d(x, x')}$$

where ε is the privacy budget (higher $\rightarrow$ less privacy)



ϵ -Geo-Indistinguishability (GeoInd)

Let X, Z be the set of all possible user locations.

A randomized mechanism $K(X)(Z)$

satisfies ϵ -GeoInd iff for all $x, x', z \subseteq Z$:

$$\frac{\Pr(K(x) = z)}{\Pr(K(x') = z)} \leq e^{\epsilon d(x, x')}$$

where ϵ is the privacy budget (higher $\rightarrow$ less privacy)

The constraints are over the probability of reporting location z when the input is x , and the other one is when the input is another location x' . It must be satisfied for all possibilities of x, x' and z .

The equation can be read as: Given a reported location z , an adversary cannot distinguish whether the user location is x or x' by a factor larger than $e^{\epsilon d(x, x')}$.



ϵ -Geo-Indistinguishability (GeoInd)

Let X, Z be the set of all possible user locations.

A randomized mechanism $K(X)(Z)$

satisfies ϵ -GeoInd iff for all $x, x', z \subseteq Z$:

$$\frac{\Pr(K(x) = z)}{\Pr(K(x') = z)} \leq e^{\epsilon d(x, x')}$$

where ϵ is the privacy budget (higher $\rightarrow$ less privacy)

The constraints are over the probability of reporting location z when the input is x , and the other one is when the input is another location x' . It must be satisfied for all possibilities of x, x' and z .

The equation can be read as: Given a reported location z , an adversary cannot distinguish whether the user location is x or x' by a factor larger than $e^{\epsilon d(x, x')}$.



A GeoInd mechanism should produce similar results when applied to locations that are geographically close.



ϵ -Geo-Indistinguishability (GeoInd)

Let X, Z be the set of all possible user locations.

A randomized mechanism $K(X)(Z)$ satisfies ϵ -GeoInd iff for all $x, x', z \subseteq Z$:

$$\frac{\Pr(K(x) = z)}{\Pr(K(x') = z)} \leq e^{\epsilon d(x, x')}$$

where ϵ is the privacy budget (higher $\rightarrow$ less privacy)

The constraints are over the probability of reporting location z when the input is x , and the other one is when the input is another location x' . It must be satisfied for all possibilities of x, x' and z .

The equation can be read as: Given a reported location z , an adversary cannot distinguish whether the user location is x or x' by a factor larger than $e^{\epsilon d(x, x')}$.



A GeoInd mechanism should produce similar results when applied to locations that are geographically close.



The uncertainty of the adversary increases as he tries to narrow down your location.

E.g. LA ok, USC not ok.



Planar Laplace Mechanism (PL)

The bi-variate pdf of PL noise mechanism is:

$$f_K(z \mid x) = \frac{\varepsilon^2}{2\pi} e^{-\varepsilon d(z, x)}$$



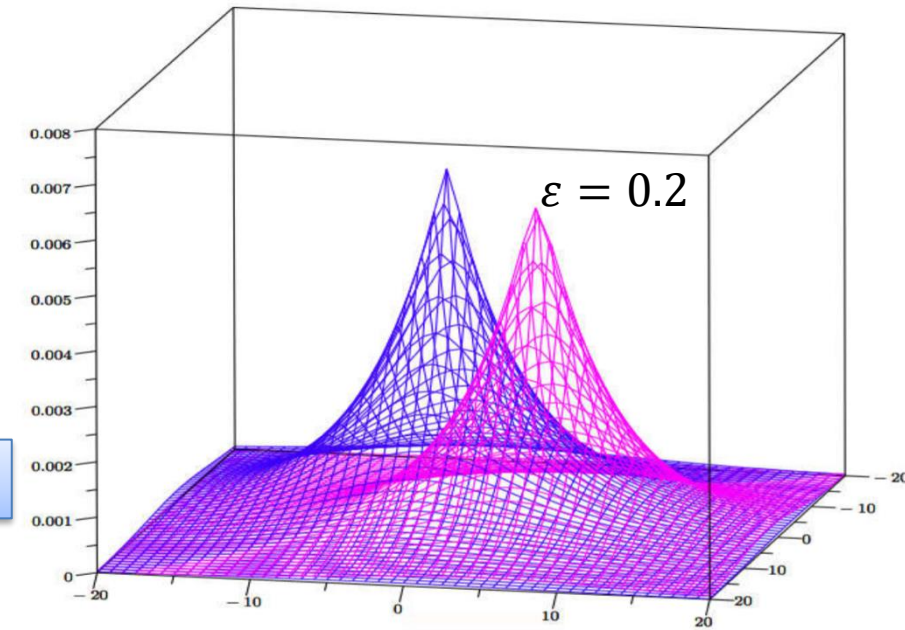
Planar Laplace Mechanism (PL)

The bi-variate pdf of PL noise mechanism is:

$$f_K(z \mid x) = \frac{\varepsilon^2}{2\pi} e^{-\varepsilon d(z, x)}$$

Normalization factor.

Distance to reported location.





Planar Laplace Mechanism (PL)

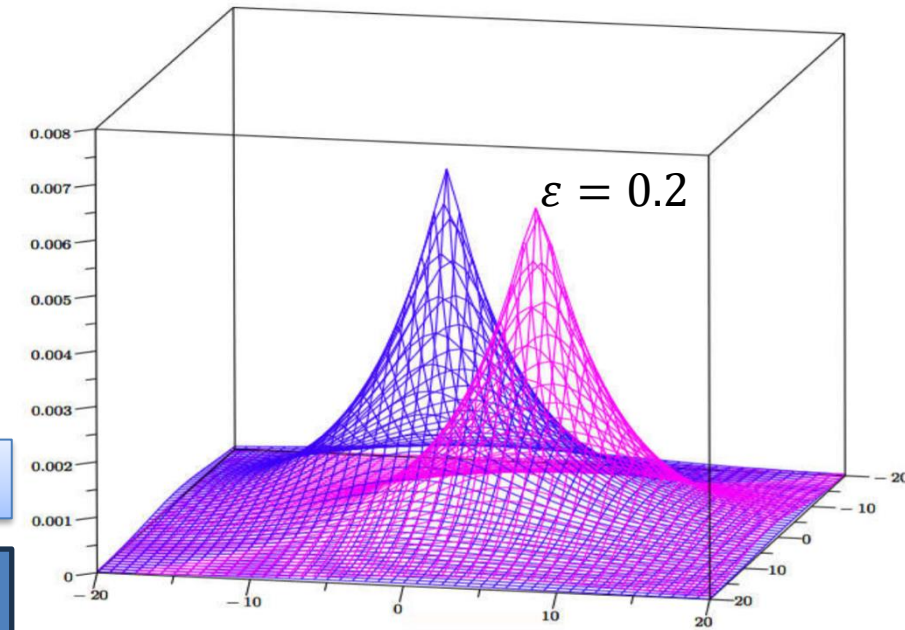
The bi-variate pdf of PL noise mechanism is:

$$f_K(z \mid x) = \frac{\varepsilon^2}{2\pi} e^{-\varepsilon d(z, x)}$$

Normalization factor.

Distance to reported location.

Noise is radially symmetric and decays exponentially with distance





Planar Laplace Mechanism (PL)

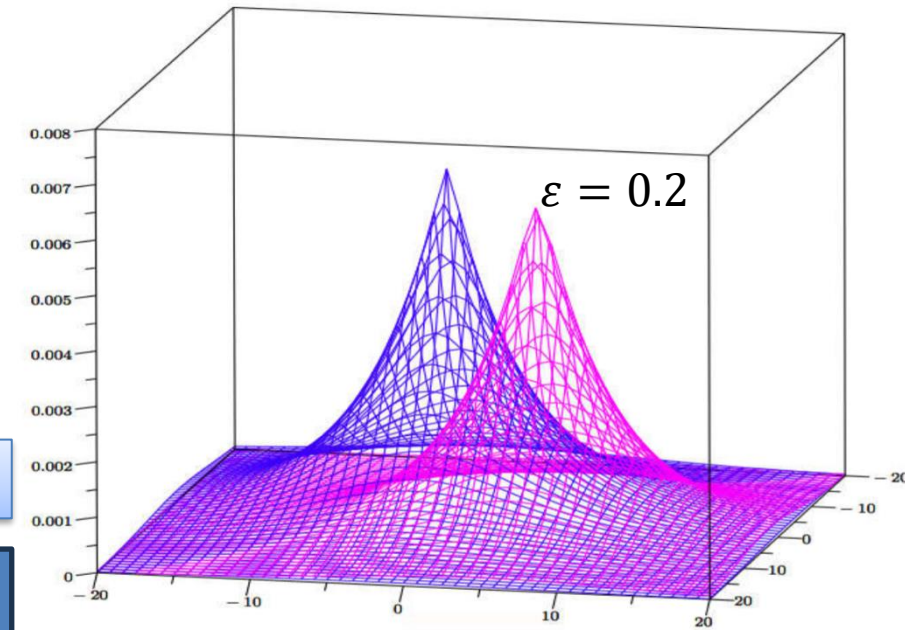
The bi-variate pdf of PL noise mechanism is:

$$f_K(z \mid x) = \frac{\varepsilon^2}{2\pi} e^{-\varepsilon d(z, x)}$$

Normalization factor.

Distance to reported location.

Noise is radially symmetric and decays exponentially with distance



Method to obtain GeoInd:

- I. Sample a 2D displacement vector $\mathbf{v}$ from the pdf.
- II. Report $z = x + \mathbf{v}$



Planar Laplace Mechanism (PL)

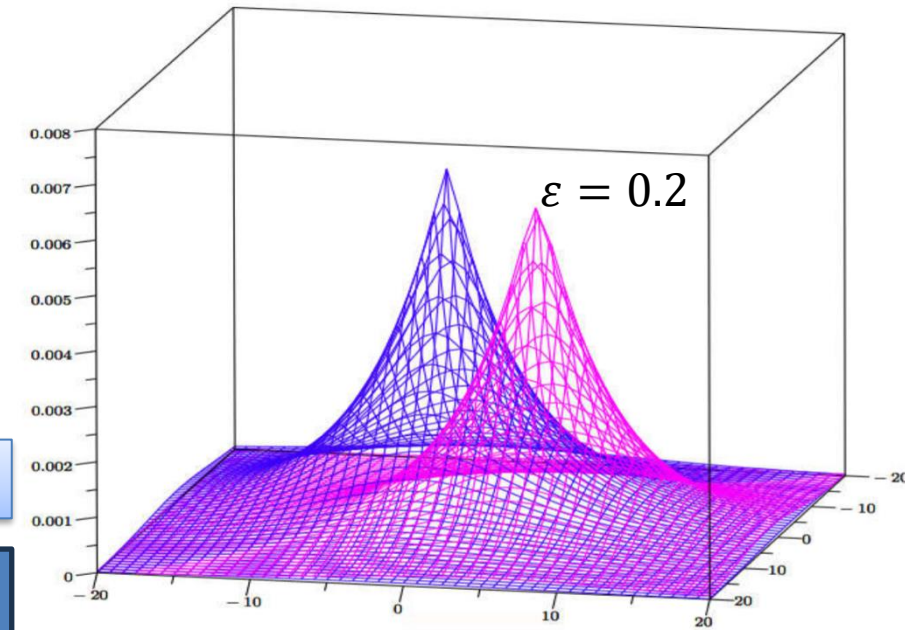
The bi-variate pdf of PL noise mechanism is:

$$f_K(z \mid x) = \frac{\varepsilon^2}{2\pi} e^{-\varepsilon d(z, x)}$$

Normalization factor.

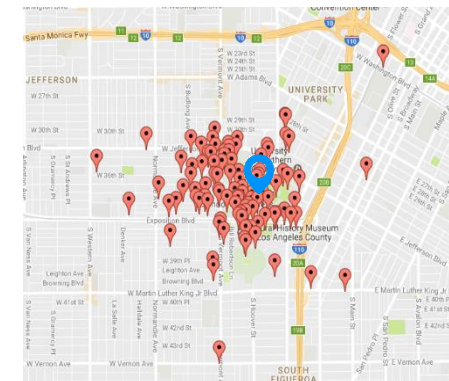
Distance to reported location.

Noise is radially symmetric and decays exponentially with distance



Method to obtain GeoInd:

- I. Sample a 2D displacement vector $\mathbf{v}$ from the pdf.
- II. Report $z = x + \mathbf{v}$



True locations

Perturbed locations

$\epsilon = 0.5$



Planar Laplace Mechanism (PL)

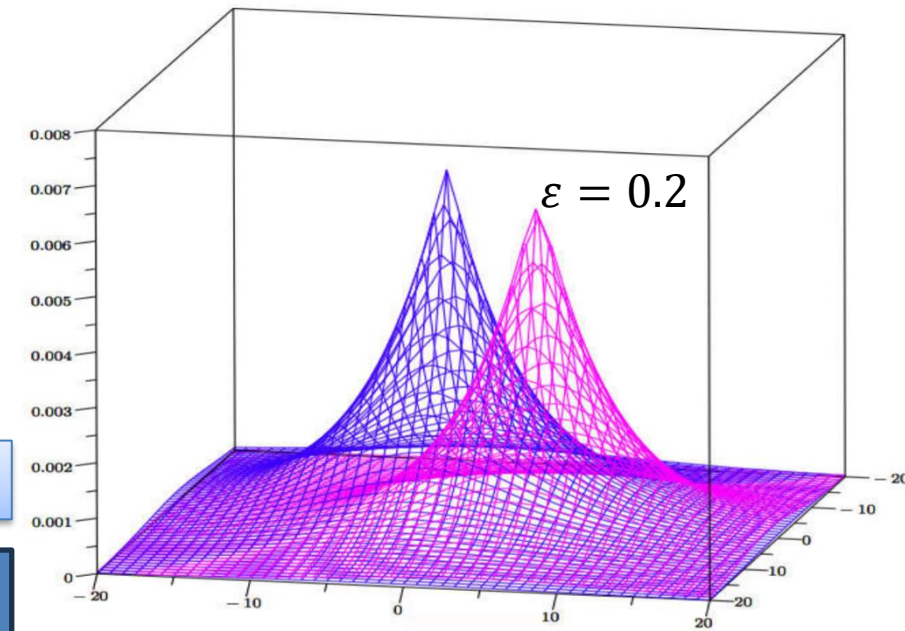
The bi-variate pdf of PL noise mechanism is:

$$f_K(z \mid x) = \frac{\varepsilon^2}{2\pi} e^{-\varepsilon d(z, x)}$$

Normalization factor.

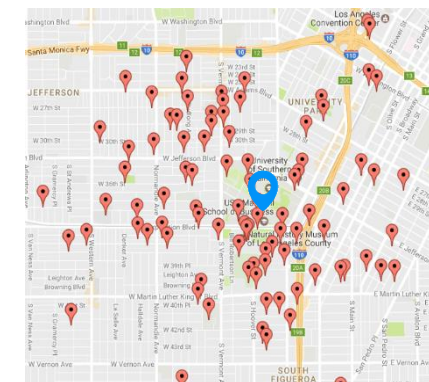
Distance to reported location.

Noise is radially symmetric and decays exponentially with distance



Method to obtain GeoInd:

- I. Sample a 2D displacement vector $\mathbf{v}$ from the pdf.
- II. Report $z = x + \mathbf{v}$



$\epsilon = 0.2$

True locations
Perturbed locations



Planar Laplace Mechanism (PL)

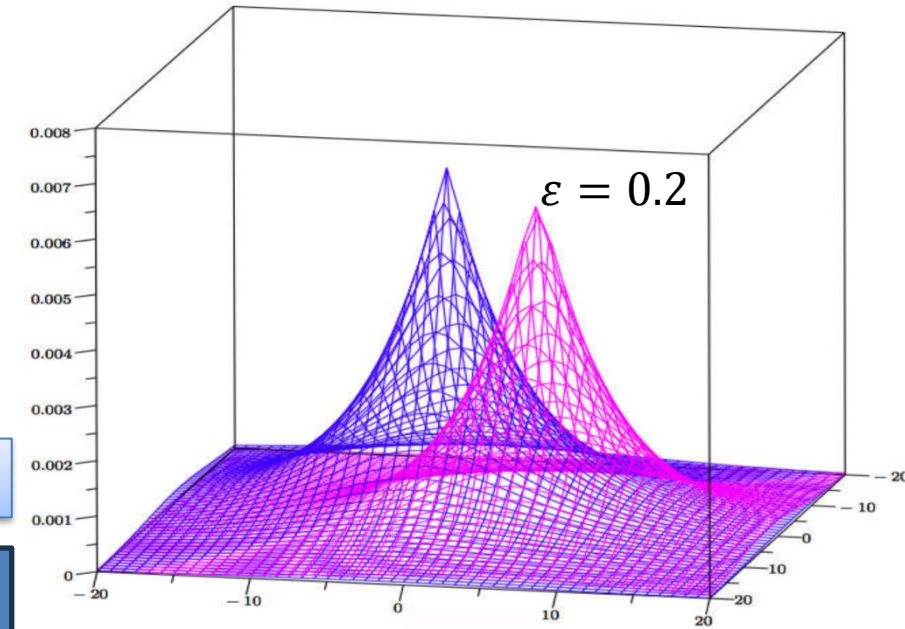
The bi-variate pdf of PL noise mechanism is:

$$f_K(z \mid x) = \frac{\varepsilon^2}{2\pi} e^{-\varepsilon d(z, x)}$$

Normalization factor.

Distance to reported location.

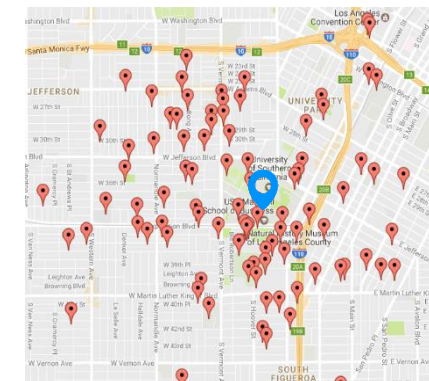
Noise is radially symmetric and decays exponentially with distance



Method to obtain GeoInd:

- I. Sample a 2D displacement vector $\mathbf{v}$ from the pdf.
- II. Report $z = x + \mathbf{v}$

How to sample ?



$\epsilon = 0.2$

True locations

Perturbed locations



Planar Laplace Mechanism (PL) contd.

- Not equivalent to generating the two coordinates independently from a standard (one dimensional) Laplace distribution.
- Correct way to sample:



Planar Laplace Mechanism (PL) contd.

- Not equivalent to generating the two coordinates independently from a standard (one dimensional) Laplace distribution.
- Correct way to sample:
 - Convert to polar coordinates $D_{\epsilon}(r, \theta) = \frac{\epsilon^2}{2\pi} r e^{-\epsilon r}$



Planar Laplace Mechanism (PL) contd.

- Not equivalent to generating the two coordinates independently from a standard (one dimensional) Laplace distribution.
- Correct way to sample:
 - Convert to polar coordinates $D_{\epsilon}(r, \theta) = \frac{\epsilon^2}{2\pi} r e^{-\epsilon r}$
 - Determine Angular and Radial Marginals:
$$D_{\epsilon,R}(r) = \int_0^{2\pi} D_{\epsilon}(r, \theta) d\theta = \epsilon^2 r e^{-\epsilon r}$$
$$D_{\epsilon,\Theta}(\theta) = \int_0^{\infty} D_{\epsilon}(r, \theta) dr = \frac{1}{2\pi}$$



Planar Laplace Mechanism (PL) contd.

- Not equivalent to generating the two coordinates independently from a standard (one dimensional) Laplace distribution.
- Correct way to sample:
 - Convert to polar coordinates $D_{\epsilon}(r, \theta) = \frac{\epsilon^2}{2\pi} r e^{-\epsilon r}$
 - Determine Angular and Radial Marginals: $D_{\epsilon,R}(r) = \int_0^{2\pi} D_{\epsilon}(r, \theta) d\theta = \epsilon^2 r e^{-\epsilon r}$
 $D_{\epsilon,\Theta}(\theta) = \int_0^{\infty} D_{\epsilon}(r, \theta) dr = \frac{1}{2\pi}$
 - Draw a point (r, θ) , by drawing separately r and θ from
 - $D(r)$ and $D(\theta)$ respectively



Planar Laplace Mechanism (PL) contd.

- The closer (geographically) two points are, the less distinguishable we would like them to be.



Planar Laplace Mechanism (PL) contd.

- The closer (geographically) two points are, the less distinguishable we would like them to be.
- The planar Laplace mechanism offers no optimality guarantees for the quality loss of the reported location



Planar Laplace Mechanism (PL) contd.

- The closer (geographically) two points are, the less distinguishable we would like them to be.
- The planar Laplace mechanism offers no optimality guarantees for the quality loss of the reported location

Efficient, BUT poor Utility in practice.

Can you achieve better utility by using some knowledge of user check-in behavior ?



What about protecting trajectories

- Let a trajectory $x = [x_1, \dots, x_n]$
- We can simply apply a noise mechanism independently to each secret x_i .

```
mechanism IM(x)  
  z := []  
  for  $i := 1$  to  $|\mathbf{x}|$   
     $z := N(\epsilon_N)(\mathbf{x}[i])$   
  z :=  $z :: \mathbf{z}$   
return z
```

Can we do better ?



What about protecting trajectories

- Let a trajectory $x = [x_1, \dots, x_n]$
- We can simply apply a noise mechanism independently to each secret x_i .
- Total privacy leakage ?

```
mechanism IM(x)  
  z := []  
  for  $i := 1$  to  $|\mathbf{x}|$   
     $z := N(\epsilon_N)(\mathbf{x}[i])$   
  z :=  $z :: \mathbf{z}$   
return z
```

Can we do better ?



What about protecting trajectories

- Let a trajectory $x = [x_1, \dots, x_n]$
- We can simply apply a noise mechanism independently to each secret x_i .

```
mechanism IM(x)  
  z := []  
  for  $i := 1$  to  $|\mathbf{x}|$   
     $z := N(\epsilon_N)(\mathbf{x}[i])$   
  z :=  $z :: \mathbf{z}$   
return z
```

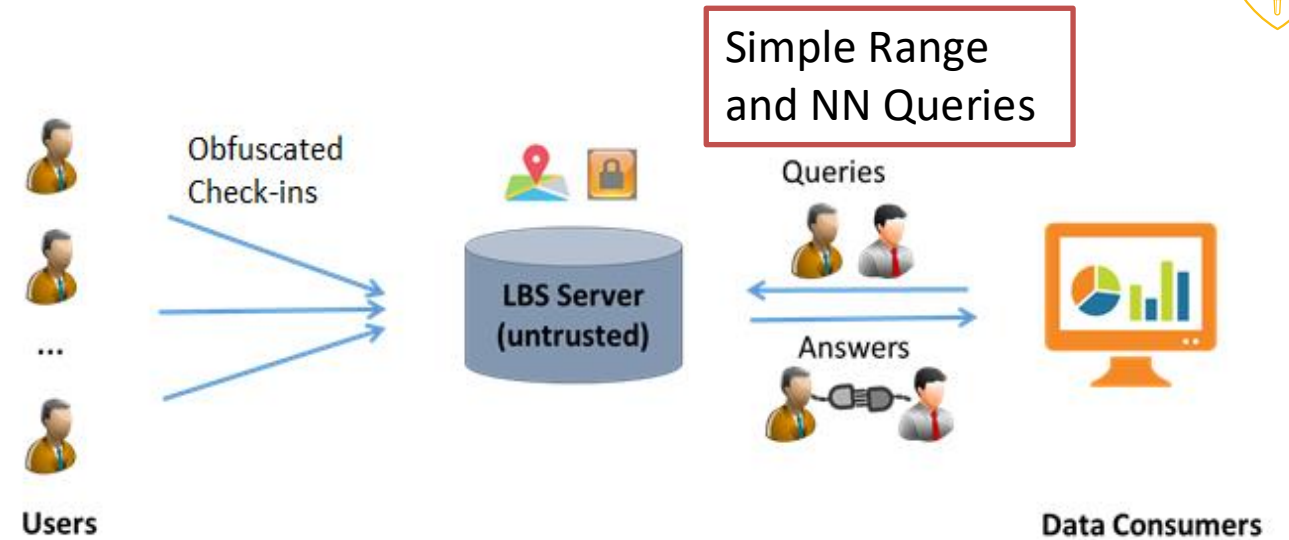
- Total privacy leakage ?
- $n\epsilon$ according to Composition Theorem of DP

Can we do better ?

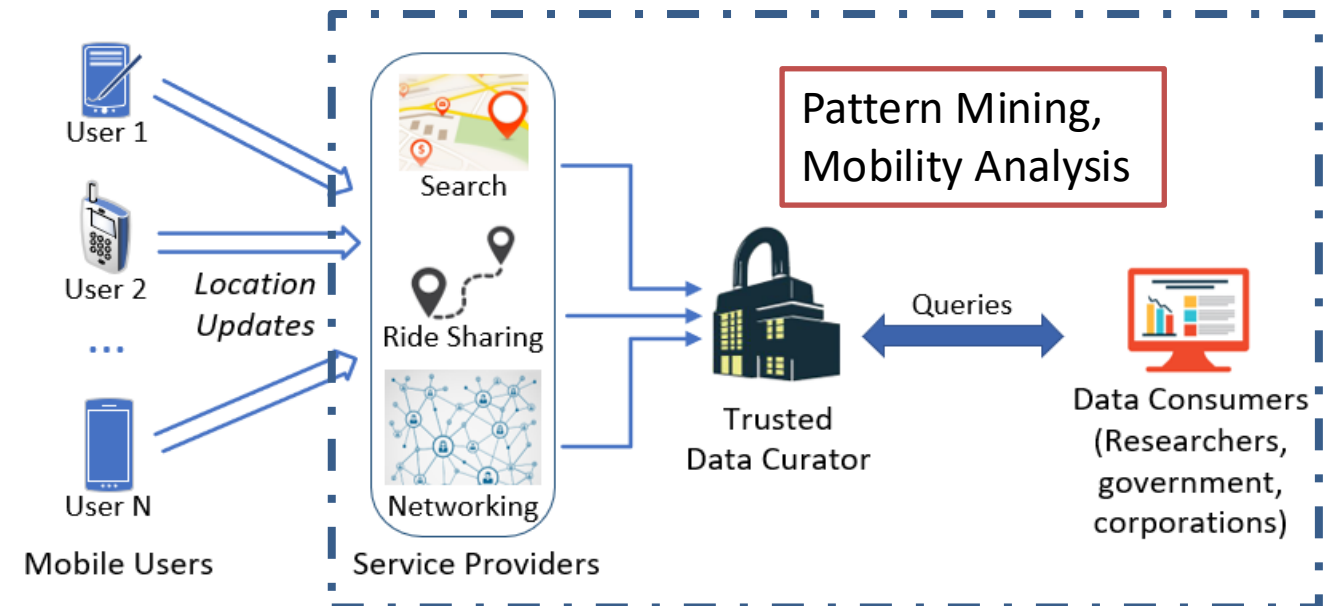


System Models

Online Setting:



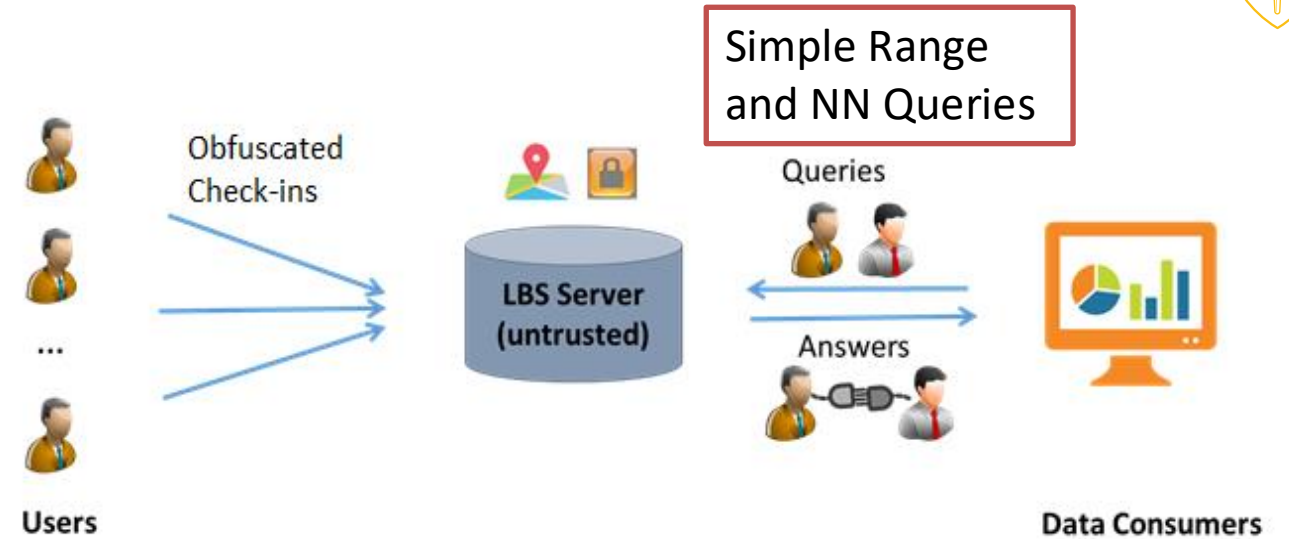
Offline Setting:



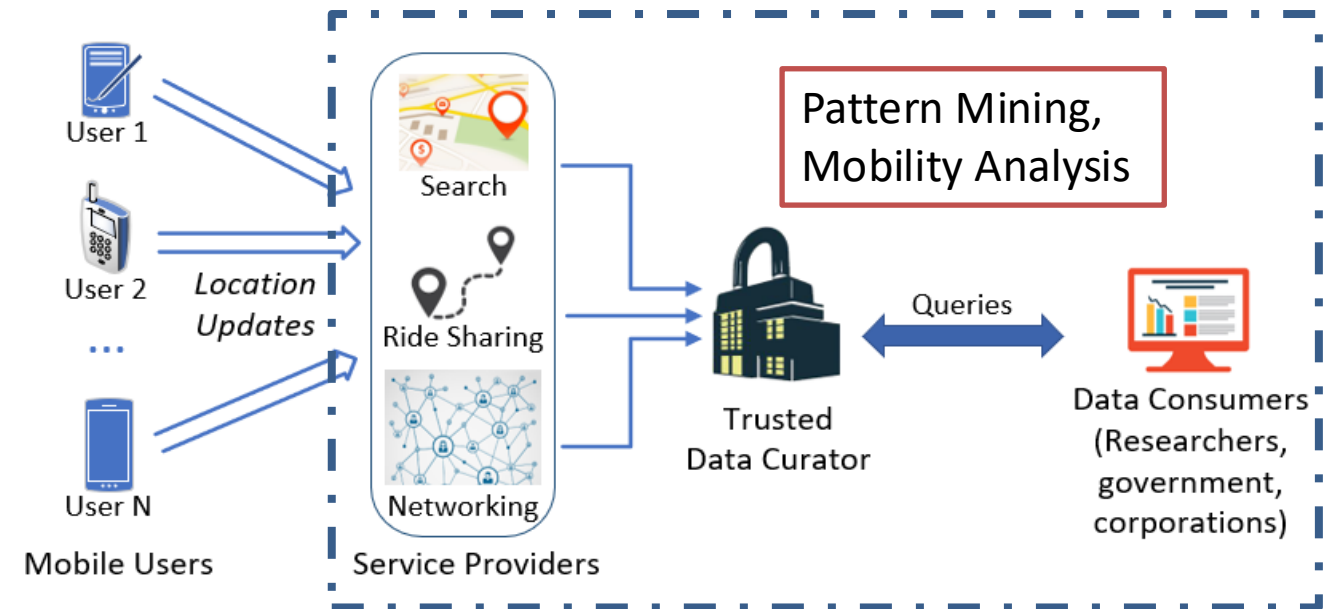


System Models

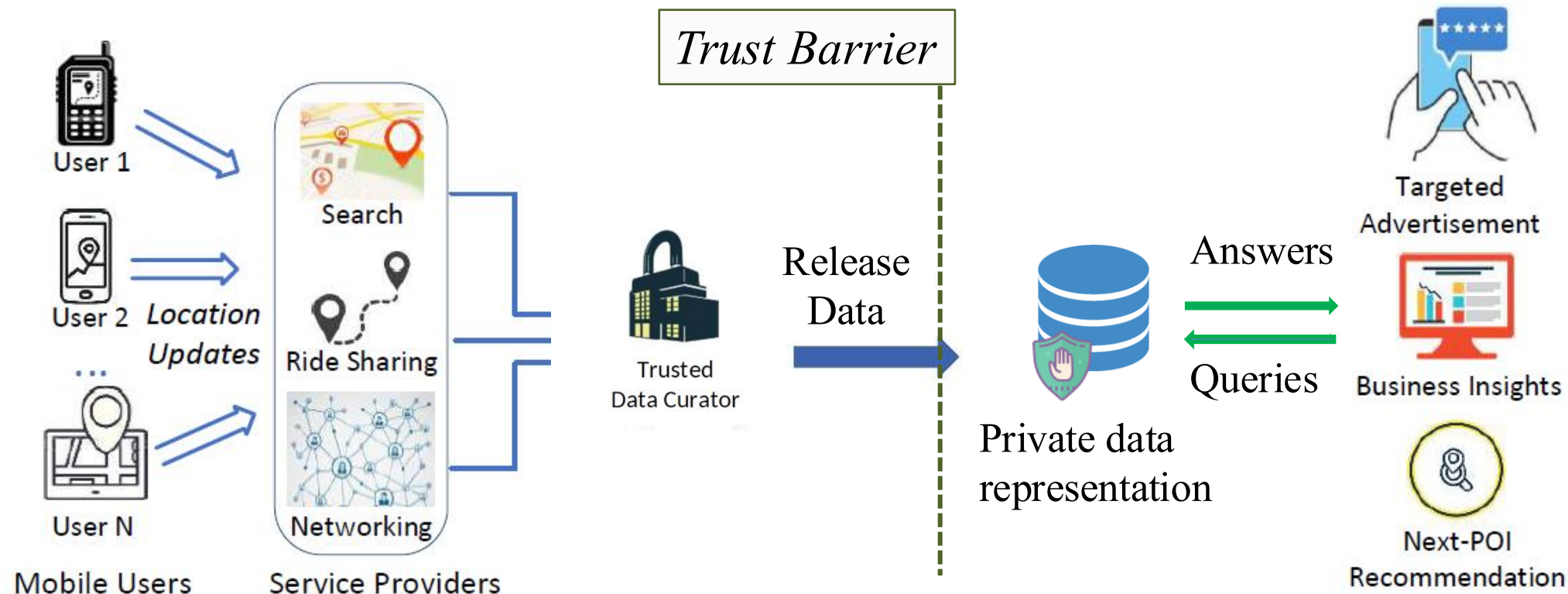
Online Setting:



Offline Setting:



Privacy-Preserving Services Offline Setting (Publishing)

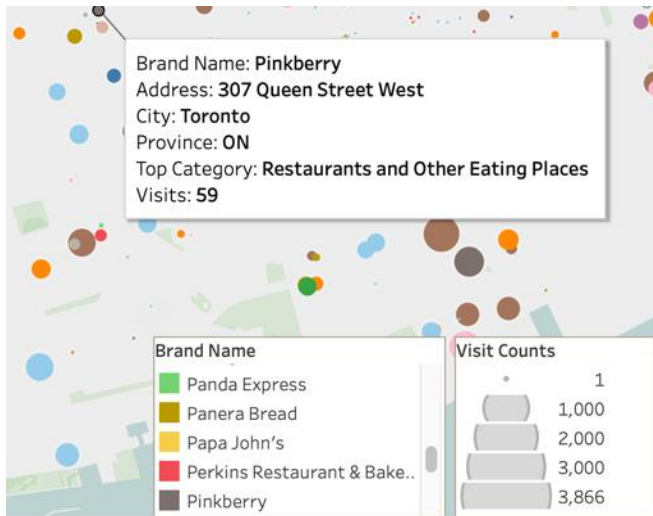


Published data representations must preserve user's privacy.

Privacy-Preserving Release of Aggregate Location Data



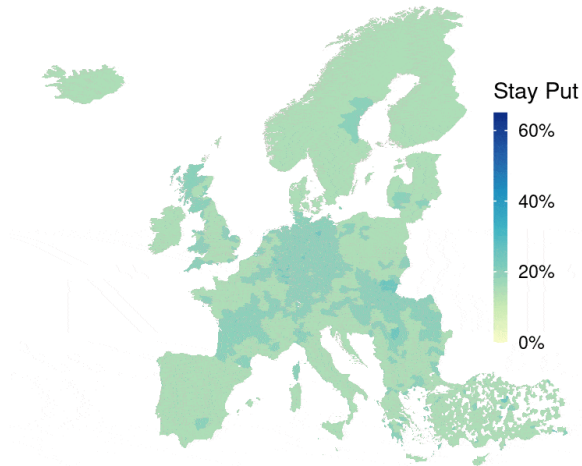
POI Visits Pattern



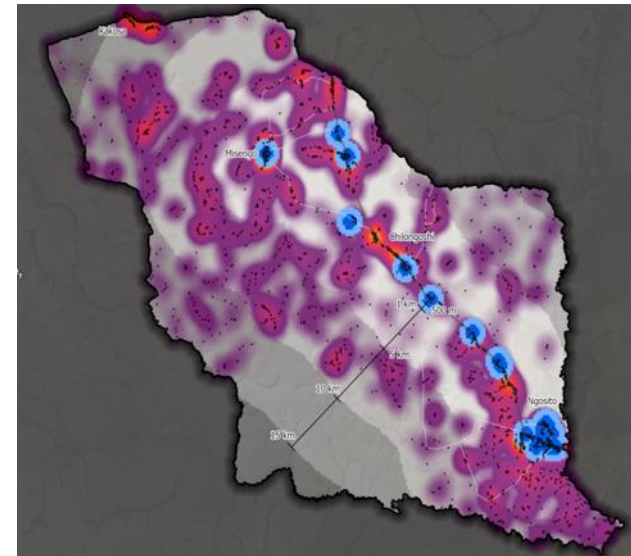
Google Mobility Reports

Europe's COVID-19 response efforts

Date: 2020-03-04



World Vision Project for Clean Water Access



United States Decennial Census



Apportionment
Redistricting
Funding allocation

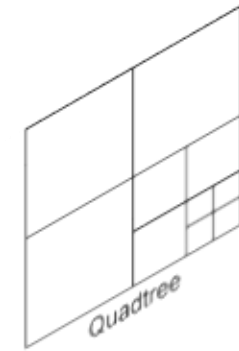
Problem and Related Work

Privately Answering RCQs



Publishes
DP
histogram

- *Given:* A spatial database D , record containing [Lat Lon]
- *Problem:* Release a Differentially Private data histogram to answer
- spatial range count queries (RCQs) accurately



Examples of
Domain
Partitioning

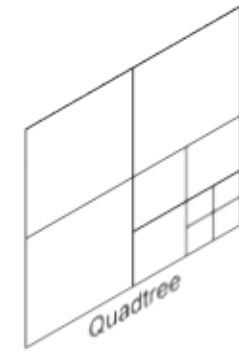
Problem and Related Work Privately Answering RCQs



Publishes
DP
histogram

- *Given:* A spatial database D , record containing [Lat Lon]
- *Problem:* Release a Differentially Private data histogram to answer
- spatial range count queries (RCQs) accurately

Noisy histogram. privacy level,
 ϵ , determines noise scale.



Examples of
Domain
Partitioning

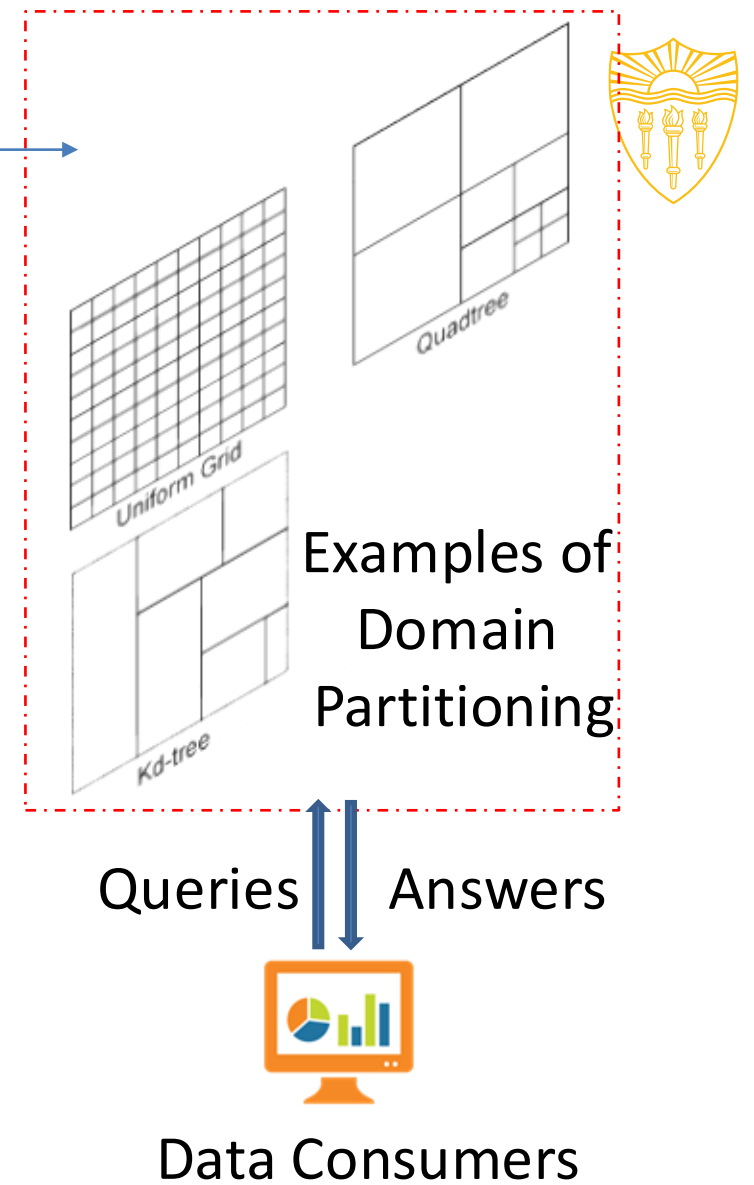
Problem and Related Work Privately Answering RCQs



Publishes
DP
histogram

- *Given:* A spatial database D , record containing [Lat Lon]
- *Problem:* Release a Differentially Private data histogram to answer
- spatial range count queries (RCQs) accurately

Noisy histogram. privacy level,
 ϵ , determines noise scale.



Problem and Related Work

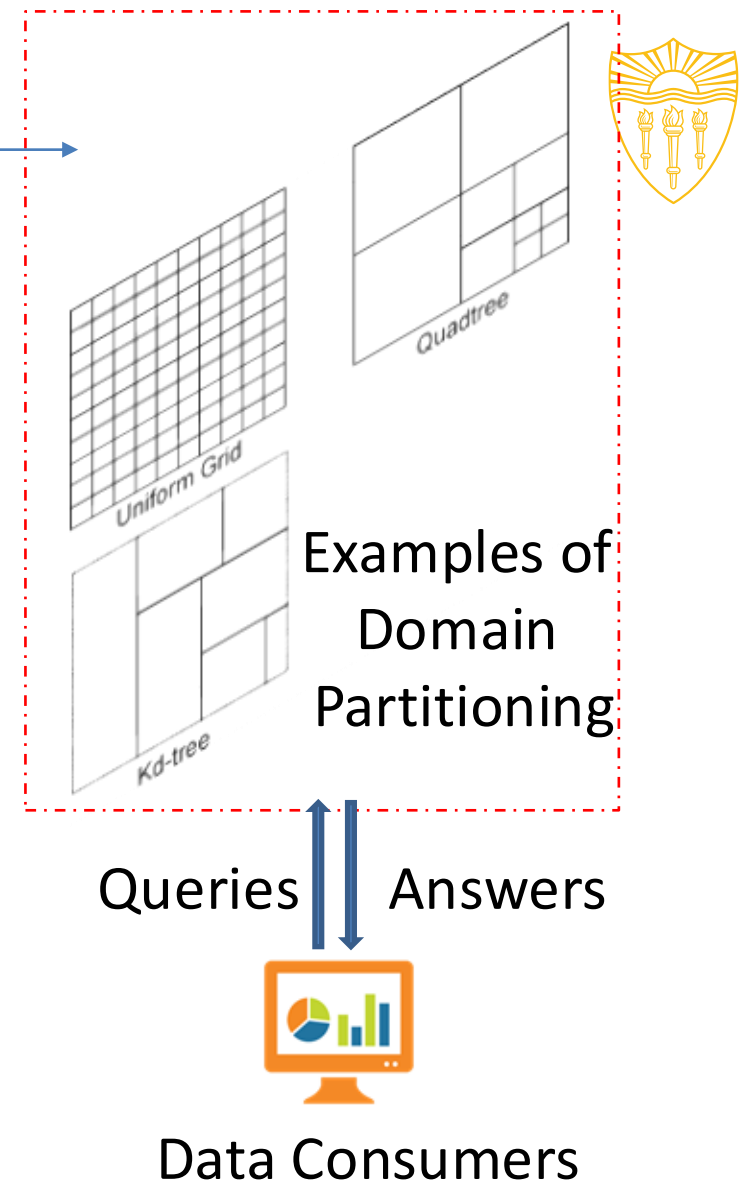
Privately Answering RCQs



Publishes
DP
histogram

- *Given:* A spatial database D , record containing [Lat Lon]
- *Problem:* Release a Differentially Private data histogram to answer
- spatial range count queries (RCQs) accurately

Noisy histogram. privacy level,
 ϵ , determines noise scale.

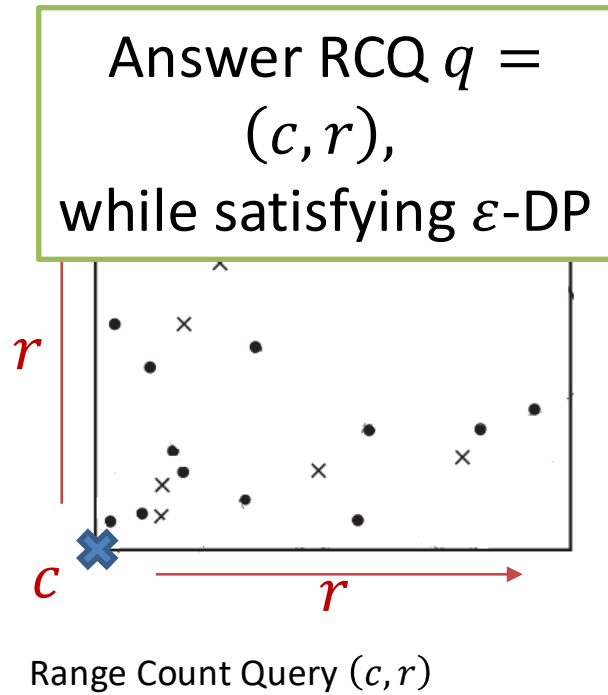


- All related work use *Domain Partitioning*.
 - Uniform Grid (UG), Adaptive Grid (AG) [ICDE'13]
 - QuadTree, kd-Tree, kd-hybrid [ICDE'12]
 - Data and Workload Aware Algorithm(DAWA) [VLDB'14]
 - PrivTree [SIGMOD'16],
 - DPCUBE [ICDE'12, TDP'13]
 - HB2D Hierarchical methods in 2D [VLDB'13]



DP location data release

An example of a domain partitioning model

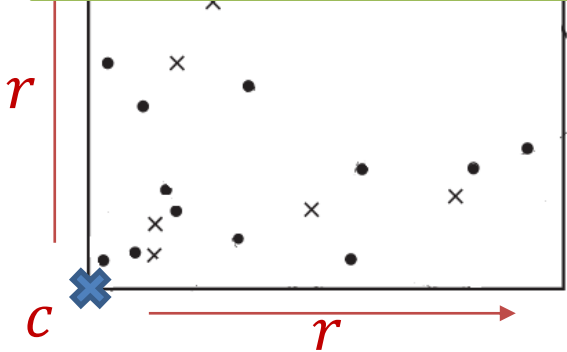




DP location data release

An example of a domain partitioning model

Answer RCQ $q = (c, r)$, while satisfying ϵ -DP



Range Count Query (c, r)



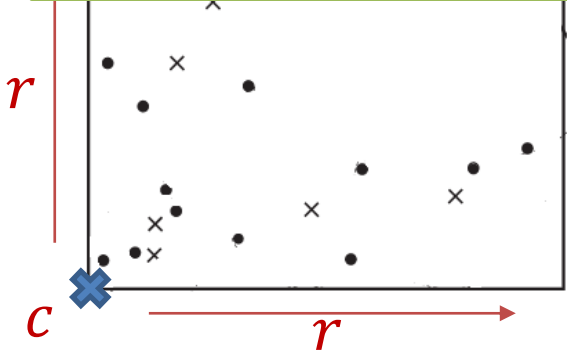
True database



DP location data release

An example of a domain partitioning model

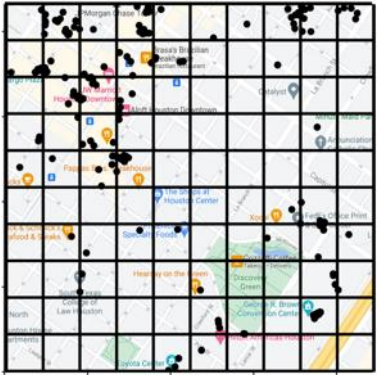
Answer RCQ $q = (c, r)$, while satisfying ϵ -DP



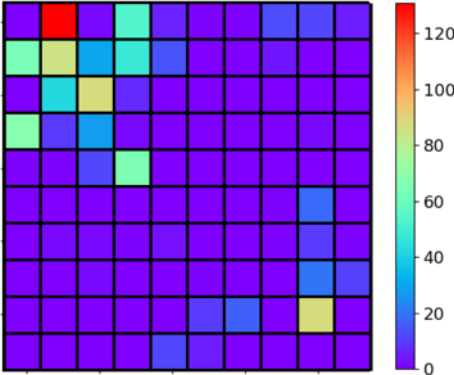
Range Count Query (c, r)



True database



2-d grid partitioning and histogram

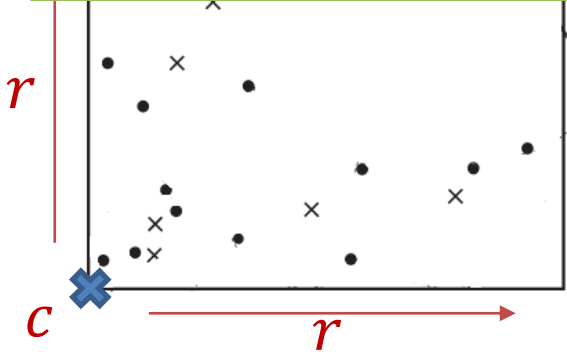




DP location data release

An example of a domain partitioning model

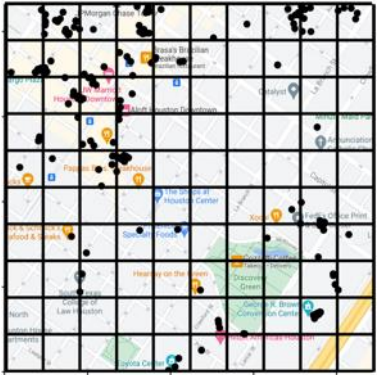
Answer RCQ $q = (c, r)$, while satisfying ϵ -DP



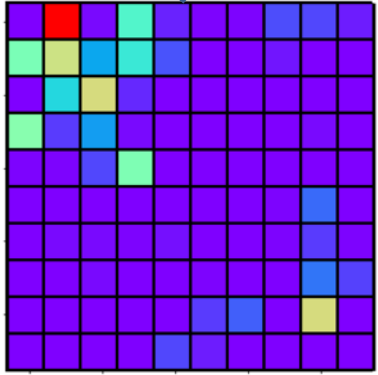
Range Count Query (c, r)



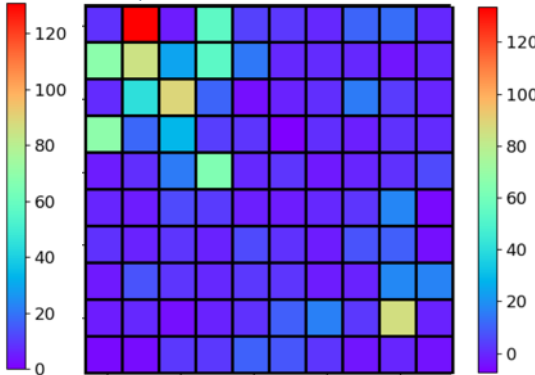
True database



2-d grid partitioning and histogram



$Lap(1/\epsilon)$



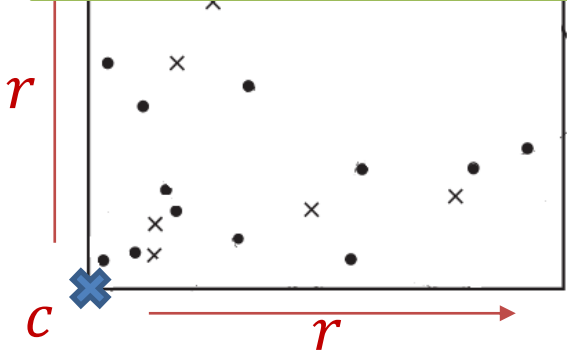
DP-compliant release



DP location data release

An example of a domain partitioning model

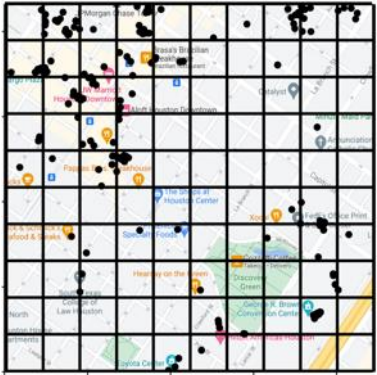
Answer RCQ $q = (c, r)$, while satisfying ϵ -DP



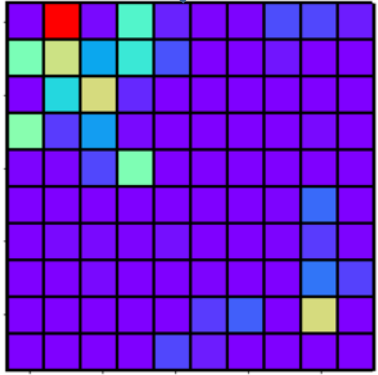
Range Count Query (c, r)



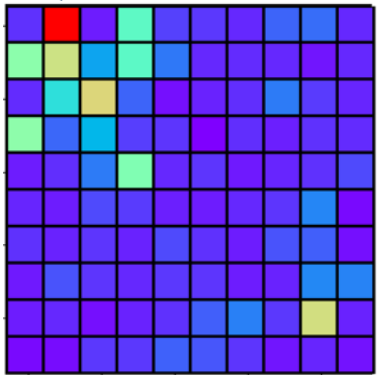
True database



2-d grid partitioning and histogram



$Lap(1/\epsilon)$



DP-compliant release

Let the true answer to q be $f(q)$ and the noisy answer be y .

Utility Metric is the Relative Error:

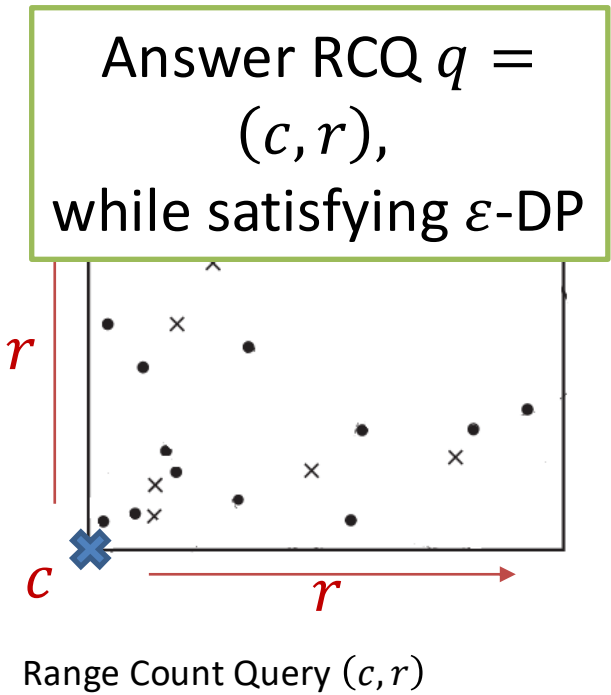
$$\Delta(y, f(q)) = \frac{|y - f(q)|}{\max\{f(q), \psi\}}$$

where ψ is a smoothing factor

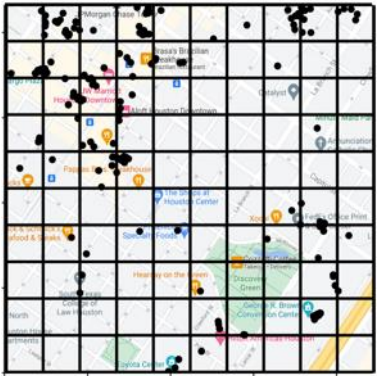


DP location data release

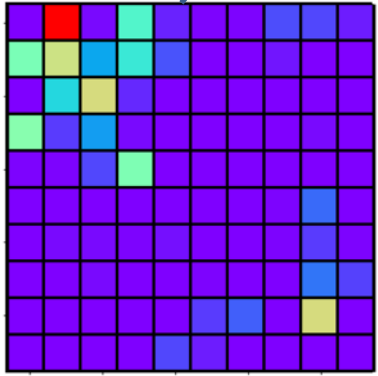
An example of a domain partitioning model



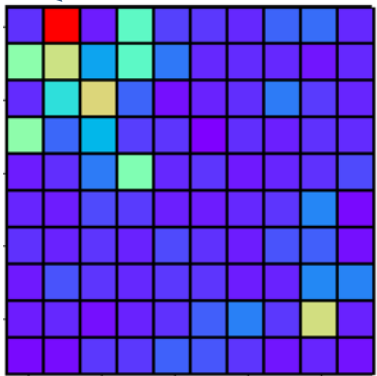
True database



2-d grid partitioning and histogram



$Lap(1/\epsilon)$



DP-compliant release

Let the true answer to q be $f(q)$ and the noisy answer be y .

Utility Metric is the Relative Error:

$$\Delta(y, f(q)) = \frac{|y - f(q)|}{\max\{f(q), \psi\}}$$

where ψ is a smoothing factor

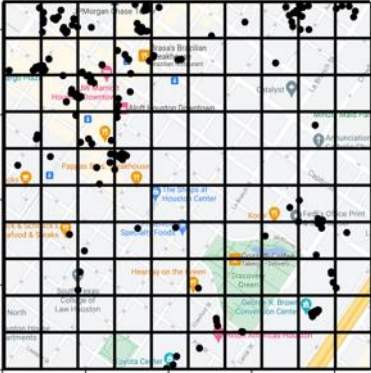


Spatial Neural Histograms (SNH)

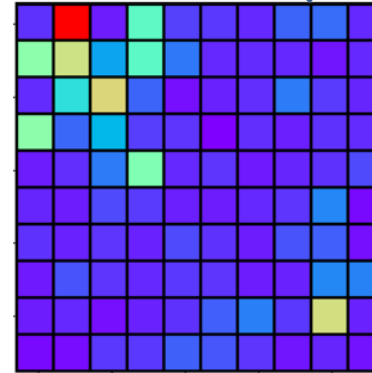
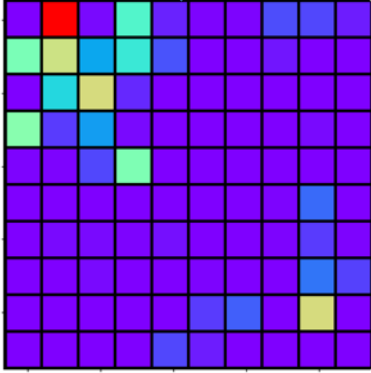
Train a neural network



True database

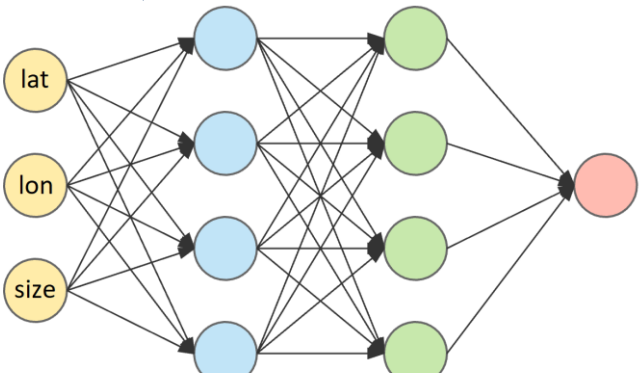


2-d grid partitioning and histogram



DP-compliant histogram

$Lap(1/\epsilon)$



Release neural network

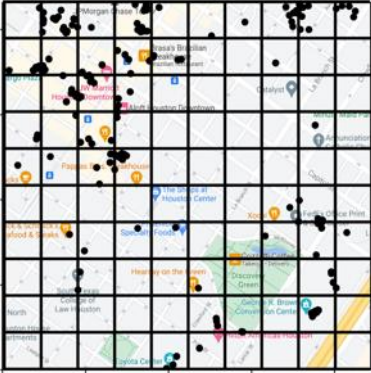


Spatial Neural Histograms (SNH)

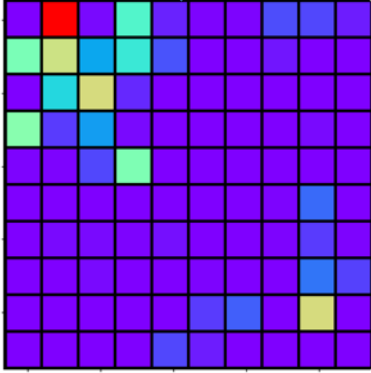
Train a neural network



True database

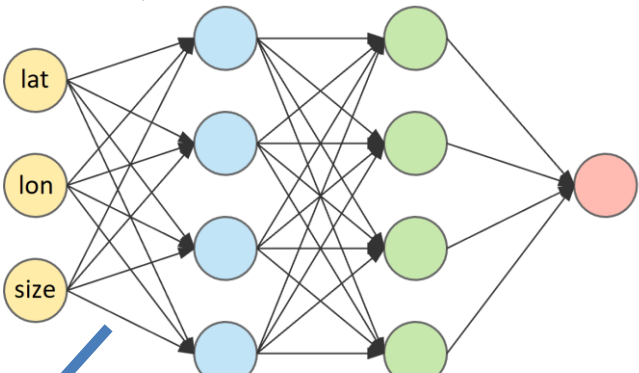
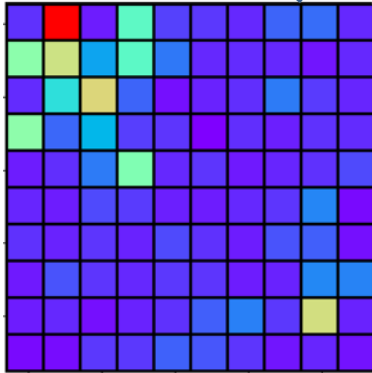


2-d grid partitioning and histogram

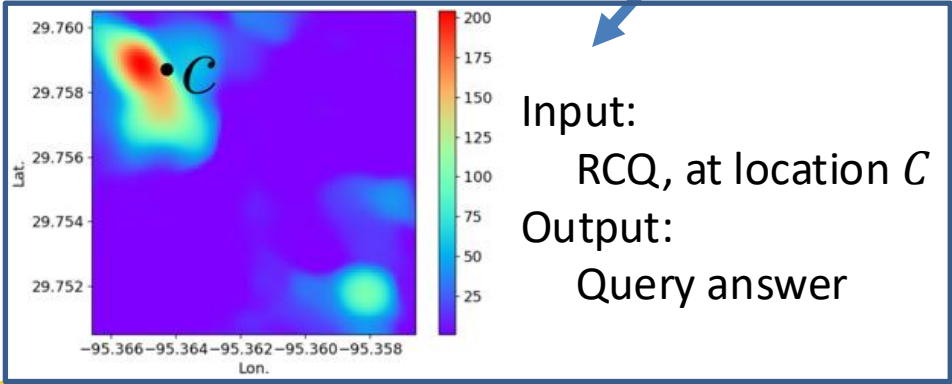


DP-compliant histogram

$Lap(1/\epsilon)$



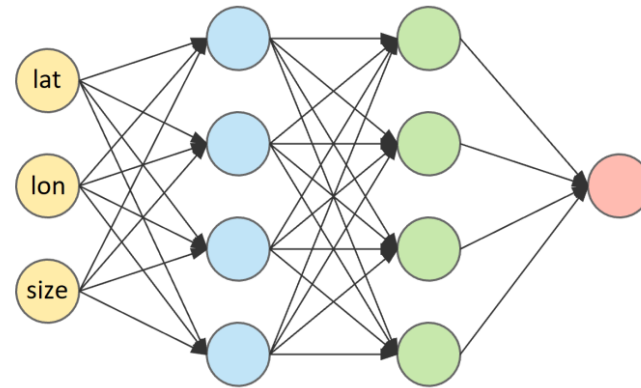
Release neural network



Neural Network Training



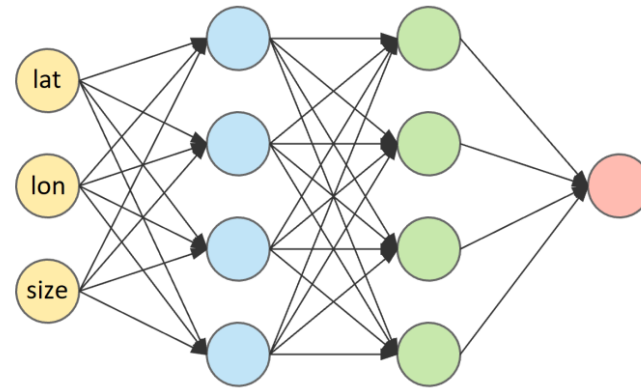
Neural Network,
Parameters θ



Neural Network Training



Neural Network,
Parameters θ



Training set
 $\{(q, f(q)), q \in Q\}$

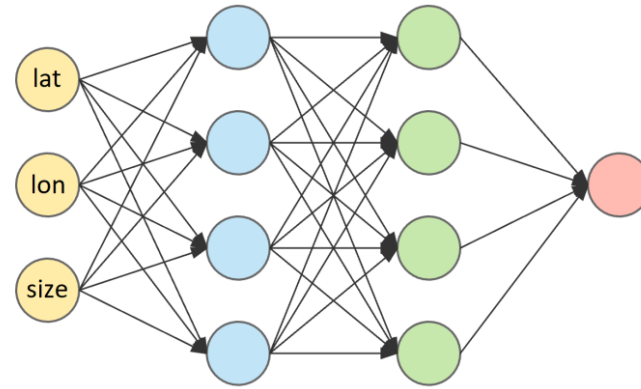
Neural Network Training



training sample $q = (c, r)$, e.g.
 $c : [34.11, -118.17]$
 $r : 65m$

Training set
 $\{(q, f(q)), q \in Q\}$

Neural Network,
Parameters θ



Neural Network Training

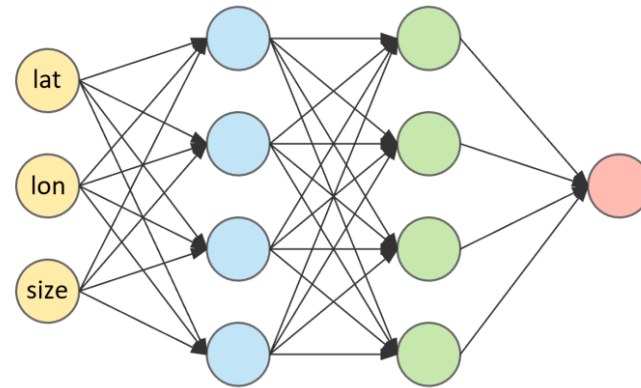


training sample $q = (c, r)$, e.g.
 $c : [34.11, -118.17]$
 $r : 65m$

Training set
 $\{(q, f(q)), q \in Q\}$

Histogram
cells

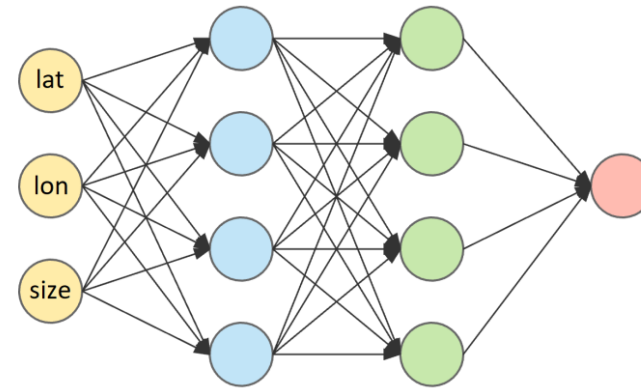
Neural Network,
Parameters θ



Neural Network Training



Neural Network,
Parameters θ



training sample $q = (c, r)$, e.g.

$c : [34.11, -118.17]$

$r : 65m$

Training set

$\{(q, f(q)), q \in Q\}$

Histogram
cells

Answer from histogram,
i.e., $\bar{f}(q) = f(q) + Lap(1/\epsilon)$

Neural Network Training



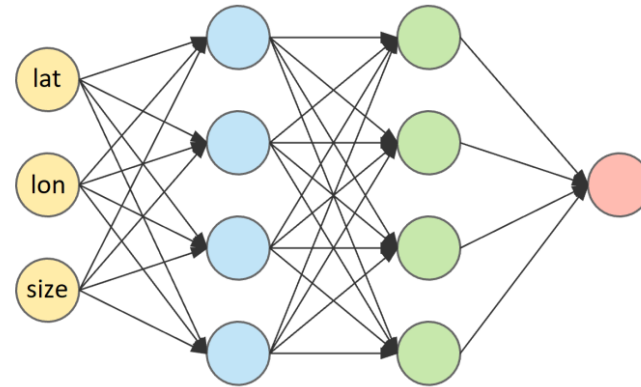
training sample $q = (c, r)$, e.g.
 $c : [34.11, -118.17]$
 $r : 65m$

Training set
 $\{(q, f(q)), q \in Q\}$

Histogram
cells

Answer from histogram,
i.e., $\bar{f}(q) = f(q) + Lap(1/\epsilon)$

Neural Network,
Parameters θ



Model output
 $\hat{f}(q; \theta) = 32$

DP histogram
answer
 $\bar{f}(q) = 45.8$

Neural Network Training



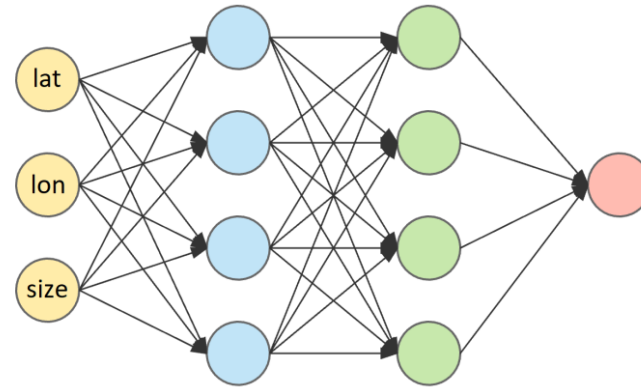
training sample $q = (c, r)$, e.g.
 $c : [34.11, -118.17]$
 $r : 65m$

Training set
 $\{(q, f(q)), q \in Q\}$

Histogram
cells

Answer from histogram,
i.e., $\bar{f}(q) = f(q) + \text{Lap}(1/\epsilon)$

Neural Network,
Parameters θ



Model output
 $\hat{f}(q; \theta) = 32$

DP histogram
answer
 $\bar{f}(q) = 45.8$

MSE Loss Function

$$\mathcal{L} = \sum_{q \in Q} \left(\hat{f}(q; \theta) - \bar{f}(q) \right)^2$$

Train with SGD

Neural Network Training



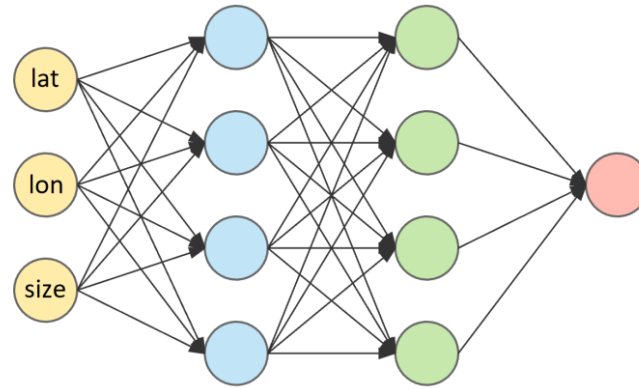
training sample $q = (c, r)$, e.g.
 $c : [34.11, -118.17]$
 $r : 65m$

Training set
 $\{(q, f(q)), q \in Q\}$

Histogram
cells

Answer from histogram,
i.e., $\bar{f}(q) = f(q) + \text{Lap}(1/\epsilon)$

Neural Network,
Parameters θ



Model output
 $\hat{f}(q; \theta) = 32$

DP histogram
answer
 $\bar{f}(q) = 45.8$

MSE Loss Function

$$\mathcal{L} = \sum_{q \in Q} \left(\hat{f}(q; \theta) - \bar{f}(q) \right)^2$$

Train with SGD

Disclaimer: Actual loss function and training set are more complicated, see paper.

Neural Network Training



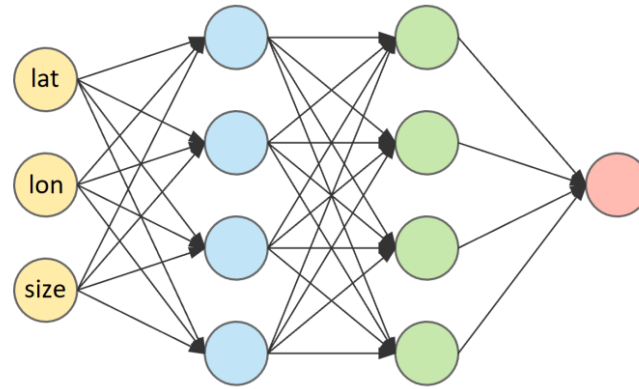
training sample $q = (c, r)$, e.g.
 $c : [34.11, -118.17]$
 $r : 65m$

Training set
 $\{(q, f(q)), q \in Q\}$

Histogram
cells

Answer from histogram,
i.e., $\bar{f}(q) = f(q) + Lap(1/\epsilon)$

Neural Network,
Parameters θ



Model output
 $\hat{f}(q; \theta) = 32$

DP histogram
answer
 $\bar{f}(q) = 45.8$

MSE Loss Function

$$\mathcal{L} = \sum_{q \in Q} \left(\hat{f}(q; \theta) - \bar{f}(q) \right)^2$$

Train with SGD

Disclaimer: Actual loss function and training set are more complicated, see paper.

A Neural Database (NeuroSketch):

- Model weights store data
- Answer query with forward pass

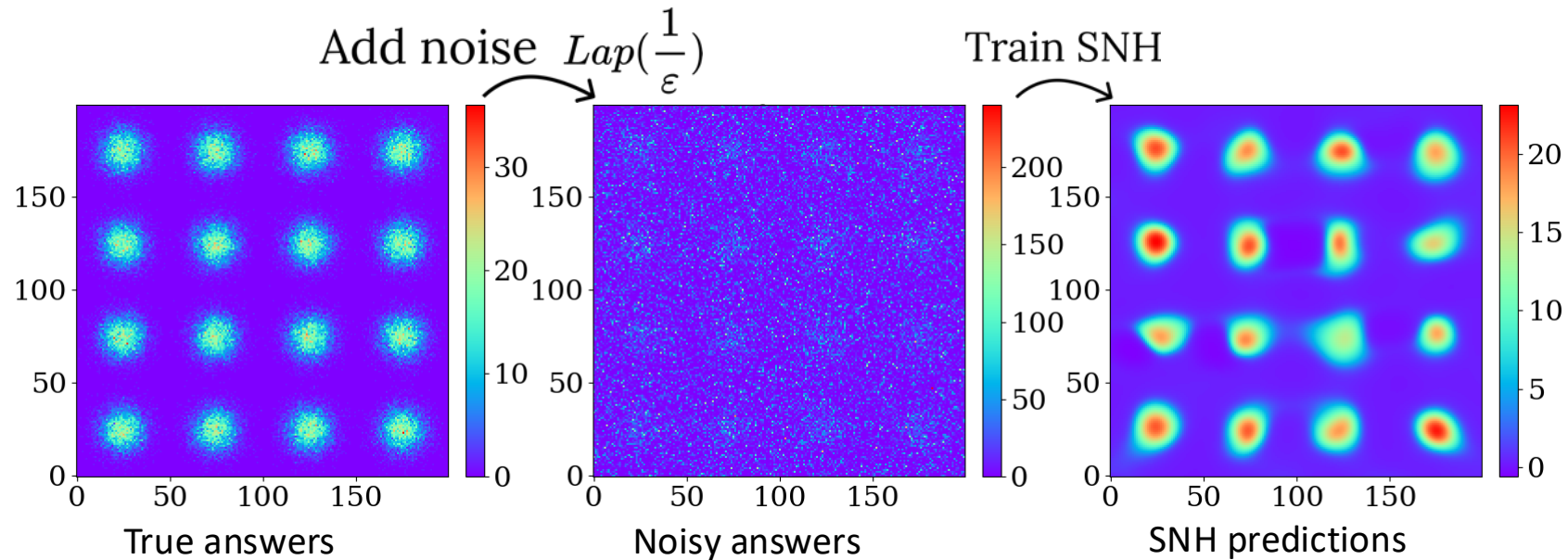
Also has applications in
non-private setting



But Why Does It Work?

Neural network fits to the patterns not noise

- Random noise difficult to fit
 - Highly non-smooth
- Neural network learns a smoother underlying function



Synthetic Dataset

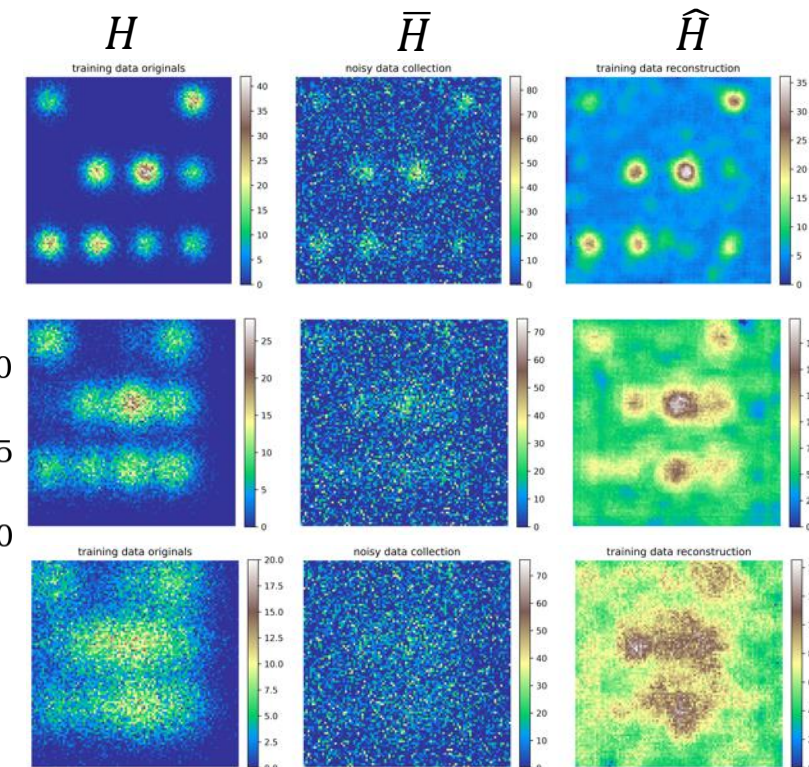
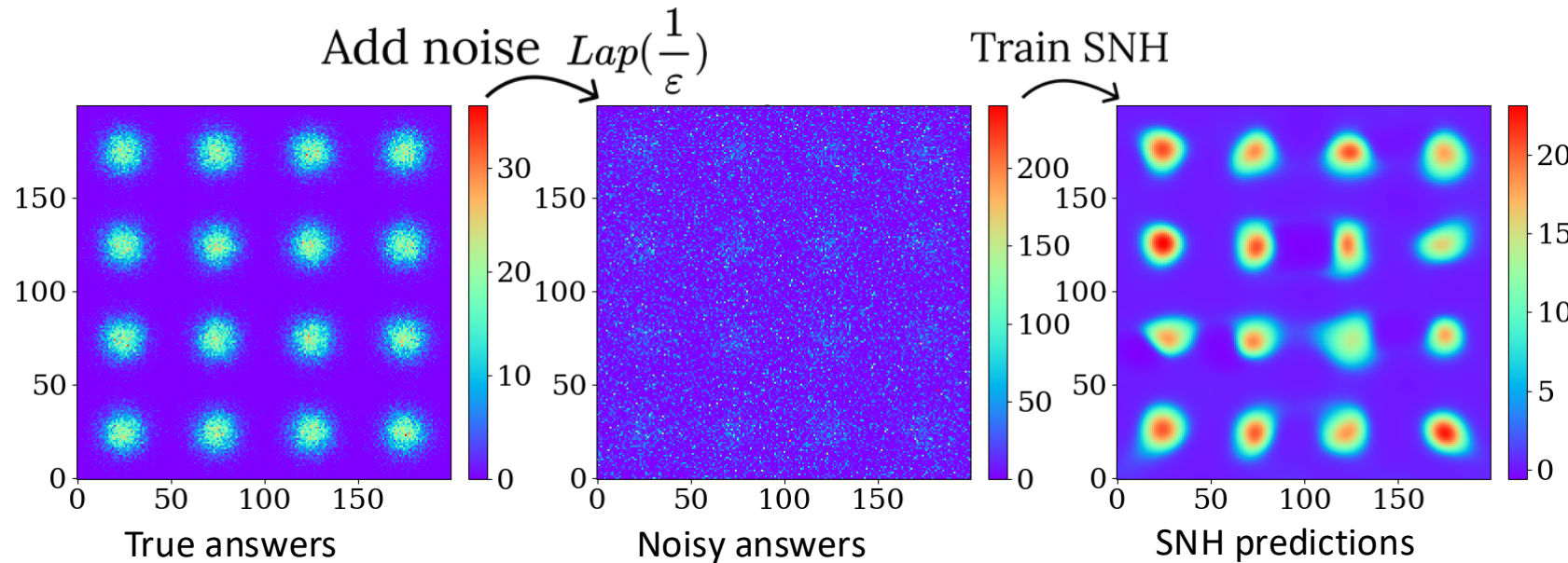
- Gaussian Mixture Models. 16 components, $\sigma^2 = 1\%$ of space
- Identity covariance. Epsilon = 0.05, 100k points
- Query set (train/test) is the 200x200 grid cells.



But Why Does It Work?

Neural network fits to the patterns not noise

- Random noise difficult to fit
 - Highly non-smooth
- Neural network learns a smoother underlying function



Synthetic Dataset

- Gaussian Mixture Models. 16 components, $\sigma^2 = 1\%$ of space
- Identity covariance. Epsilon = 0.05, 100k points
- Query set (train/test) is the 200x200 grid cells.

Experimental Evaluation

Datasets



Veraset (VS)

- Covers 10% of U.S. mobile devices 2019
- 2.5B check-ins from 1.2M devices per day

Gowalla (GW)

- 6.4M records from 200k users
- From Feb 2009 – Oct 2010

San Francisco-CABS

- GPS coordinates of approximately 250 taxis collected over 30 days in San Francisco

SPD-VS

- Veraset dataset with StayPoint Detection algorithm to retrieve POI visits of users.

$\leq 1000/\text{sq mi}$	$> 1000, \leq 4000/\text{sq mi}$	$> 4000/\text{sq mi}$
Low Pop. density	Medium Pop. density	High Pop. density
Fargo [46.877, -96.789]	Phoenix [33.448 -112.073]	Miami [25.801, -80.256]
Kansas City [39.09, -94.59]	Los Angeles [34.02, -118.29]	Chicago [41.880, -87.70]
Salt Lake [40.73, -111.926]	Houston [29.747, -95.365]	SF [37.764, -122.43]
Tulsa [36.153, -95.992]	Milwaukee [43.038, -87.910]	Boston [42.360 -71.058]

Default city

Wide range of location datasets, with application scenarios ranging for location networks, POI visitations, taxis, etc.



Experimental Evaluation Parameters

Query Specification

- 5000 RCQs centered at uniformly random positions, size = [25 m to 200 m].
- Metric: relative error, with smoothing factor $\psi = 0.1\%$ of n

Workload Queries

- 2000 RCQ more sampled from same distribution.



Experimental Evaluation Parameters

Query Specification

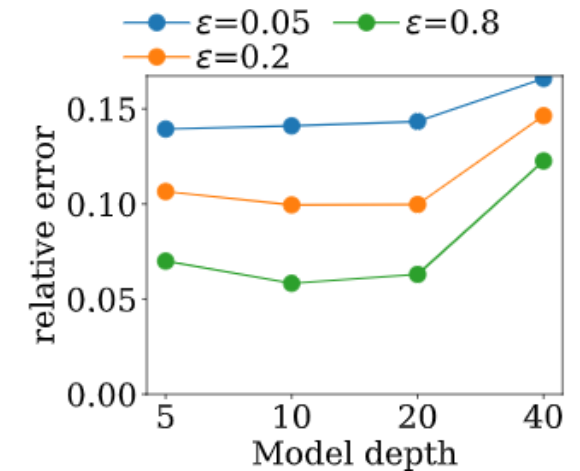
- 5000 RCQs centered at uniformly random positions, size = [25 m to 200 m].
- Metric: relative error, with smoothing factor $\psi = 0.1\%$ of

Workload Queries

- 2000 RCQ more sampled from same distribution.

SNH model specification

- Fully connected neural networks is set to 20 layers of 80 units each

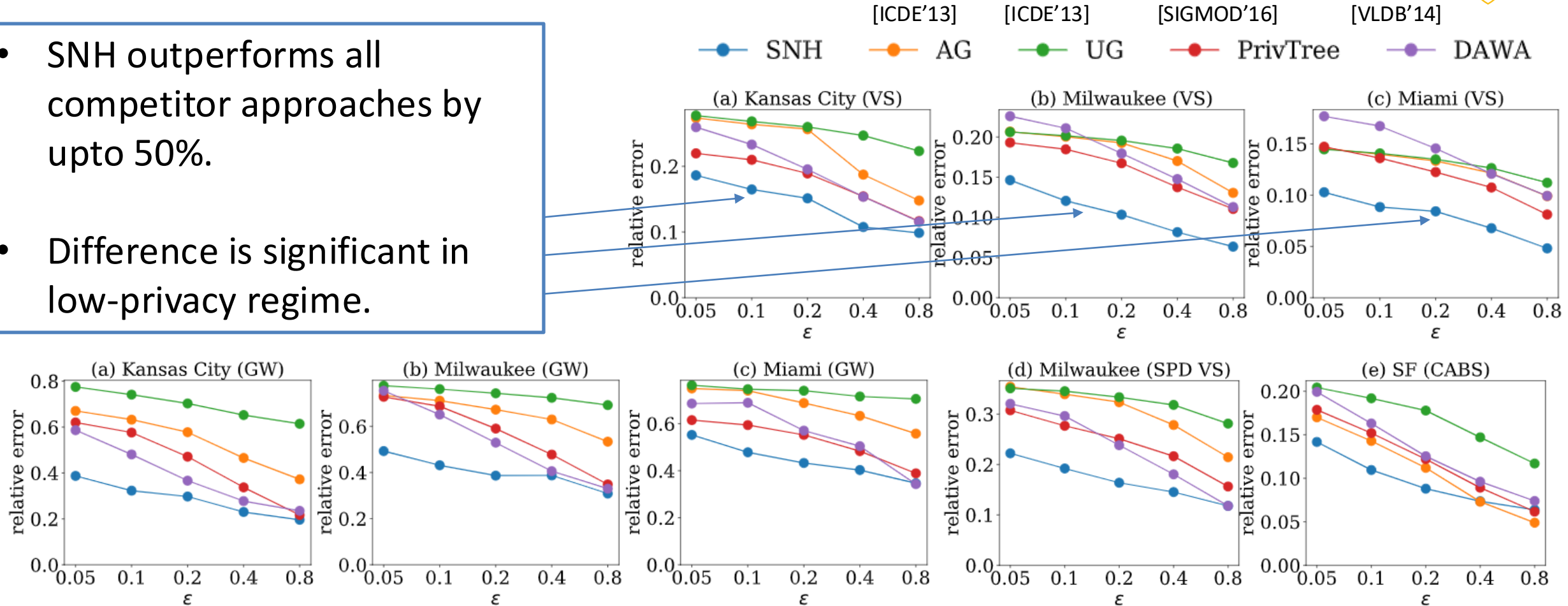


Comparison with baselines

Impact of Privacy Budget ϵ



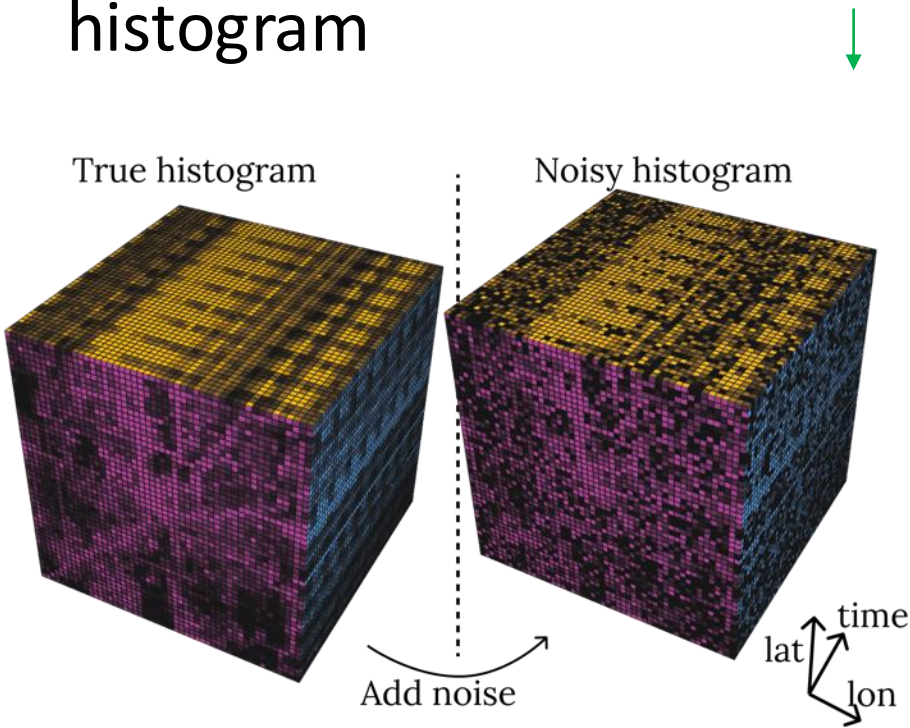
- SNH outperforms all competitor approaches by upto 50%.
- Difference is significant in low-privacy regime.



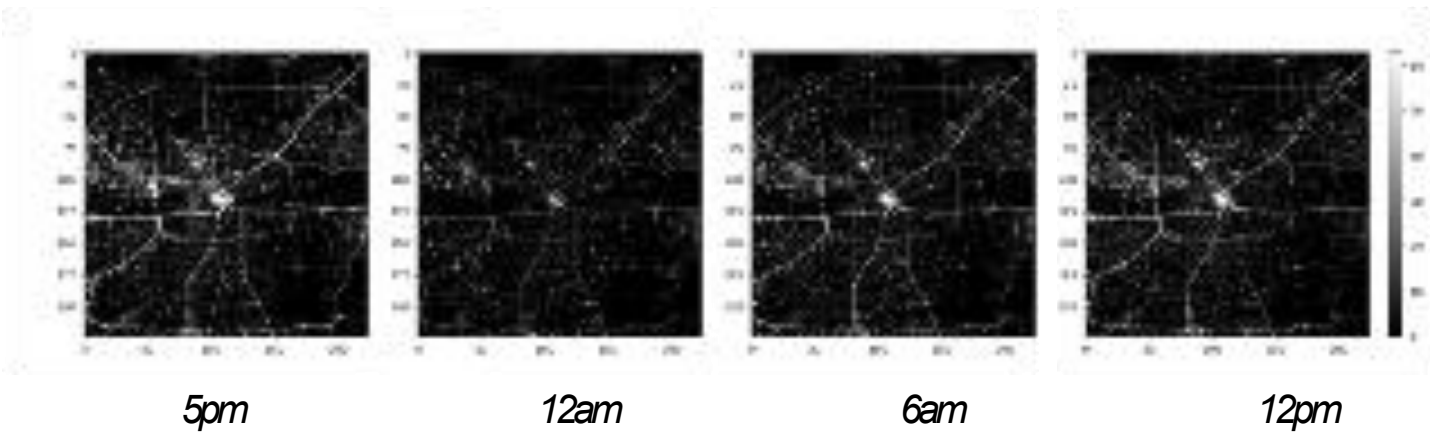


Spatio-Temporal Data Release

- Release a differentially private 3-dimensional histogram



User_id	Latitude	Longitude	Timestamp
John	37.7920	-122.3927	10/11 20:32
Kyle	37.7930	-122.3827	10/11 20:33
John	37.7936	-122.3224	10/11 21:45
...	...		...
John	37.7143	-122.3687	10/11 23:50

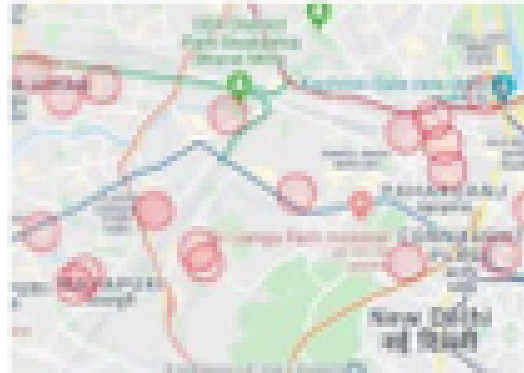


Variational Autoencoder-Based Density Release (VDR)



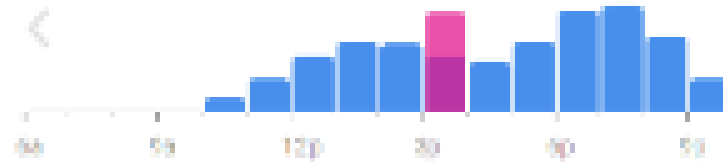
- Allows arbitrary query types, e.g., Range Count Queries at time instances and more:

Hotspot
discovery



Popular Times: Wednesdays +

LIVE Bigger than usual



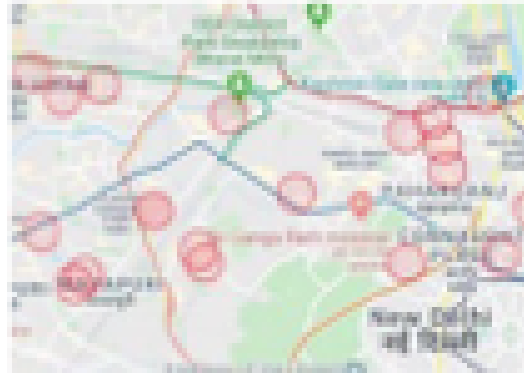
POI visits
forecasting

Variational Autoencoder-Based Density Release (VDR)



- Allows arbitrary query types, e.g., Range Count Queries at time instances and more:

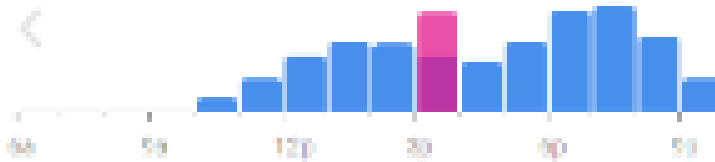
Hotspot
discovery



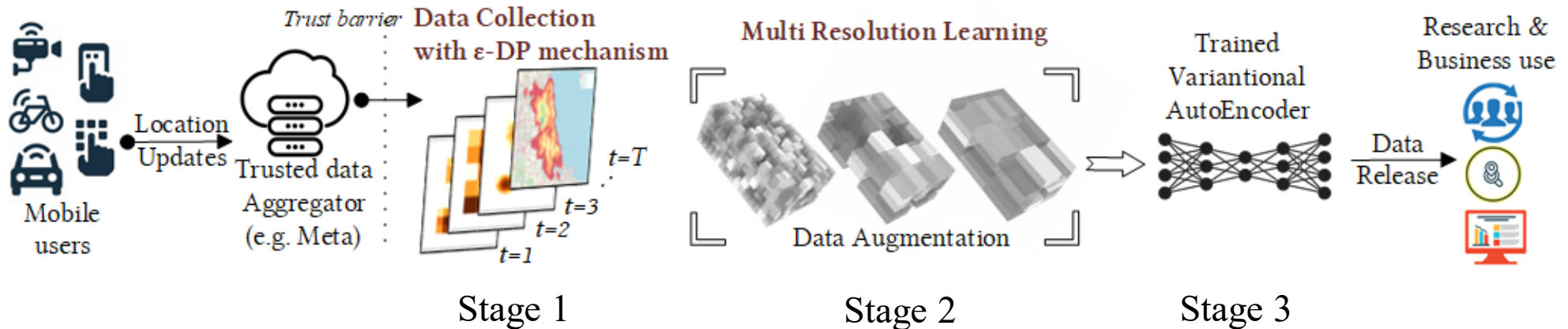
Popular Times: Wednesday

LIVE

Bustier than usual

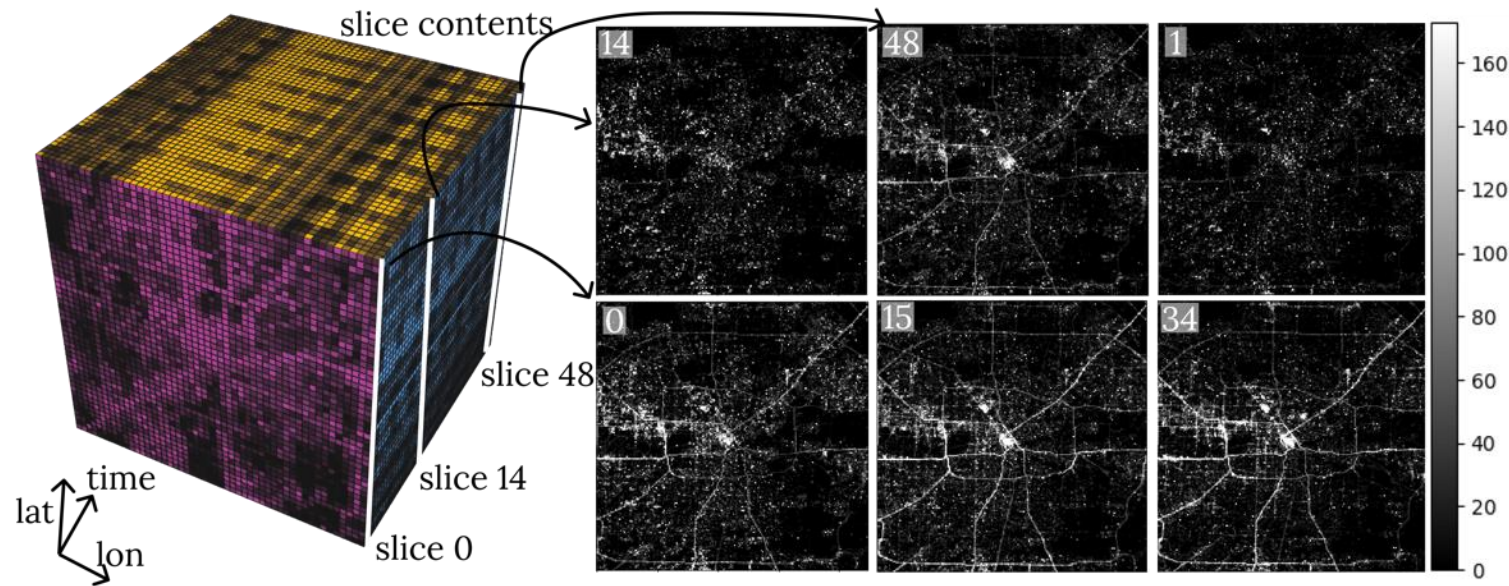


POI visits
forecasting



Stage 3: Learned Denoising

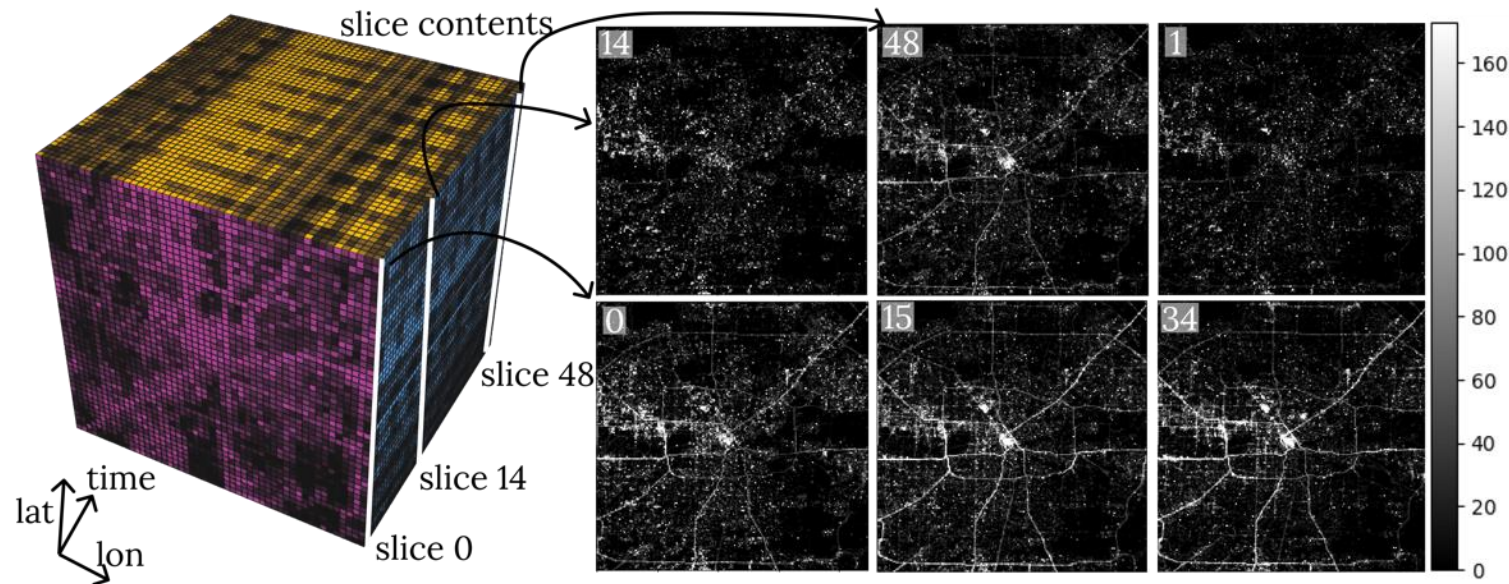
Spatial Patterns as Visual patterns



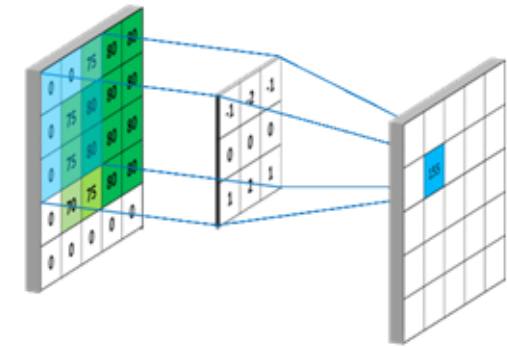
Spatio-temporal location data can be viewed as a series of images.
We utilize lessons from image feature extraction literature.

Stage 3: Learned Denoising

Spatial Patterns as Visual patterns



Spatio-temporal location data can be viewed as a series of images.
We utilize lessons from image feature extraction literature.

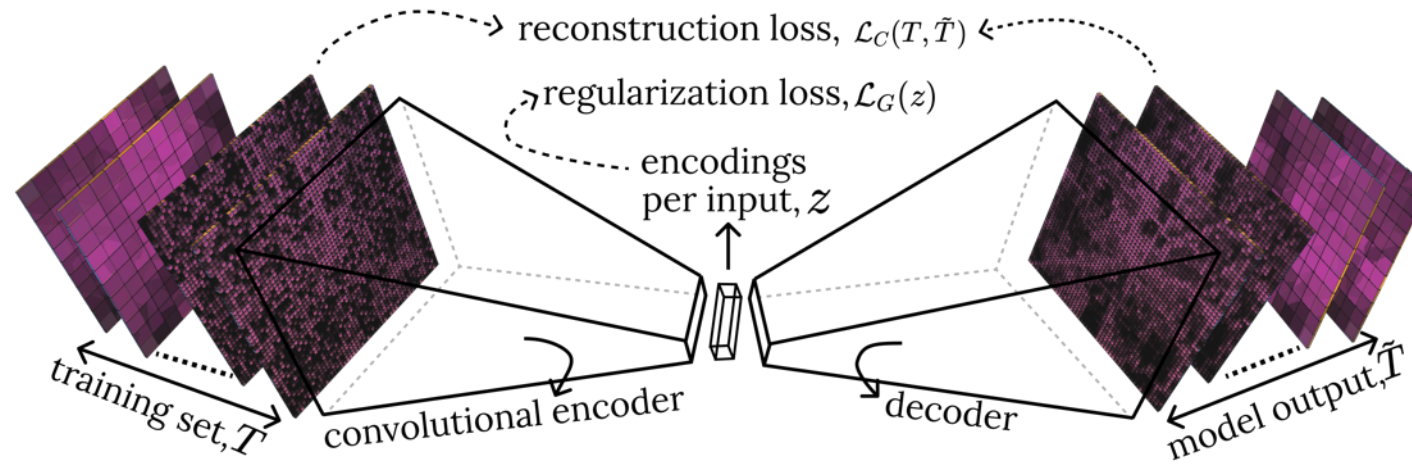


Stage 3 : Learned Denoising

Given: Noisy histogram $\bar{H}$

Constraint: With no knowledge of true histogram H

Objective: Obtain a histogram $\hat{H}$ such that with lower error than $\hat{H}$, $\| \hat{H} - H \| < \| \bar{H} - H \|$



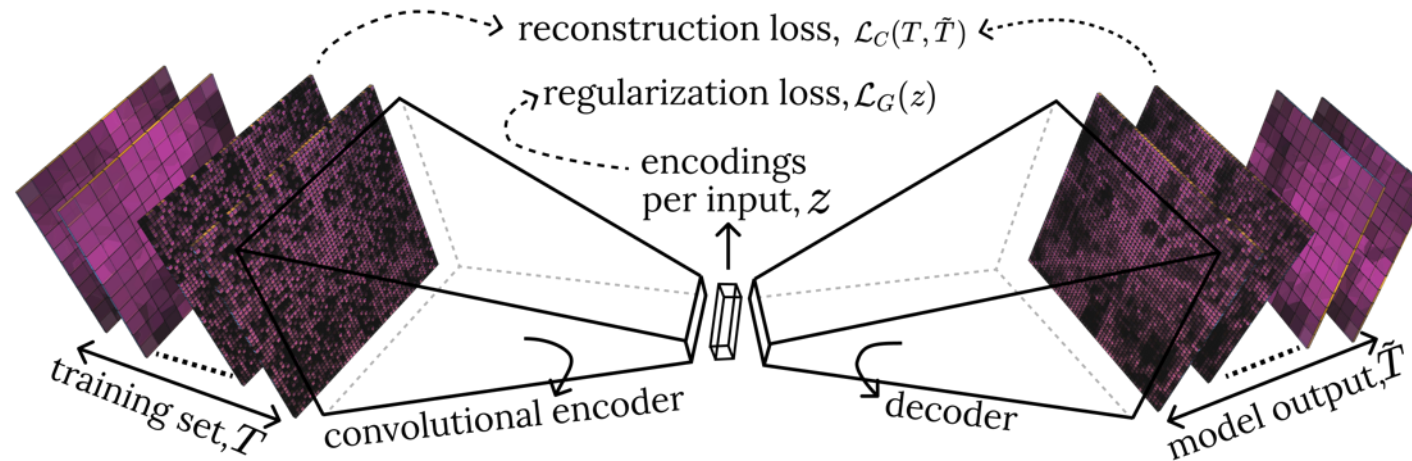
Stage 3 : Learned Denoising

Given: Noisy histogram $\bar{H}$

Constraint: With no knowledge of true histogram H

Final histogram
released publicly

Objective: Obtain a histogram $\hat{H}$ such that with lower error than $\hat{H}$, $\| \hat{H} - H \| < \| \bar{H} - H \|$



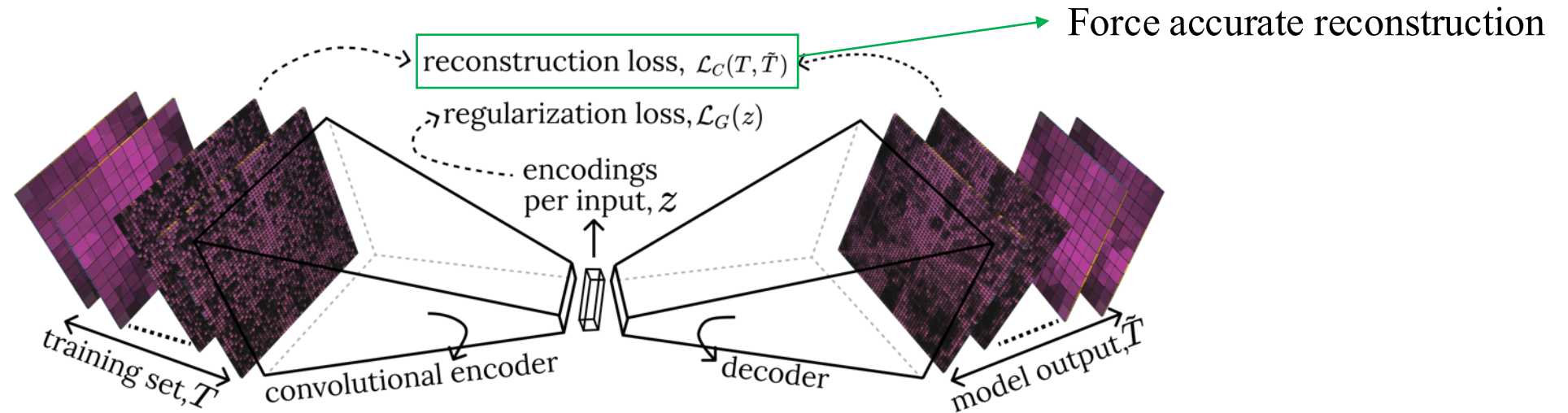
Stage 3 : Learned Denoising

Given: Noisy histogram $\bar{H}$

Constraint: With no knowledge of true histogram H

Final histogram
released publicly

Objective: Obtain a histogram $\hat{H}$ such that with lower error than $\hat{H}$, $\|\hat{H} - H\| < \|\bar{H} - H\|$



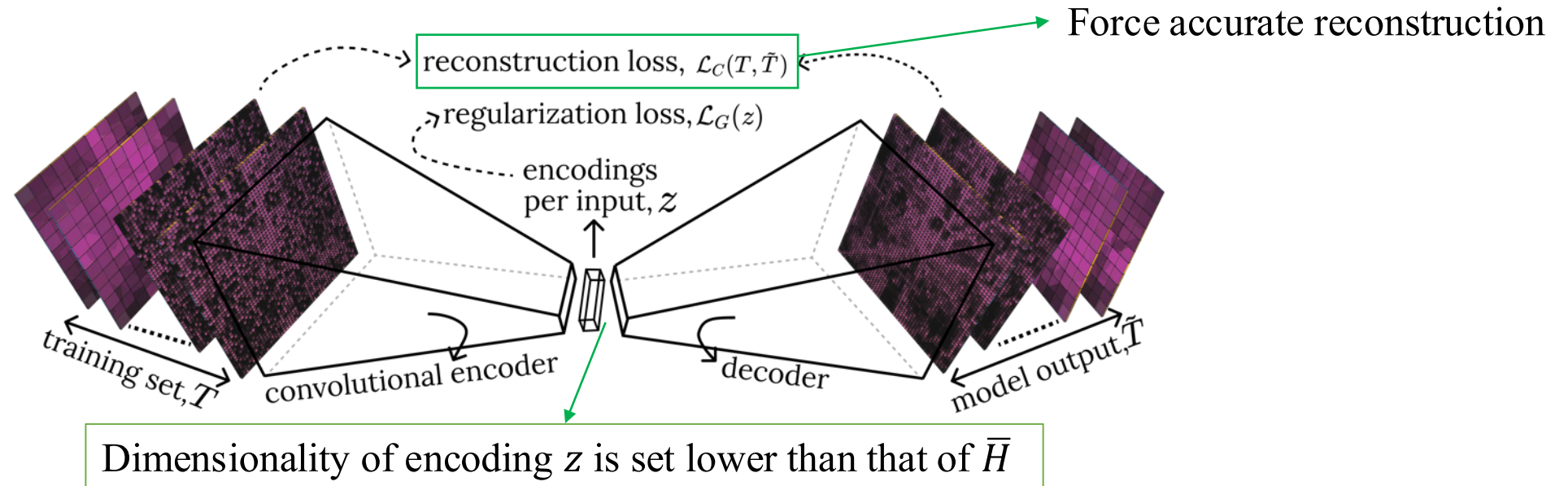
Stage 3 : Learned Denoising

Given: Noisy histogram $\bar{H}$

Constraint: With no knowledge of true histogram H

Final histogram
released publicly

Objective: Obtain a histogram $\hat{H}$ such that with lower error than $\hat{H}$, $\| \hat{H} - H \| < \| \bar{H} - H \|$



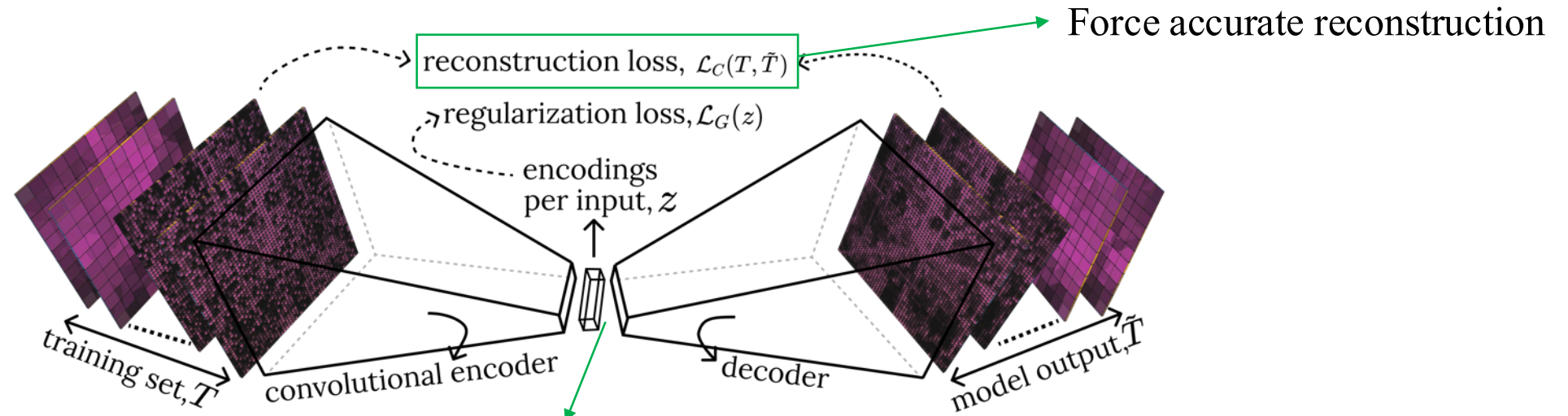
Stage 3 : Learned Denoising

Given: Noisy histogram $\bar{H}$

Constraint: With no knowledge of true histogram H

Final histogram
released publicly

Objective: Obtain a histogram $\hat{H}$ such that with lower error than $\hat{H}$, $\|\hat{H} - H\| < \|\bar{H} - H\|$



Dimensionality of encoding z is set lower than that of $\bar{H}$

z cannot contain as much information as $\bar{H}$ (especially DP-noise i.i.d. added to each cell)

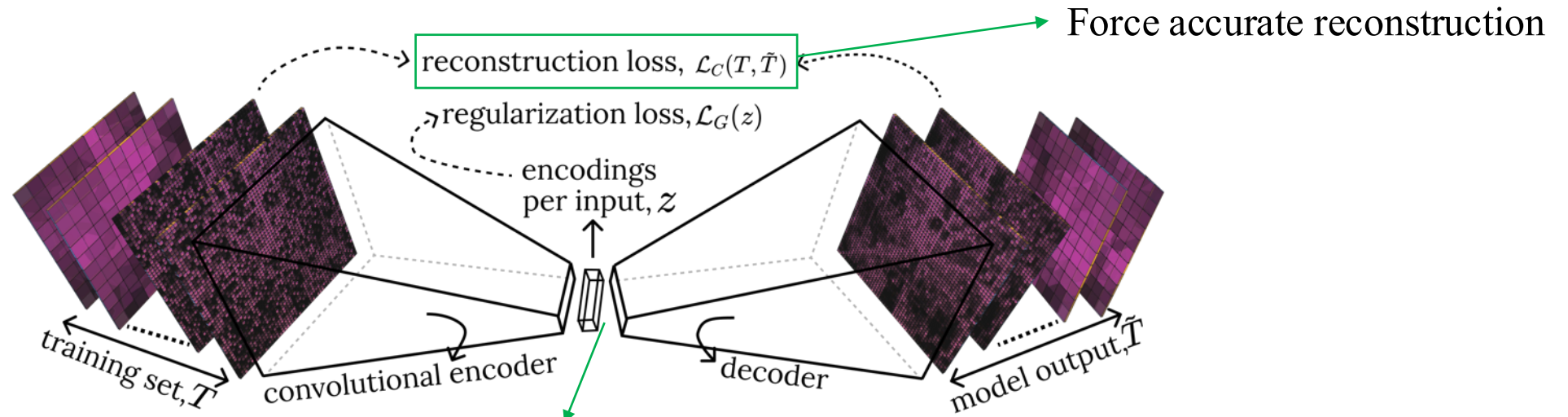
Stage 3 : Learned Denoising

Given: Noisy histogram $\bar{H}$

Constraint: With no knowledge of true histogram H

Final histogram
released publicly

Objective: Obtain a histogram $\hat{H}$ such that with lower error than $\hat{H}$, $\|\hat{H} - H\| < \|\bar{H} - H\|$



Dimensionality of encoding z is set lower than that of $\bar{H}$

z cannot contain as much information as $\bar{H}$ (especially DP-noise i.i.d. added to each cell)

Need to learn repeatable patterns to maximize accuracy

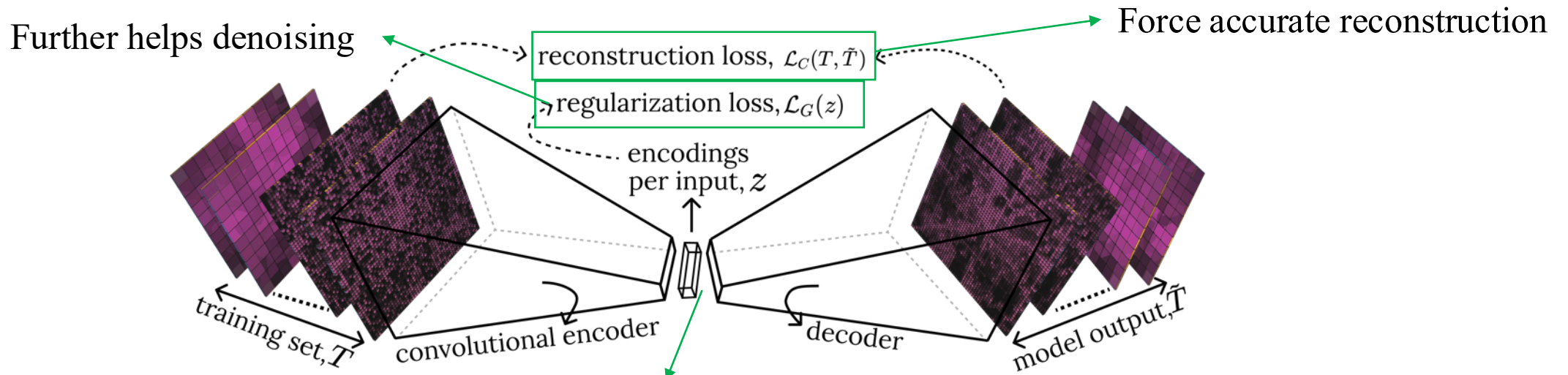
Stage 3 : Learned Denoising

Given: Noisy histogram $\bar{H}$

Constraint: With no knowledge of true histogram H

Final histogram
released publicly

Objective: Obtain a histogram $\hat{H}$ such that with lower error than $\bar{H}$, $\|\hat{H} - H\| < \|\bar{H} - H\|$



Dimensionality of encoding z is set lower than that of $\bar{H}$

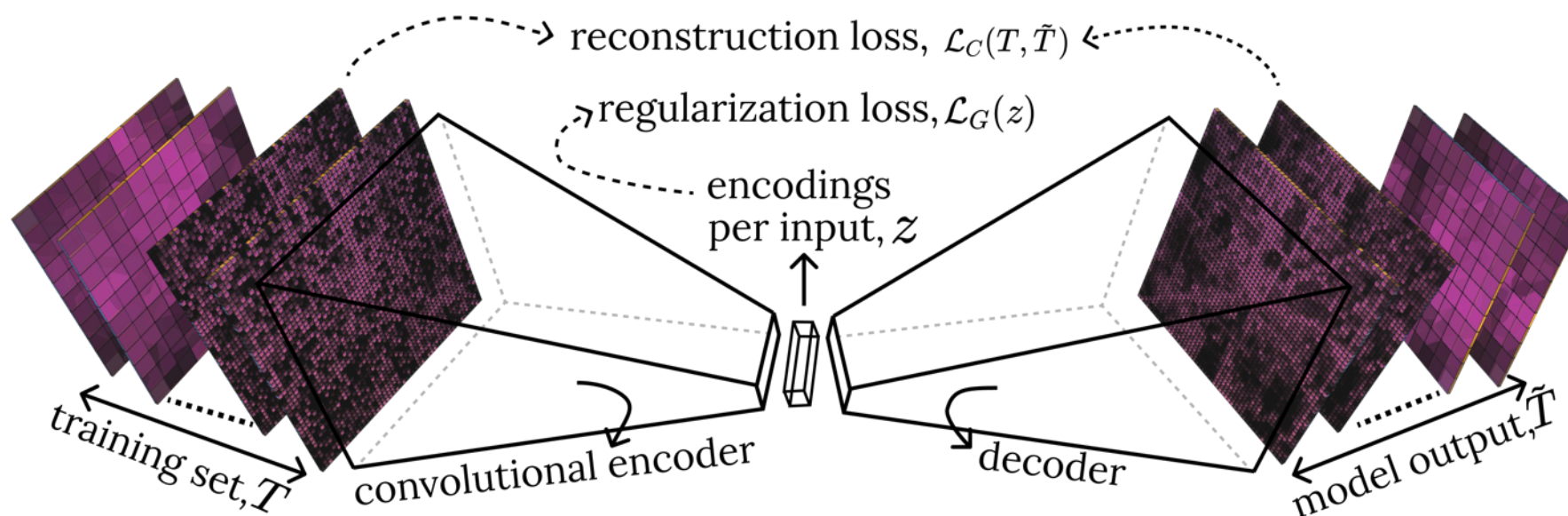
z cannot contain as much information as $\bar{H}$ (especially DP-noise i.i.d. added to each cell)

Need to learn repeatable patterns to maximize accuracy



VDR Features

- Does not introduce bias from complex domain partitioning
- Exploit spatial patterns to reduce variance (i.e., denoise) by learning a VAE
- Explicitly account for user-level privacy (compared with event-level privacy)



Q&A



Thanks!



References

- [Bordenabe et. al., CCS 2014] Bordenabe et. al. “Optimal Geo-Indistinguishable Mechanisms for Location Privacy”. CCS 2014
- [Andres et. al., CCS 2013] Andres et. al. “Geo-indistinguishability: differential privacy for location-based systems” CCS 2013
- Sepanta Zeighami, Ritesh Ahuja, Gabriel Ghinita, Cyrus Shahabi: A Neural Database for Differentially Private Spatial Range Queries. In VLDB 2022
- Sepanta Zeighami, Ritesh Ahuja, Gabriel Ghinita, Cyrus Shahabi: A Neural Approach to Spatio-Temporal Data Release with User-Level Differential Privacy, In SIGMOD 2023
- Some slides borrowed from ICDM08 tutorial by Mohamed F. Mokbel